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A thesis presented for the degree of
Bachelor of Science (Honours) (Physics)

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August 18, 2026

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Tous les fichiers svg :

Dedication

For those who hate looking at a template with 500 lines of code and an extra 300 lines commented out.

Declaration

Acknowledgements

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Introduction

sur optical clocks: Degenerate quantum gases of strontium de Schreck (et même un peu tout le contexte) Quantum simulation field has grown the past few decades with the emergence of second quantization. Plenty of groups of cold atoms experiments have exploited the quantum properties of atoms to simulate condensed matter systems, quantum magnetism, and many-body physics.

Chapter 1

Gas preparation and spin control

1.1 ^{87}Sr Properties and interest

Strontium is an alkaline-earth-like element with several stable isotopes well documented in the literature (e.g., [5, 6, 7, 8, 9]). The majority of these isotopes are bosonic (^{88}Sr, ^{86}Sr, and ^{84}Sr) with zero nuclear spin ($I = 0$).

In our work, we focus on the fermionic isotope ^{87}Sr, which has a natural abundance of 7.0%. In contrast to its bosonic counterparts, ^{87}Sr possesses a large nuclear spin $I = 9/2$, providing access to rich multi-level quantum dynamics ($SU(N)$ manifolds).

In the ground state $^1S_0$, the total electronic angular momentum is zero ($J = 0$), making the atomic state almost insensitive to stray magnetic field fluctuations—a major advantage for mitigating environmental decoherence. However, this absence of an electronic magnetic moment also implies that magnetic fields alone cannot drive direct spin transitions via standard Zeeman manipulation. Specific optical techniques must therefore be developed to circumvent this limitation.

Furthermore, the decoupling between the nuclear spin and the electronic shell in the ground state leads to an $SU(N)$ collisional symmetry: the elastic scattering properties are strictly independent of the nuclear spin state m_F. The spin manipulation techniques developed by our group [10] exploit this symmetry to yield a highly controllable quantum system whose internal spin dynamics remain decoupled from the spatial dynamics of the cloud once thermalized.

Preparation of the cold gas follows the standard scheme of ultracold gas experiments. For $t_{\mathrm{BlueMot}} = 8.2$ s, 2D transverse cooling (TC) collimates atoms prior to a Zeeman slowing stage, followed by a blue Magneto-Optical Trap (MOT), operating on the broad $^1S_0 \to {}^1P_1$ transition (see Fig. 1.1). Following this blue stage, repumping light is applied for $t_{\mathrm{Repump}} =$ time, swept across multiple hyperfine states of the $^3P_2 \to {}^3D_2$ transition to recover approximately 25% of the atoms trapped in the metastable $^3P_2$ state. The atoms are then cooled to temperatures close to $T =$ temperature by a dynamic red MOT for $t^{\mathrm{tot}}_{\mathrm{Red}} =$ time on the narrow $^1S_0 \to {}^3P_1$ transition. Finally, the gas is loaded and evaporated inside an Optical Dipole Trap (ODT).

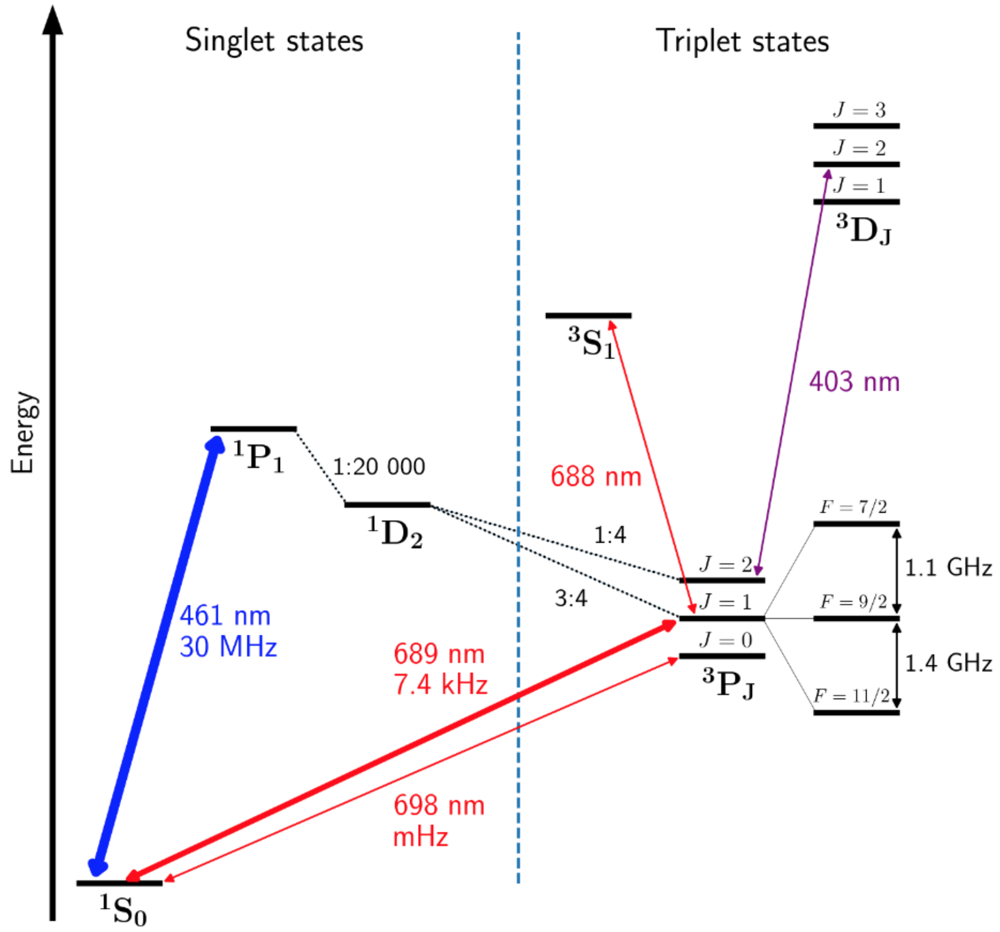


Figure 1.1: Relevant low-lying electronic energy levels and optical cooling transitions for ^{87}Sr.

In this chapter, I describe each stage of the cooling and loading process from a practical "user guide" perspective, concluding with an explanation of the spin manipulation schemes tailored to our experimental goals: coherent control of quantum systems, precision measurement, and photoassociation spectroscopy of ^{87}Sr.

1.2 Fermi gas preparation

Ajouter ce qui est judicieux d'optimiser à chaque étape : nombre d'atomes, T, T/T_F, nombre d'atomes final chargé à chaque étape, vitesse.

Solid strontium is vaporized in an oven heated to 490°C, outputting an atomic beam through a cluster of fine capillaries that provide initial spatial collimation.

1.2.1 Blue optical setup

The first cooling stages operate on the main $^1S_0 \to {}^1P_1$ transition using a master-slave laser architecture (see Fig. 1.2). The master laser is an Extended-Cavity Diode Laser (ECDL, New Focus Vortex) frequency-stabilized via saturated absorption spectroscopy on ^{88}Sr [11]. It seeds three high-power slave laser diodes, forcing them to inherit the master's narrow linewidth and central frequency while delivering the high output power needed for the cooling beams. This stage brings the atomic beam to a mean longitudinal velocity of v at an equivalent beam temperature of $T =$.

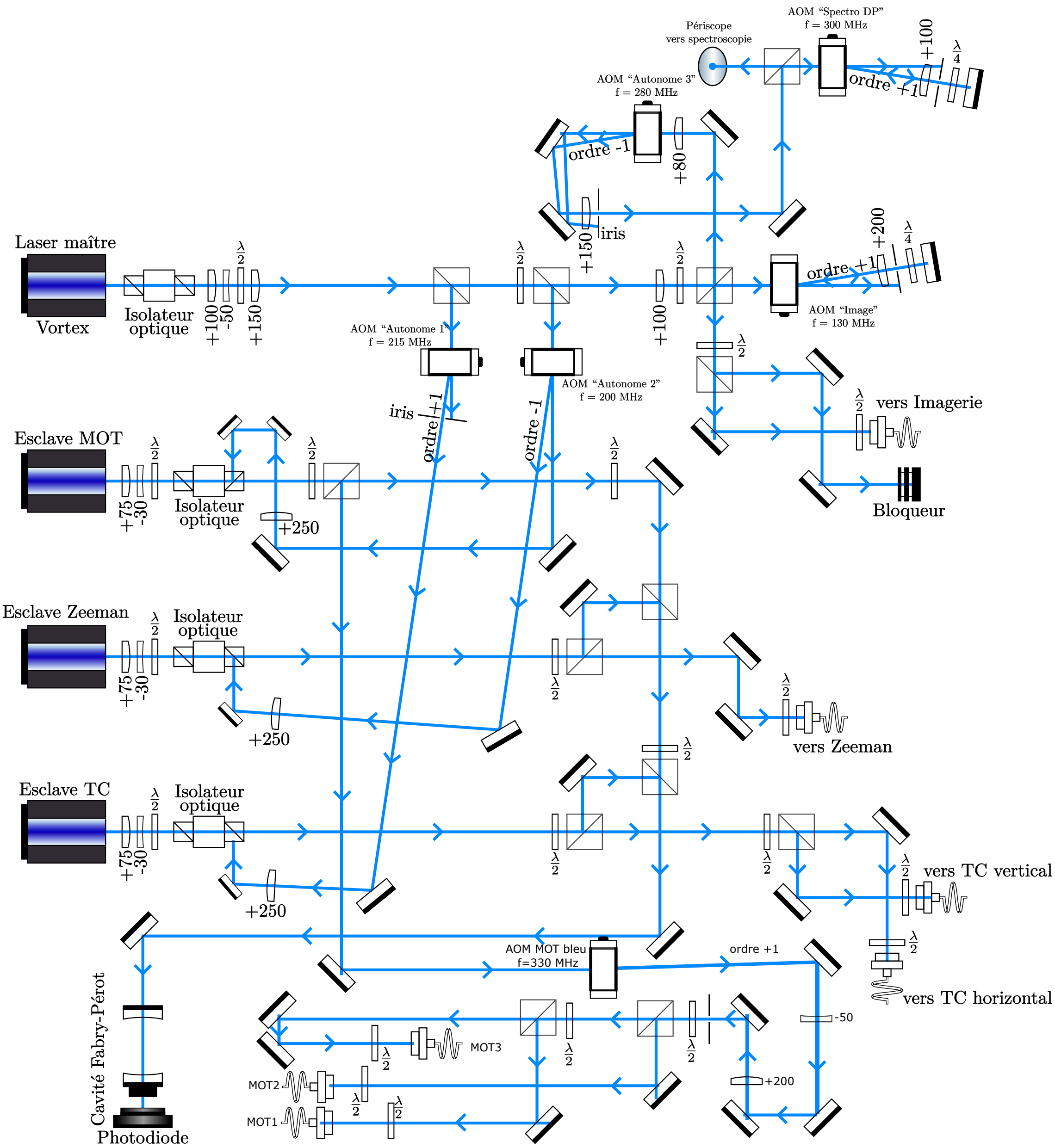


Figure 1.2: Optical layout of the 461 nm blue laser system for transverse cooling, Zeeman slowing, and blue MOT.

Ajouter schémas des fréquences.

1.2.2 Transverse cooling

Following extraction from the oven capillaries, the atomic beam is collimated using 2D transverse cooling (TC). This stage employs two orthogonal pairs of counter-propagating laser beams perpendicular to the main longitudinal atomic propagation axis (z-axis), reducing the transverse velocity spread $v_\perp$ via photon recoil recoil momentum ($\hbar k_\perp$).

1.2.2.1 Operating principle and radiation oressure

The collimation relies on Doppler cooling. An atom moving with a transverse velocity $\vec{v}_\perp$ experiences a Doppler shift $\Delta_D = -\vec{k}_\perp \cdot \vec{v}_\perp$. As illustrated in Fig. 1.3, atomic velocity classes observe shifted transition resonances centered at $\omega_0 \pm \vec{k}_\perp \cdot \vec{v}_\perp$. When the laser frequency is red-detuned ($\Delta < 0$) with respect to the atomic transition frequency (ω_0), the atom preferentially absorbs photons from the counter-propagating beam, generating a net damping force.

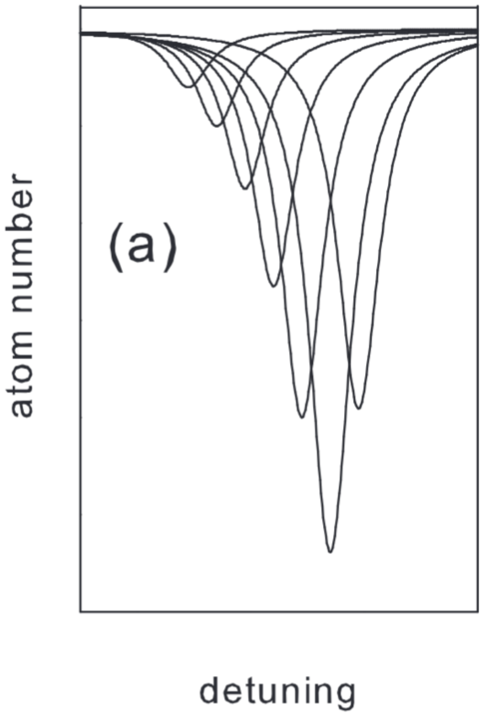


Figure 1.3: Illustration of Doppler cooling: velocity classes are shifted relative to the atomic resonance frequency ω_0.

The underlying radiation pressure force acting on the atoms is given by:

$$F_{\text{rad}}(v_\perp) = \frac{\hbar k_\perp \Gamma}{2} \frac{s}{1 + s + 4\left(\frac{\Delta_{\text{eff}}(v_\perp)}{\Gamma}\right)^2} \tag{1.1}$$

where Γ is the natural linewidth of the transition, $s = I/I_{\text{sat}}$ is the saturation parameter, and $\Delta_{\text{eff}} = \Delta - \vec{k}_\perp \cdot \vec{v}_\perp$ is the effective detuning ($\Delta = \omega_{\text{laser}} - \omega_0$). In the low-intensity limit ($s \ll 1$), standard Doppler cooling theory predicts a maximum damping force gradient at a detuning of $\Delta_{\text{eff}} = -\Gamma/2$.

1.2.2.2 Experimental setup and laser configuration

For this stage, we drive the broad-linewidth ${}^1S_0 \rightarrow {}^1P_1$ transition ($\Gamma/2\pi = 30.5$ MHz, $I_{\text{sat}} \approx 40$ mW/cm^2). The laser frequency f_{TC} is generated by shifting the main blue master frequency using an acousto-optic modulator (AOM) driven at $f_{\text{TC}}^{\text{RF}}$:

$$f_{\text{TC}} = f_{\text{Blue,Master}} + f_{\text{TC}}^{\text{RF}} \tag{1.2}$$

Due to anisotropic geometric constraints of our atomic beam, the intensity in the horizontal arm is set roughly three times higher than in the vertical arm, puissance pour nI_{sat} though both remain below saturation ($I \ll I_{\text{sat}}$) over an interaction area of approximately 10 cm^2. The TC laser remains active throughout subsequent blue-cooling stages to ensure continuous atomic flux collimation.

1.2.2.3 Detuning and power optimization

While ideal two-level theory predicts an optimal detuning of $\Delta = -\Gamma/2$, experimental constraints alter this optimum:

- **Power limitations over large beam areas:** Operating near saturation ($I \sim I_{\text{sat}}$) would maximize the velocity capture range through power broadening. However, illuminating the 10 cm^2 interaction area at saturation would require over 400 mW of optical power. Consequently, the system operates in the low-intensity regime ($I \ll I_{\text{sat}}$).
- **Optimal detuning shift:** Experimental characterization reported in F. Faisant's thesis [1] demonstrated that maximum collimation efficiency (measured by downstream atom yield) is achieved at $\Delta = -\Gamma/4$ rather than $-\Gamma/2$, as shown in Fig. 1.4. This plot also illustrates how the maximum gain improves with total beam power.

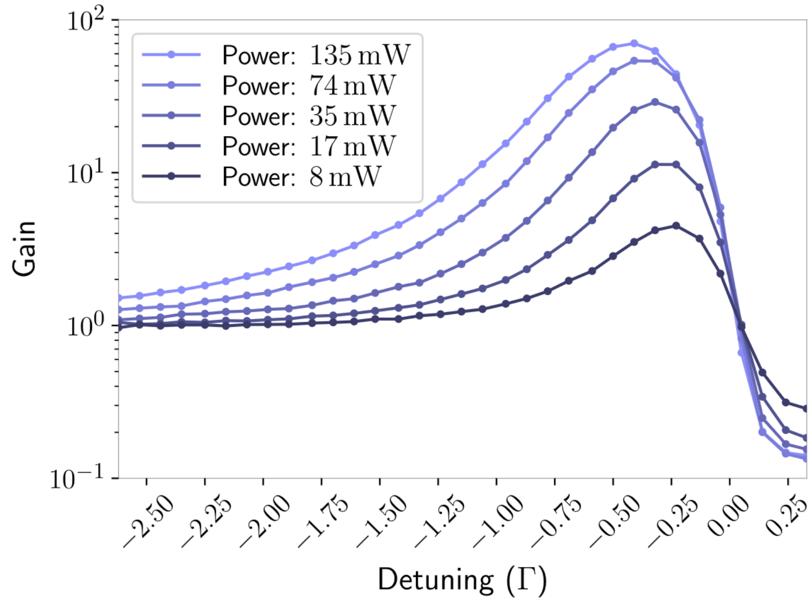


Figure 1.4: Transverse cooling performance as a function of laser detuning and power (adapted from [1]).

Future upgrades in progress include replacing the current TC slave diode with a higher-power diode and re-optimizing Δ_{eff} to further enhance atom capture.
Following transverse collimation, the atomic beam enters the Zeeman slower stage along the main longitudinal axis.

1.2.3 Zeeman cooling

Valeur du gradient de champ B? Following transverse collimation, atoms are decelerated longitudinally using a Zeeman slower [12]. The mechanism relies on a counter-propagating laser beam maintained on resonance with the decelerating atoms as they travel through a spatially varying magnetic field gradient $B(z)$ generated by a tapered solenoid.

To compensate for the decreasing Doppler shift $kv(z)$ as atoms slow down, the magnetic field $B(z)$ shifts the Zeeman sub-levels to satisfy the resonance condition along the entire deceleration path:

$$\Delta_{\text{eff}}(z) = \omega_{\text{laser}} - \omega_0 + kv(z) - \frac{\mu_B g_J B(z)}{\hbar} = 0 \tag{1.3}$$

At the slower entrance, atoms exhibit a mean velocity of $v_0 = 434$ m/s [13]. As the field $B(z)$ drops toward zero near the main cell center to avoid disturbing downstream trapping, atoms above the capture velocity can pass through this zero-field zone. To capture these remaining fast atoms, supplementary coils with increasing radii produce an additional field gradient that shifts Zeeman levels back into resonance, ensuring deceleration down to the final capture velocity.

1.2.4 Blue MOT

The Magneto-Optical Trap (MOT) provides simultaneous spatial confinement and cooling.

The blue MOT combines a magnetic quadrupole field (generated by anti-Helmholtz coils centered on the vacuum chamber) with three orthogonal pairs of counter-propagating 461 nm laser beams carrying opposite circular polarizations (σ^+/σ^-) [14].

The position-dependent Zeeman shift breaks the spatial degeneracy of the sub-levels:

$$\Delta E(z) = -m_F g_F \mu_B B(z) \tag{1.4}$$

Combined with red-detuned laser beams, this spatial shift creates a restoring force toward the trap center ($z = 0$) alongside the velocity-dependent radiation pressure force.

Gravity compensation and vertical spatial confinement are provided by the magnetic field gradient itself. Because the quadrupole coils produce a steeper gradient along their symmetry axis (z) than in the horizontal plane (x, y), half as much intensity is delivered along the vertical axis compared to the horizontal axes. This ratio achieves a balanced, quasi-spherical trap while minimizing power broadening and optical density effects.

During this stage, the cloud reaches the Doppler temperature limit [15]:

$$k_B T_D = \frac{\hbar \Gamma_{^1S_0 \to ^1P_1}}{2} \tag{1.5}$$

Due to the broad linewidth of the 461 nm transition ($\Gamma/2\pi = 30.5$ MHz), this Doppler temperature ($T_D \approx 770\ \mu$K) remains significantly higher than the single-photon recoil limit T_r.

1.2.5 Repumper

To sustain the blue MOT cycle, a repumping laser system is required: approximately 1 out of every 150,000 excitations to $^1P_1$ decays into the metastable $^3P_2$ state (lifetime $\tau \approx 500$ s) via the intermediate $^1D_2$ state. Without repumping, atoms become dark to the 461 nm light and are lost from the cooling cycle.

1.2.5.1 Hyperfine structure of Strontium-87

Because fermionic ^{87}Sr possesses a nuclear spin $I = 9/2$, its electronic states exhibit a rich hyperfine structure that must be addressed for efficient repumping.

Repumping is provided by a 403 nm external-cavity diode laser (TOPTICA). To address multiple F hyperfine sub-levels (see Fig. 1.5), the laser frequency is rapidly swept over a 3 GHz bandwidth.

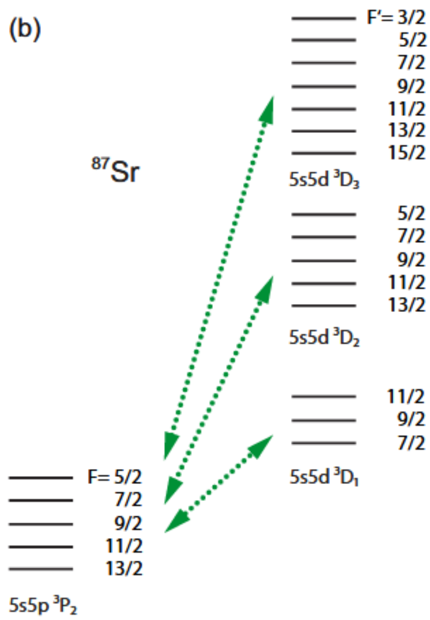


Figure 1.5: Energy level diagram of ^{87}Sr showing the primary 461 nm cooling transition and the 403 nm repumping pathway from $^3P_2$ to $^3D_2$.

1.2.5.2 Optimization of the repumper sweep

The $^3P_2$ hyperfine levels are unequally populated during blue MOT decay due to their differing g_F Landé factors, affecting their magnetic trapping efficiency [16].

The repumper frequency scan range is optimized around 3 GHz to selectively cover the $F = 11/2$ and $F = 13/2$ states. Scanning a wider range reduces total atom recovery by decreasing the dwell time on these critical transitions. At the end of the blue MOT phase, the 461 nm beam is extinguished while the 403 nm repumper remains active for $\approx$ 200 ms at $I \approx 15$ mW/cm^2, returning trapped metastable atoms to the $^1S_0$ ground state before starting the red MOT.

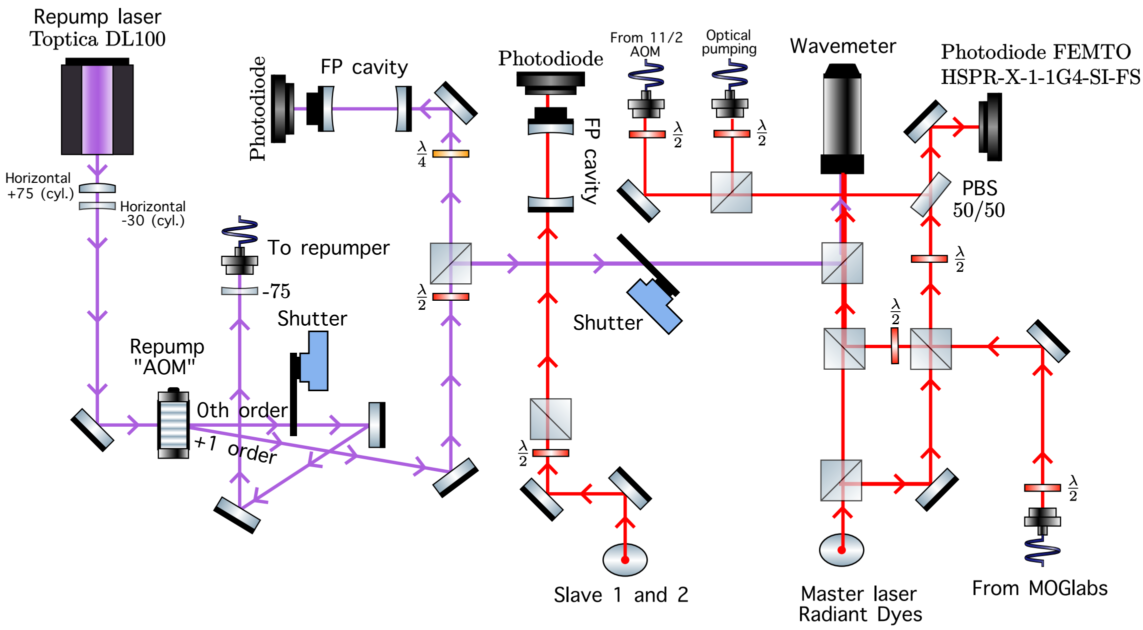


Figure 1.6: Summary of repumping beam frequencies and timing sequence.

1.2.6 Red MOT

1.2.6.1 Optical setup

Because the blue MOT temperature is bounded by $T_D \approx 770$ μK, further cooling requires transferring atoms to the narrow intercombination transition $^1S_0 \rightarrow {}^3P_1$ at 689 nm ($\Gamma = 2\pi \times 7.4$ kHz).

For this transition, the formal Doppler limit ($T_D = \hbar\Gamma/2k_B = 179$ nK) falls below the single-photon recoil limit ($T_r = \hbar^2 k^2/mk_B = 283$ nK). Consequently, the ultimate red MOT temperature is bounded by the photon recoil limit.

The 689 nm light is generated by a commercial laser locked to an ultra-stable optical cavity. The frequency is offset-shifted by $f = -600$ MHz using a first AOM before injection into a master-slave system. A double-pass AOM fine-tunes the frequency before splitting into three spatial counter-propagating beam pairs.

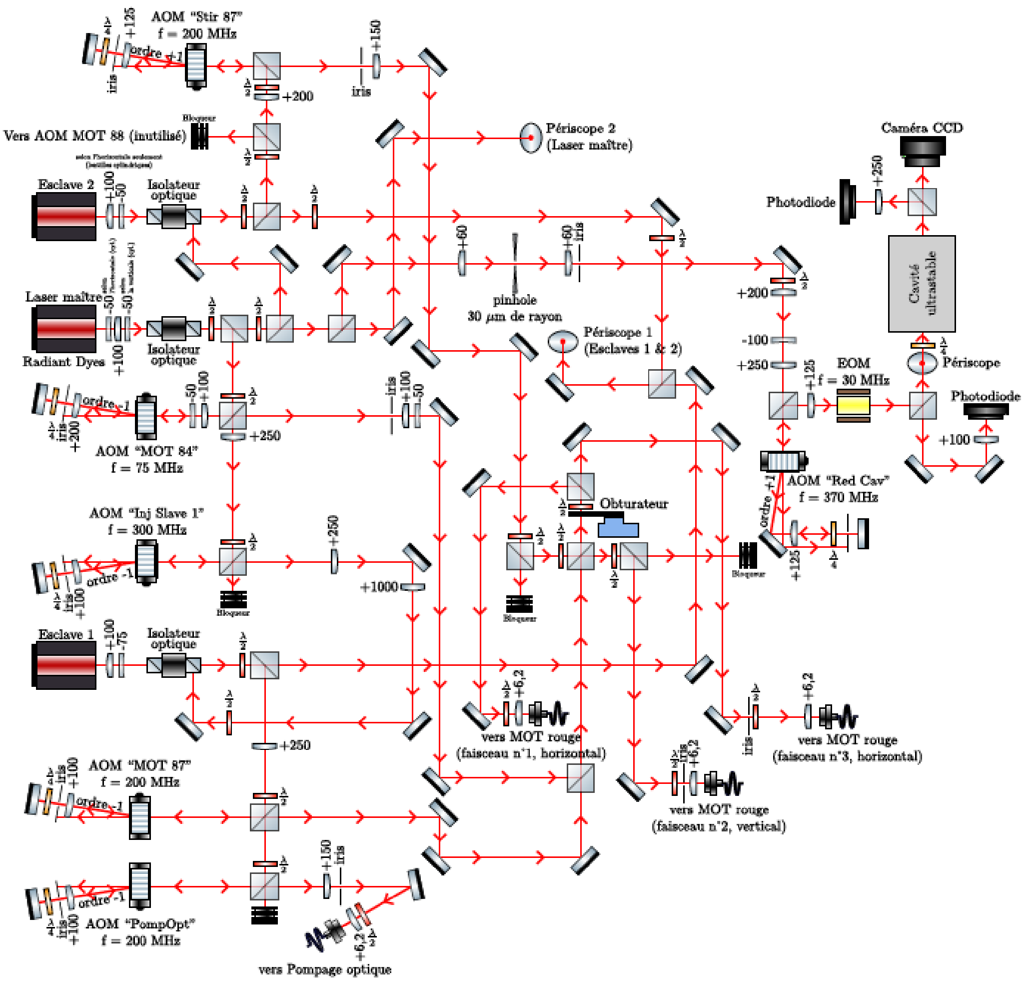


FIGURE 14 – Préparation des différents lasers rouges

Figure 1.7: Optical layout of the 689 nm laser cooling system.

1.2.6.2 Broadband MOT

At the end of the blue MOT, the atomic cloud's spatial spread (σ = value) and thermal energy exceed the capture velocity of a single-frequency 689 nm MOT.

To capture these thermal atoms, a broadband MOT is implemented on the ${}^1S_0(F = 9/2) \to {}^3P_1(F = 11/2)$ transition under a weak magnetic field gradient. The laser spectrum is artificially broadened by applying a sawtooth frequency modulation red-detuned from resonance. As the cloud cools and compresses, both the modulation amplitude Δf and total optical power are ramped down.

1.2.6.3 Stirring laser

Simultaneously, a "stirring" laser beam is applied to redistribute populations among magnetic sub-levels. Because the $F = 9/2 \to F = 11/2$ cooling transition contains weak or non-absorbing Zeeman sub-states at the cloud edges, atoms accumulated in these states experience reduced radiation pressure and escape the trap (Fig. 1.8).

The stirring laser drives the ${}^1S_0(F = 9/2) \to {}^3P_1(F = 9/2)$ transition to optically pump and randomize m_F populations (Fig. 1.9), restoring efficiency to the cooling cycle.

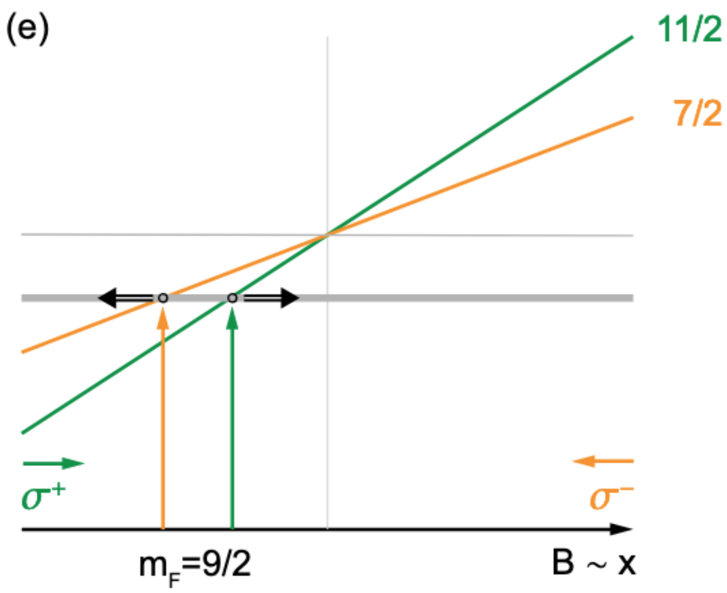


Figure 1.8: Mechanism of population trapping in extreme Zeeman sub-states during the red MOT.

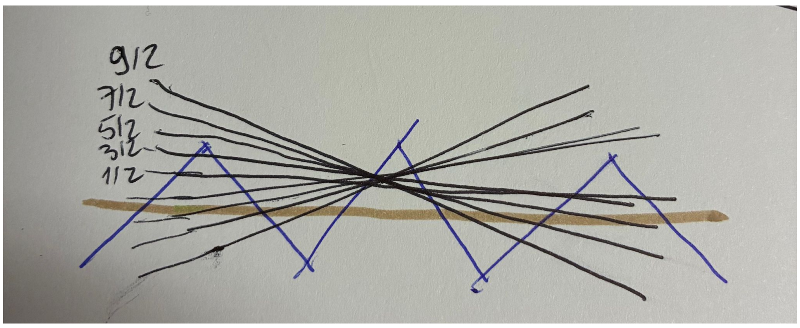


Figure 1.9: Randomization of m_F populations induced by the 689 nm stirring laser.

Because the narrow linewidth of the intercombination transition ($\Gamma = 2\pi \times 7.4$ kHz) strictly bounds the saturation radiation pressure force to a theoretical maximum of $F_{\text{max}} = \hbar k \Gamma/2 \approx 1.6\, mg$, gravity plays a critical role during this stage.

In practice, to minimize power broadening and reach microkelvin temperatures, the red MOT operates at low saturation parameters ($s \lesssim 1$). Under these conditions, a symmetric beam intensity would yield a radiation pressure force weaker than gravity ($F_{\text{rad}} < mg$), causing the atoms to fall out of the trap. To overcome this limitation, a significantly higher intensity is delivered specifically along the vertical axis to maintain $F_{\text{rad}} > mg$ and support the cloud against gravity, while lower intensities are maintained on the horizontal axes to prevent unnecessary power broadening and transverse heating.

1.2.6.4 Narrowband MOT (Single-frequency MOT)

Once the cloud is sufficiently cold, frequency modulation is switched off to operate a single-frequency (monofrequency) MOT. This stage reduces the temperature to 1–2 μK (near the recoil limit) with a peak density of $n =$ value and an RMS width of $\sigma =$ value.

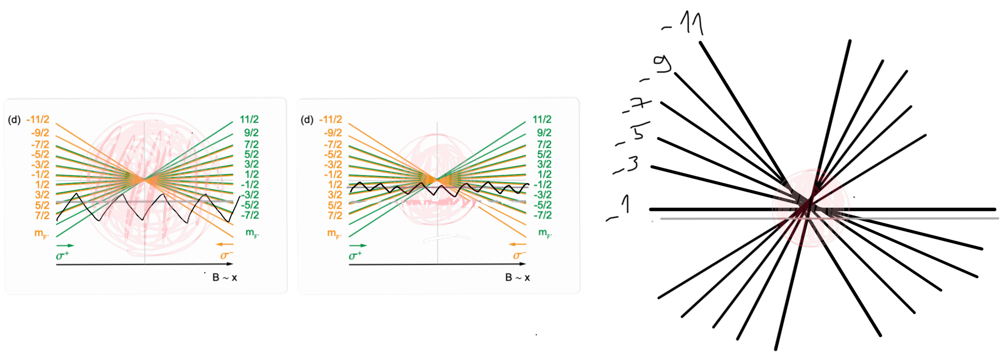


Figure 1.10: Sequential stages of the 689 nm red MOT: broadband capture, compression, and single-frequency cooling (adapted from [**?**]).

1.2.6.5 Optimization of narrowband MOT Parameters

Cooling efficiency during the single-frequency stage depends on precise frequency, intensity, and timing parameters:

- **Broadband modulation amplitude:** The initial sweep covers $\Delta f = 3$ MHz at the RF source to capture the Doppler distribution from the blue MOT.
- **Intensity ramp and power broadening:** High intensity provides a deep trap for initial capture, but causes power broadening ($\Gamma_{\text{eff}} = \Gamma\sqrt{1 + I/I_s}$), raising the achievable Doppler limit. For single-frequency cooling, intensity is adiabatically ramped down over 45 ms to reduce power broadening while maintaining sufficient force to support the cloud against gravity.
- **Detuning optimization:** Setting the laser too close to resonance causes reheating via photon re-absorption in dense clouds. During the single-frequency MOT, the laser detuning is fixed at $\Delta f = -600$ kHz, optimized experimentally to minimize final cloud temperature.
- **Magnetic field bias and quantization axis:** To preserve spin polarization, the magnetic field is never allowed to cross zero during transitions between cooling phases. Maintaining non-zero bias components along at least two orthogonal directions (e.g., $|B_x|$ and $|B_y|$) preserves the quantization axis and prevents spin depolarization.

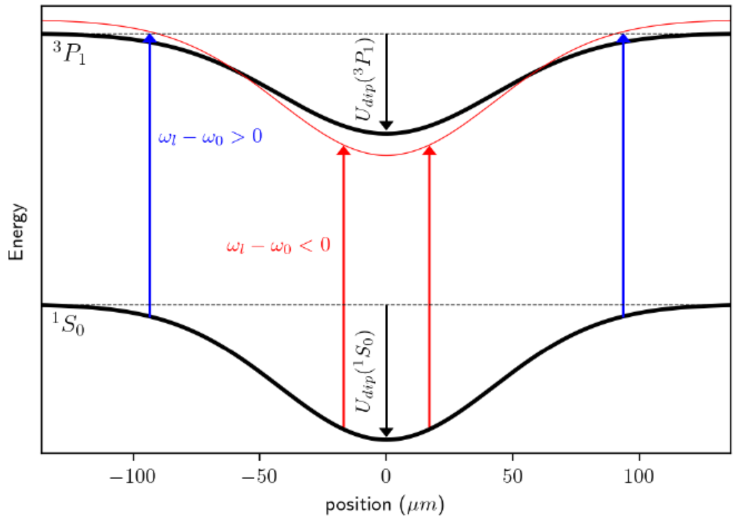


Figure 1.11: Simulation extracted from [2]. Principle of localized 689 nm MOT loading inside the crossed optical dipole trap due to the differential AC Stark shift between $^1S_0$ and $^3P_1$.

1.2.7 Optical Dipole Trap (ODT) and evaporative cooling

1.2.7.1 Loading the crossed ODT

The final cooling stage before optical trapping consists in transferring the atoms into an Optical Dipole Trap (ODT) operating at 1070 nm, followed by forced evaporative cooling.

Atoms are adiabatically loaded into the ODT while decreasing the intensity of the single-frequency 689 nm red MOT during a ramp lasting temps, intensity ramp. From second-order perturbation theory, the light shift experienced by the energy levels yields the optical dipole potential:

$$U_{\text{dip}} = -\frac{\hbar|\Omega|^2}{4\bar{\Delta}} \tag{1.6}$$

[2], where $1/\bar{\Delta} = 1/(\omega_{\text{laser}} - \omega_{\text{atomic}}) - 1/(\omega_{\text{laser}} + \omega_{\text{atomic}})$ is the effective detuning factor, and $\Omega^2 = \frac{\Gamma^2}{2}\frac{I}{I_{\text{sat}}}$ is the Rabi frequency.

As seen from these expressions, the AC Stark shift induced by the dipole laser is state-dependent. As illustrated in Fig. 1.11, the potential estimated in [2] is deeper for the ground state $^1S_0$ ($U_{\text{dip}}(^1S_0) \simeq -1.5$ MHz) than for the excited state $^3P_1$ ($U_{\text{dip}}(^3P_1) \simeq -1.2$ MHz) at the center of the trap, resulting in a differential red-shift of the $^1S_0 \to {}^3P_1$ transition.

This differential shift creates a localized spatial resonance condition: atoms at the center of the trap remain in resonance with the red MOT beams, where they are continuously cooled and trapped. Conversely, energetic atoms at the periphery experience a negligible light shift and remain blue-detuned from the local resonance, preventing unwanted re-heating.

Following loading, forced evaporative cooling is performed to reach deep Fermi degeneracy, down to temperatures on the order of $T =, T/T_F =$ calculer avec les valeurs du temps d'husain. Evaporative cooling is achieved by adiabatically lowering the trap depth: the most energetic atoms overcome the potential barrier and escape the trap, while the remaining ones re-thermalize to lower temperatures via elastic collisions (Fig. 1.12).

The results presented in the following two chapters were obtained with an atomic cloud located at the intersection of a horizontal beam, called the "Reservoir" (waist $w_0^H = 150\ \mu$m), and a vertical beam, called the "Dimple" (waist $w_0^V = 80\ \mu$m). The vertical beam axis is tilted at an angle of $\simeq 30°$ relative to the vertical to facilitate the escape of the hottest atoms from the trap.

In Chapter 2, the horizontal arm is tilted by $\simeq 1°$. For the photoassociation results in Chapter 3, this tilt was canceled: ajouter des arguments et refs pour motiver ça.

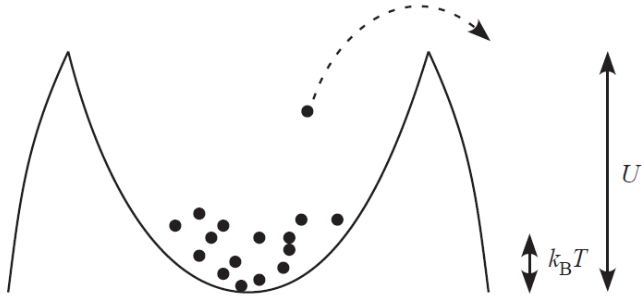


Figure 1.12: Evaporative cooling mechanism in an inclined crossed optical dipole trap, illustrating the escape of high-energy atoms and subsequent re-thermalization.

1.2.7.2 Optimization of the evaporation ramps

For a degenerate Fermi gas, the key parameter to optimize is the phase-space density (PSD), which scales as PSD $\propto N\bar{\omega}^3/T^3$, where N is the atom number, $\bar{\omega}$ the geometrically averaged trap frequency, and T the temperature. While both arms of the dipole trap influence these parameters, their main physical contributions differ:

- **Tight vertical beam ("Dimple"):** Writing the atom number as $N = nV$ (where n is the spatial density and V the trap volume), the elastic collision rate scales as $\Gamma_{\mathrm{el}} = n\sigma_{\mathrm{el}}v_{\mathrm{th}}$. The tightly focused waist of the dimple beam mainly dictates the local confinement, thereby enhancing collision rates for fast re-thermalization. Moreover, ramping down the dimple intensity directly lowers the effective trap depth, setting the final temperature T of the cloud.
- **Broad horizontal beam ("Reservoir"):** Due to its large waist and capture volume, the reservoir beam primarily serves to maximize the initial atom number N loaded from the MOT. While modifying its intensity also reshapes the cloud and affects thermalization dynamics, its primary role remains maintaining a high-capacity atomic reservoir.
- **Ramp duration and adiabaticity:** The duration and adiabaticity of the evaporative ramps depend on the targeted state. Short evaporations are preferred when aiming solely for a large atom number at moderate temperatures when it is necessary to detect atom number fluctuations, such as for our experiments on photoassociation (see Chapt.3). Conversely, reaching deep quantum

degeneracy requires a multi-stage ramp lasting approximately 5 seconds which was necessary for the results of Chapt.2.

For further details on loading atoms into the optical crossed dipole trap, the reader is referred to the thesis of A. Litvinov [2]. An additional optical lattice, formed by a 1064 nm ALS laser, was initially implemented to enhance the phase-space density (PSD) of the atomic cloud prior to forced evaporative cooling. While this configuration was employed for the experiments presented in Chapter 2, it was subsequently discarded due to intensity fluctuations in the laser that induced significant heating.

1.2.7.3 Optical setup

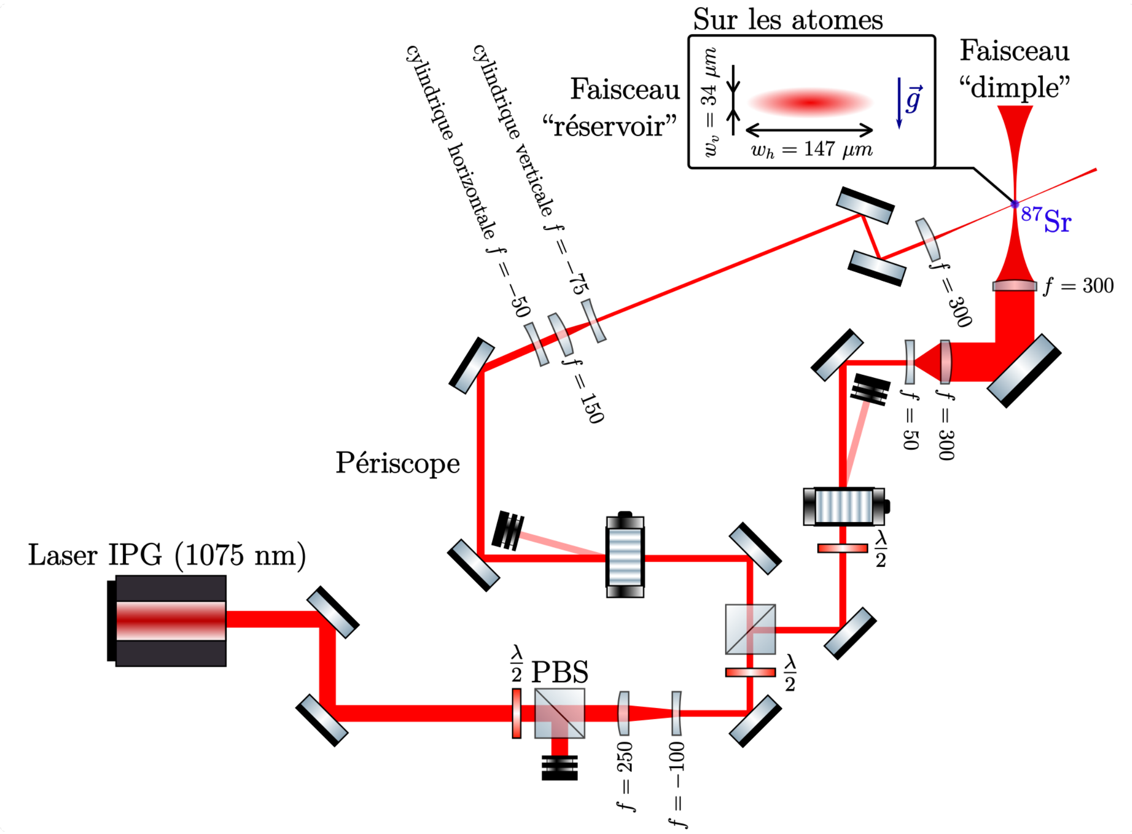


Figure 1.13: Optical layout of the 1075 nm crossed dipole trap laser system.

1.3 Spin Selection

1.3.1 Optical Pumping

Optical pumping is implemented prior to the final evaporation stages to prepare the atomic gas in a tailored mixture of nuclear spin states (m_F). Due to the $SU(N)$ symmetry of ^{87}Sr, elastic s-wave collisions between identical fermions in the same m_F state are forbidden by the Pauli exclusion principle. Therefore, at least two distinct spin components must be present to enable elastic collisions and ensure efficient re-thermalization during evaporative cooling. Furthermore, $SU(N)$ symmetry guarantees that elastic collisions conserve the nuclear spin state, preventing spin-exchange collisions during evaporation.

To perform state preparation, a bias magnetic field of 6.05 ± 0.05 G is applied to lift the degeneracy of the ${}^3P_1, F = 11/2$ hyperfine sub-levels via the linear Zeeman effect. The cloud is illuminated by a σ^--polarized optical pumping beam tuned to the ${}^1S_0(F = 9/2) \to {}^3P_1(F = 11/2)$ transition. Its frequency is defined as:

$$f_{11/2} = f_{\text{Radiant Dyes}} \pm f^{\text{RF}}_{\text{Slave 1}} + f^{\text{RF}}_{11/2} \tag{1.7}$$

The laser frequency is swept across ajouter amplitude rampe the targeted ${}^3P_1, F = 11/2, m'_F$ sub-levels to deplete specific ground-state populations in ${}^1S_0, F = 9/2, m_F$. Governed by electric dipole selection rules ($\Delta m_F = 0, \pm 1$), spontaneous emission redistributes the excited atoms into adjacent lower ground-state sub-levels, resulting in a progressive accumulation in the target spin states.

Note: Add efficiency comments, pulse duration $t =$ value, and intensity $I =$ value.

To prepare a specific mixture (e.g., a 2-spin mixture), unwanted spin populations are sequentially pumped out. One spin component, typically $m_F = -9/2$, is intentionally retained alongside the target mixture to maintain two-body elastic collisions for thermalization during evaporation. Once evaporative cooling is complete, this auxiliary component is selectively removed using a short **'blast'** pulse. Under σ^- excitation on the $F = 9/2 \to F' = 11/2$ transition, the $m_F = -9/2$ state couples exclusively to $m'_F = -11/2$. Because no $m_F = -11/2$ ground state exists, spontaneous emission from $m'_F = -11/2$ can only decay back to $m_F = -9/2$, forming a closed cycling transition. The resulting continuous photon scattering exerts a strong radiation pressure that expels the $m_F = -9/2$ population from the trap without disturbing the target mixture.

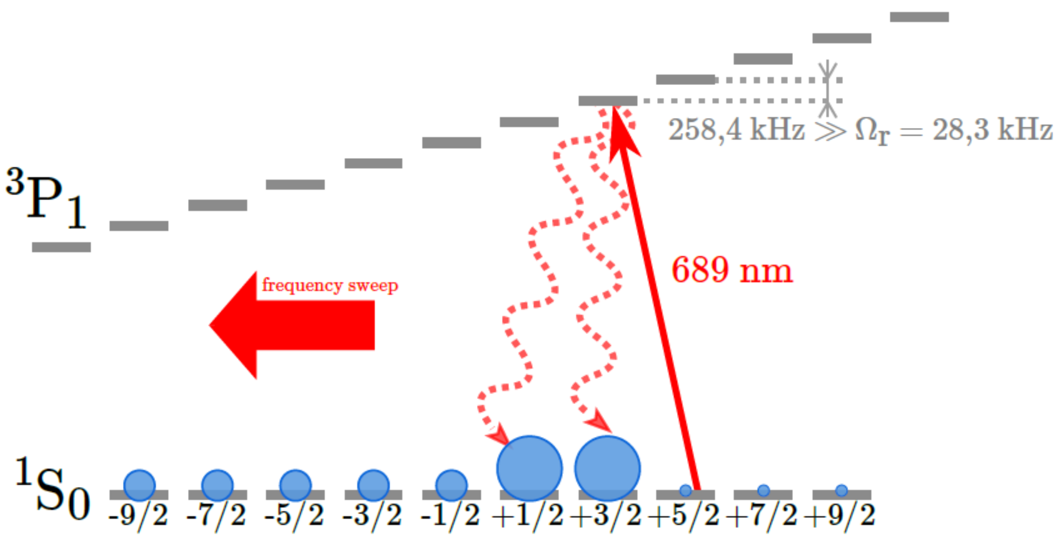


Figure 1.14: Optical pumping scheme on the ${}^1S_0(F = 9/2) \to {}^3P_1(F = 11/2)$ transition used to prepare a specific nuclear spin mixture.

1.4 Spin Manipulation

We developed a method demonstrated in [13] for coherent control within the ground state manifold of zeeman sublevels. it relies on tensor shifts enacted by a laser operating near 689 nm, and will be described in detail in chapter 2. At a later stage, we performed photoassociation experiments in the vicinity of the intercombination manifold, see chapter 3. Both experiments actually rely on the same laser system, that I will present here.

1.4.1 External Cavity Diode Laser (ECDL)

To lift the degeneracy of the Zeeman sub-levels and allow state-selective addressing, a Tensorial Light Shift (TLS) is induced using an External Cavity Diode Laser (ECDL) from MOGlabs operating near 689 nm. The laser frequency is selected by an interference filter placed at the center of a cat's-eye configuration, yielding a linewidth below 100 kHz. An internal PID controller robustly compensates for thermal and mechanical drifts of the cavity length as well as diode intensity fluctuations. The laser is subsequently beat-locked to a master optical reference adapted to the specific experimental requirements.

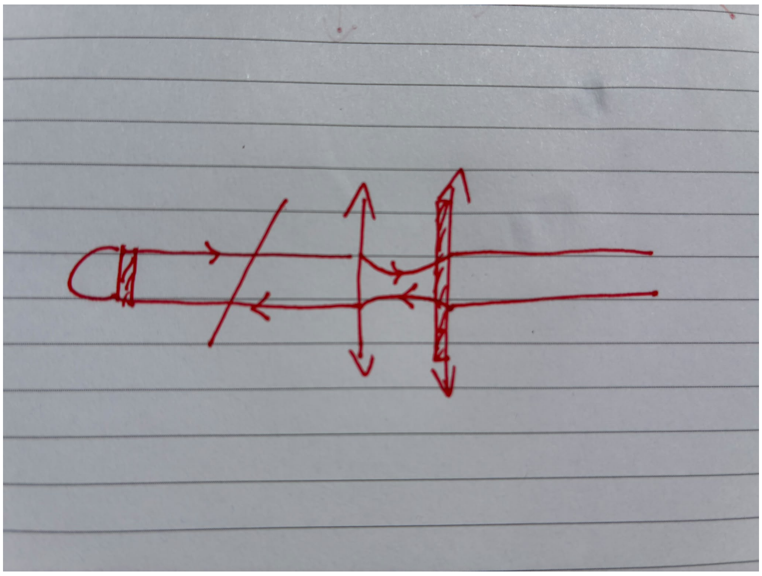


Figure 1.15: MOGlabs External Cavity Diode Laser (ECDL) used for spin manipulation.

1.4.2 Raman Transitions

1.4.2.1 Physical Concept

The second chapter of this thesis focuses on spin manipulation within the $SU(10)$ manifold of ^{87}Sr. This operation is realized via a two-photon Raman process driven by two laser beams red-detuned from the $^1S_0 \to {}^3P_1$ intercombination line.

Controlling the dimension of the effective Hilbert space from 1 to 10 requires independent addressing of individual spin states. The two-photon Rabi frequency Ω coupling two ground-state sub-levels $|g, m_F\rangle$ and $|g, m'_F\rangle$ through intermediate excited states $|e\rangle$ takes the form:

$$\hbar\Omega = \sum_e \frac{\langle g, m'_F|\, \vec{d} \cdot \vec{E}_2^* \,|e\rangle \langle e|\, \vec{d} \cdot \vec{E}_1 \,|g, m_F\rangle}{\Delta_e} \tag{1.8}$$

where $\vec{d}$ is the dipole operator, $\vec{E}_1, \vec{E}_2$ are the electric field amplitudes, and Δ_e is the single-photon detuning relative to the intermediate state $|e\rangle$.

By applying the TLS, the energy difference between two consecutive spin projections m_F and $m_F + 1$ becomes non-linear:

$$\Delta E_{m_F}^{m_F+1} = q(2m_F + 1) + \beta \tag{1.9}$$

where q parameterizes the quadratic (tensor) light shift and β represents the linear Zeeman contribution. This explicit dependence on m_F lifts the energy degeneracy between adjacent transitions, allowing spectral isolation and individual control of any given $m_F \leftrightarrow m_F + 1$ pair.

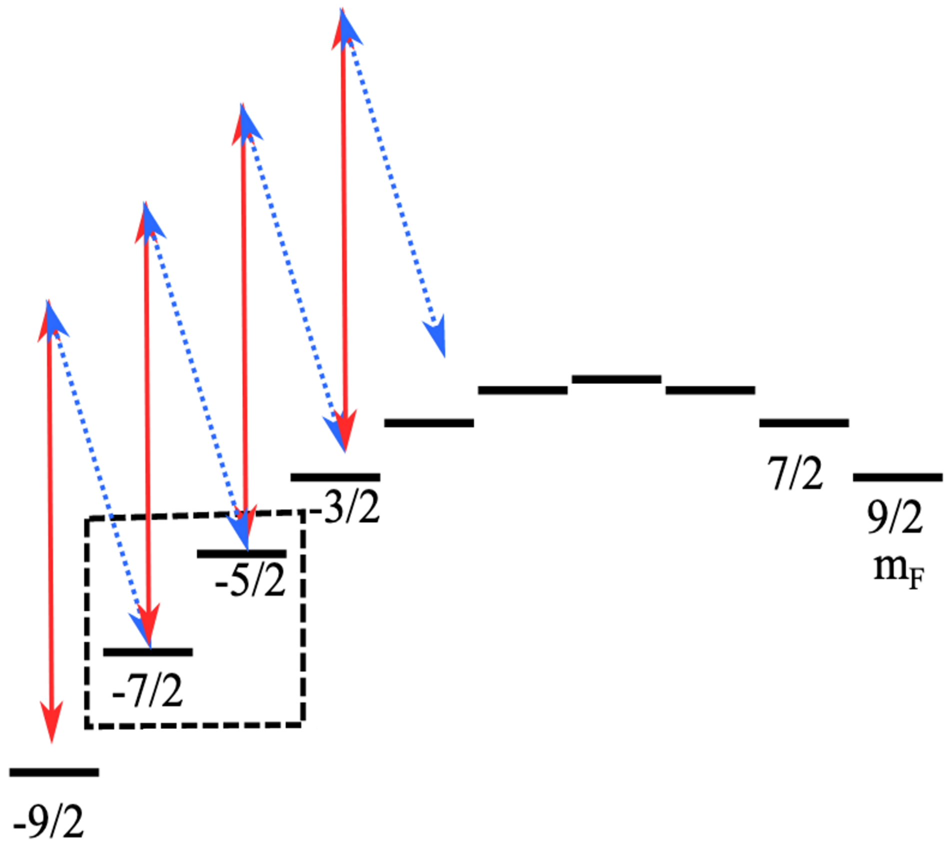


Figure 1.16: Energy level diagram illustrating the Tensorial Light Shift (TLS) and state-selective two-photon Raman transitions.

For more details this process is discussed in the second chapter of this thesis and also in the article of the group [13].

1.4.2.2 Experimental Setup of the Raman Process

The two-photon process is driven by two distinct beams: a first beam derived from the "AOM lattice 689" laser (shown in red in Fig. 1.18) generates the TLS and supplies the first photon. A second beam, frequency-shifted by the "AOM Raman" (shown in blue), completes the Raman transition.

This configuration selectively couples a single pair of states—for instance, $|^1S_0, F = 9/2, m_F = -7/2\rangle$ – $|^1S_0, F = 9/2, m_F = -5/2\rangle$—while remaining far off-resonance for all other m_F states. The complete optical path for this operation is detailed in Fig. 1.17. The magnetic field is close to zero when the ratio between the two lasers is set $I_R << I_D$ since we mostly want to separate the states.

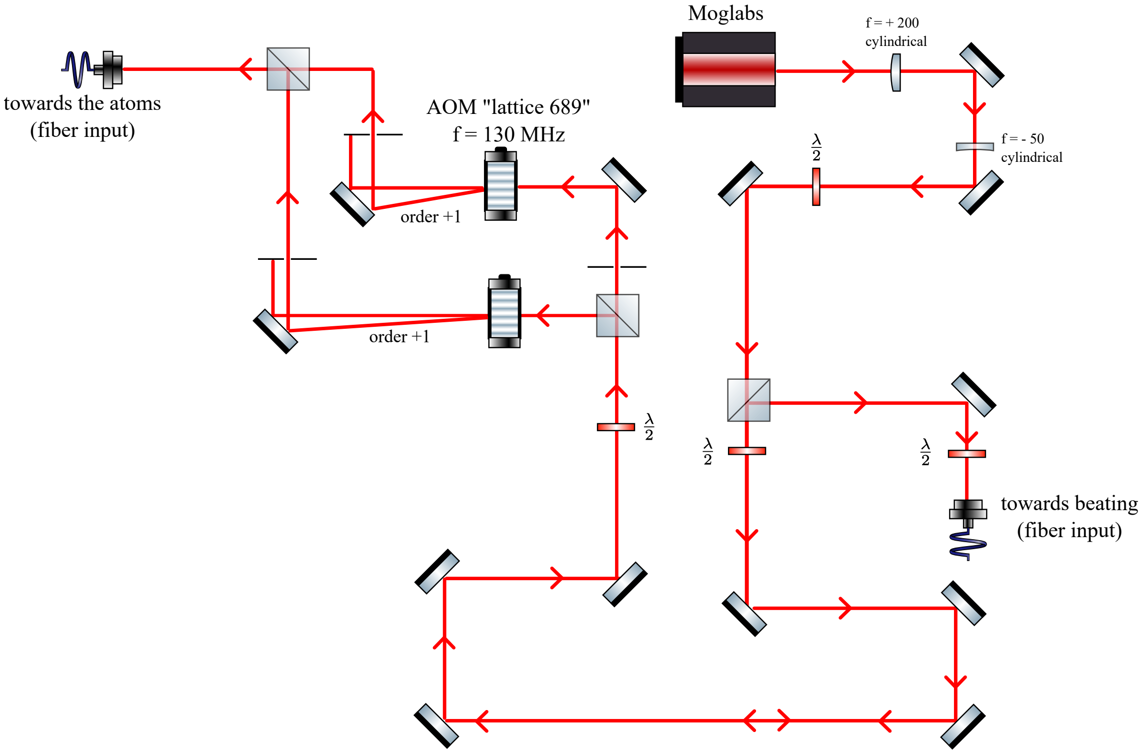


Figure 1.17: Optical layout of the MOGlabs laser chain configured for two-photon Raman transitions.

1.4.3 Photoassociation

Chapter 3 investigates photoassociation spectroscopy below the intercombination line. For these measurements, the full power of the laser is sent through the "AOM lattice 689" instead of being split with the Raman AOM.

For the dataset presented in Chapter 3, the optical chain incorporates a custom-built confocal Fabry-Pérot cavity stabilized using the Pound-Drever-Hall (PDH) technique [3], yielding the final setup depicted in Fig. 1.18. The beam first passes through an Electro-Optic Modulator (EOM) to generate phase modulation side-bands. A lens with focal length $f = 150$ mm mode-matches the beam to the fundamental cavity mode. Upon reflection from the cavity, the light is directed onto a fast photodiode and demodulated with an RF source to produce the PDH error signal.

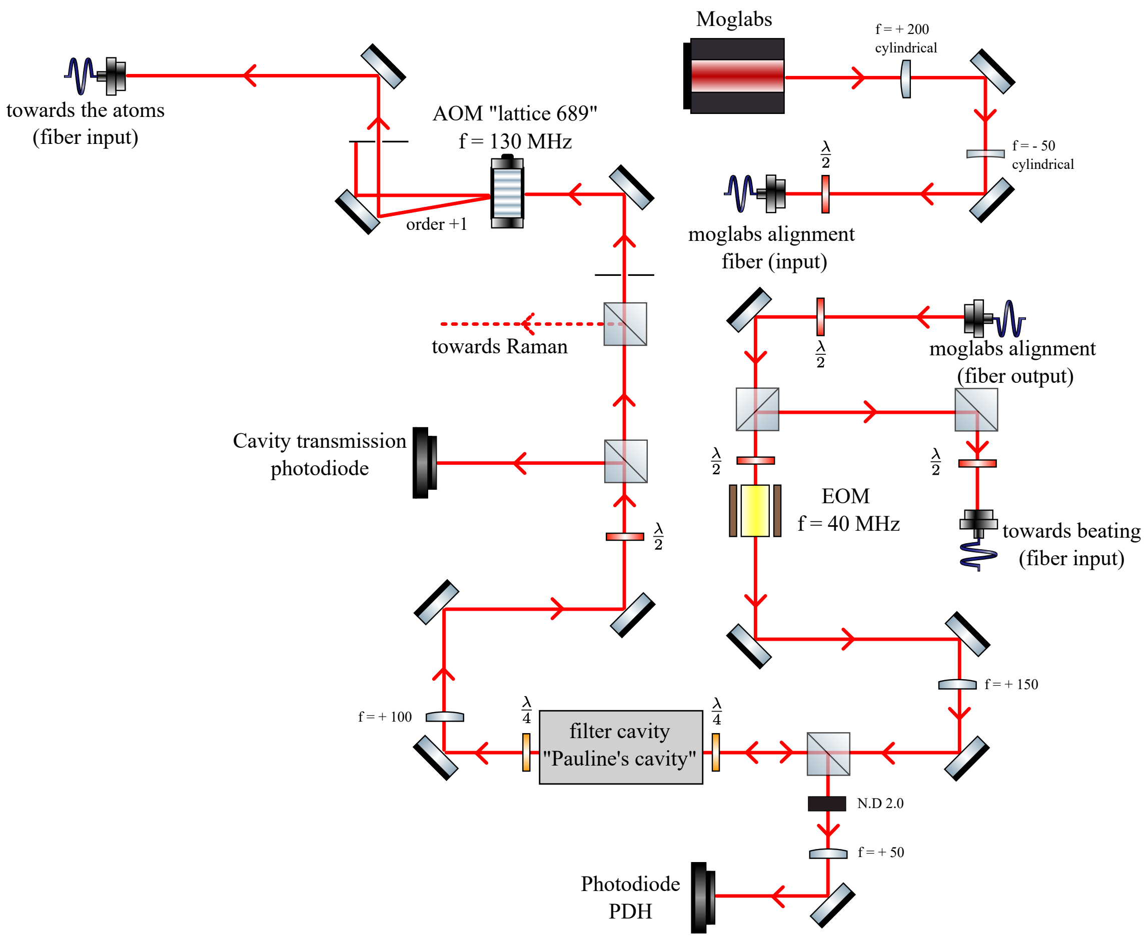


Figure 1.18: Complete optical setup including the Fabry-Pérot cavity and PDH stabilization loop.

This cavity filters amplified spontaneous emission (ASE) from the laser source, thereby reducing stray light, minimizing photon-recoil heating.

1.4.4 Fabry-Pérot cavity and PDH Locking system

1.4.4.1 Cavity design and assembly

A primary objective of the cavity injection scheme in our optical setup is to maximize overall transmission for spin manipulation. To achieve this, we constructed a confocal cavity configuration ($d_0 = R_C$, where d_0 is the mirror separation and R_C is their radius of curvature), ensuring that all odd transverse modes are degenerate and resonate at the same frequency. If these modes were non-degenerate, off-resonance spatial components would be reflected at the input mirror, thereby reducing the total cavity transmission efficiency.

In addition to the confocal geometry, longitudinal mode resonance requires that the cavity length d_0 satisfies the condition:

$$d_0 = n\frac{\lambda}{2} \tag{1.10}$$

where n is an integer and λ is the wavelength of the injection laser.

To fulfill these requirements, we selected curved mirrors (**PD080.31** from PICMA) with a diameter of $D = 4.5 \pm 0.15$ mm and a radius of curvature matching d_0. The manufacturer specified a nominal mirror transmission of $T_{\text{mirror}} = 1.28\%$, though the effective system performance can be influenced by mechanical mounting constraints.

Proper cavity operation requires the mirrors to be aligned precisely parallel to each other and perpendicular to the optical propagation axis. In our initial implementation (Fig. 1.19), the mirrors were permanently glued into place. However, this approach introduced major technical issues over time: excess glue migrated onto the optical surfaces, degrading overall transmission, while alignment errors during gluing introduced an uncorrected angular tilt relative to the beam axis.

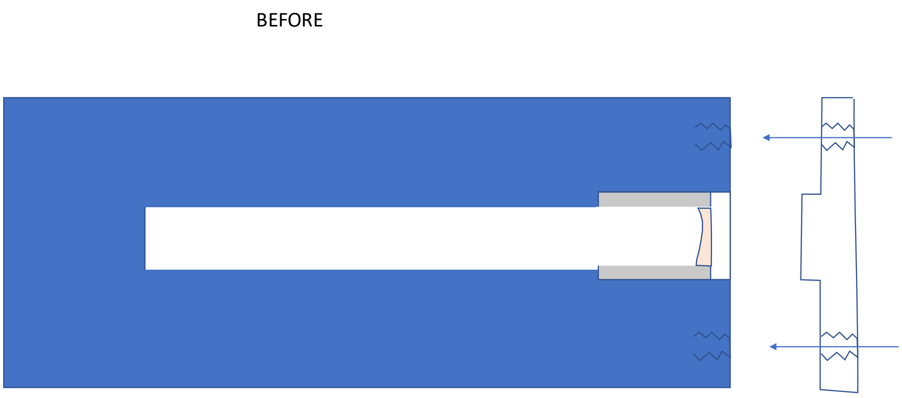


Figure 1.19: Initial fixed cavity assembly using glued mirrors, lacking mechanical alignment degrees of freedom.

Because the original design provided no mechanical degrees of freedom to correct these misalignments, we redesigned the mounting mechanism. The updated cavity assembly is presented in Fig. 1.20. In this configuration, three adjustment knobs on a circular outer mount apply controlled pressure against the central cylindrical spacer housing the mirrors, with intermediate washers providing mechanical compliance. This allows for precise tilt control and fine adjustment of the mirror separation.

Figure 1.20: Upgraded cavity mount featuring three adjustment screws and compliant washers for precise angular and longitudinal alignment.

Once locked to the injection laser, the optimized cavity achieves a total power transmission efficiency of $T = 0.63$. The Free Spectral Range (FSR) is given by:

$$\mathrm{FSR} = \frac{c}{2d_0} = 3 \text{ GHz} \tag{1.11}$$

with a measured finesse of:

$$\mathcal{F} = \frac{\mathrm{FSR}}{\delta\nu} = 310 \pm 15 \tag{1.12}$$

where $\delta\nu$ represents the full width at half maximum (FWHM) of the transmitted resonance peak measured on a photodiode for the odd transverse modes.

1.4.4.2 Pound-Drever-Hall locking system and PID controller analysis

The cavity is frequency-locked to the laser using the Pound-Drever-Hall (PDH) technique, as detailed by Black [3] and schematized in Fig. 1.21. The scheme utilizes an Electro-Optic Modulator (EOM) to generate sidebands at frequencies $f_0 \pm f_M$ around the optical carrier f_0, with a modulation frequency of $f_M = 40.48$ MHz. The modulated electric field incident on the cavity is expressed as:

$$E_{\mathrm{inc}}^{\mathrm{mod}} \approx E_0 \left[J_0(\beta) e^{i2\pi f_0 t} + J_1(\beta) e^{i2\pi (f_0 + f_M)t} - J_1(\beta) e^{i2\pi (f_0 - f_M)t} \right] \tag{1.13}$$

where $J_0(\beta)$ and $J_1(\beta)$ are the zeroth- and first-order Bessel functions of the first kind, and β is the modulation index.

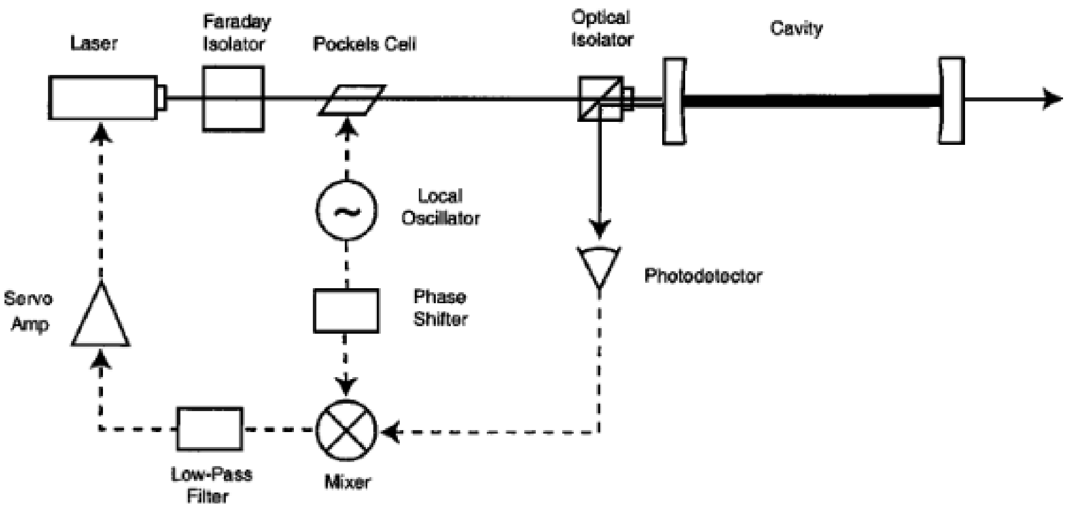


Figure 1.21: Schematic of the PDH locking system, extracted from [3].

The carrier enters the cavity and acquires a steep, frequency-dependent phase shift near resonance, whereas the sidebands are promptly reflected from the input mirror. The interference between the reflected carrier and the sidebands is detected on a fast photodiode, demodulated at f_M, and low-pass filtered to extract the dispersive error signal (Fig. 1.22). The sharp linear slope around resonance provides a robust error signal highly sensitive to the sign of the frequency excursion. From this graph, one measures a signal-to-noise ratio of SNR =. Interestingly, the sideband resonance features were not distinctly visible on the scanning error signal, even when driving the EOM with a high modulation depth to maximize the J_1/J_0 ratio.

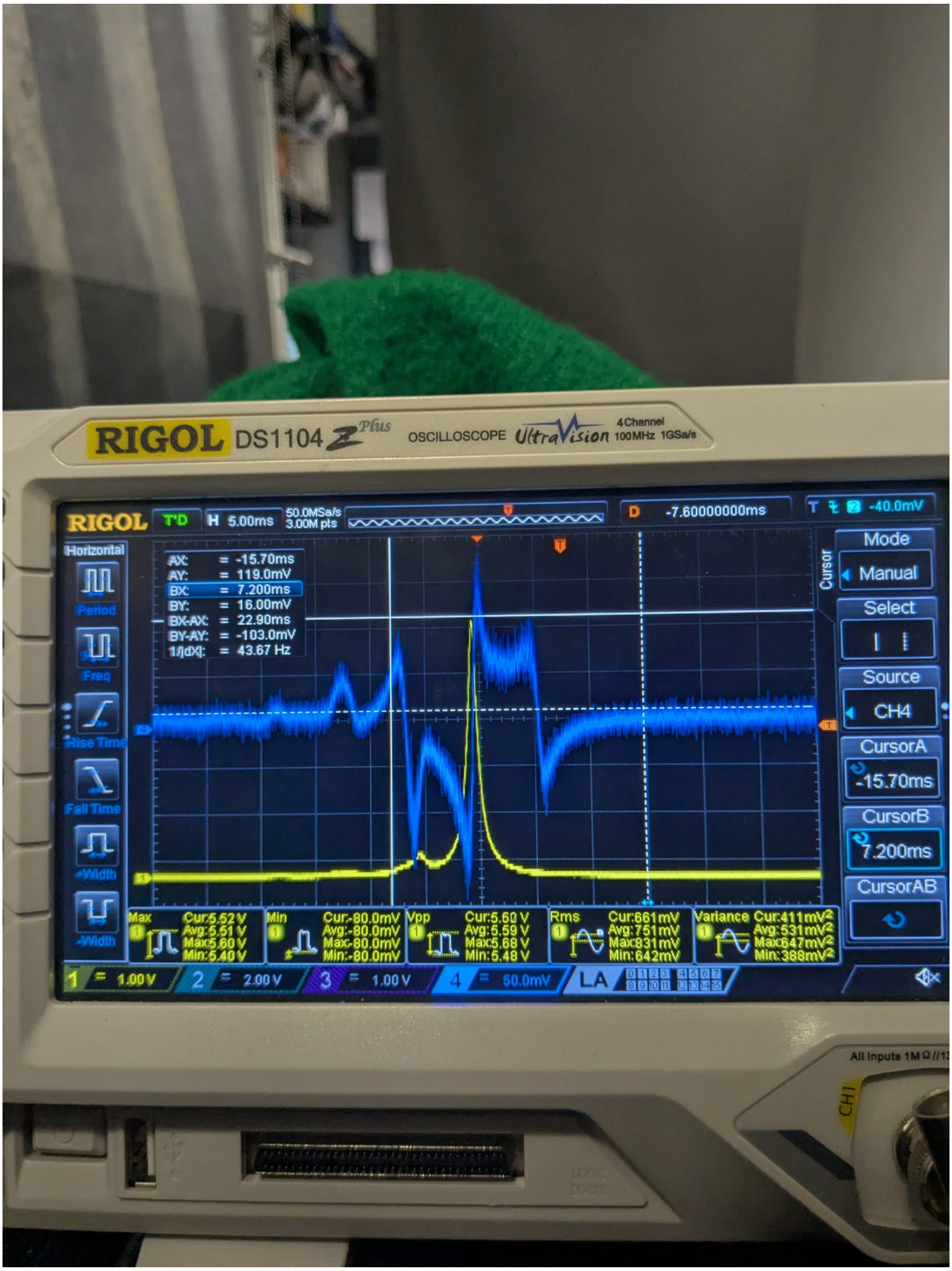


Figure 1.22: Typical Pound-Drever-Hall (PDH) error signal obtained by scanning the cavity length across resonance.

Digital feedback control and noise analysis were implemented primarily using a Red Pitaya STEMlab board. The PID parameters were optimized by injecting a broadband electronic noise signal into the cavity piezo driver and measuring the closed-loop transfer function. As seen in the frequency response plot (Fig. 1.23), operating the system with specific gain values excites mechanical resonances. We interpret the primary peak at 35 kHz as the fundamental mechanical resonance of the piezo actuator, accompanied by other higher-frequency modes. The noise injection analysis also revealed the low-frequency behavior of the loop. Consequently, the final feedback loop acting on the piezo crystal incorporates a low-pass filter with a cut-off frequency of $f_c = 40$ kHz, a proportional gain strictly bounded between $1 < K_p < 10$, and an integral gain configured between $0 < K_i < 8\ \text{s}^{-1}$.

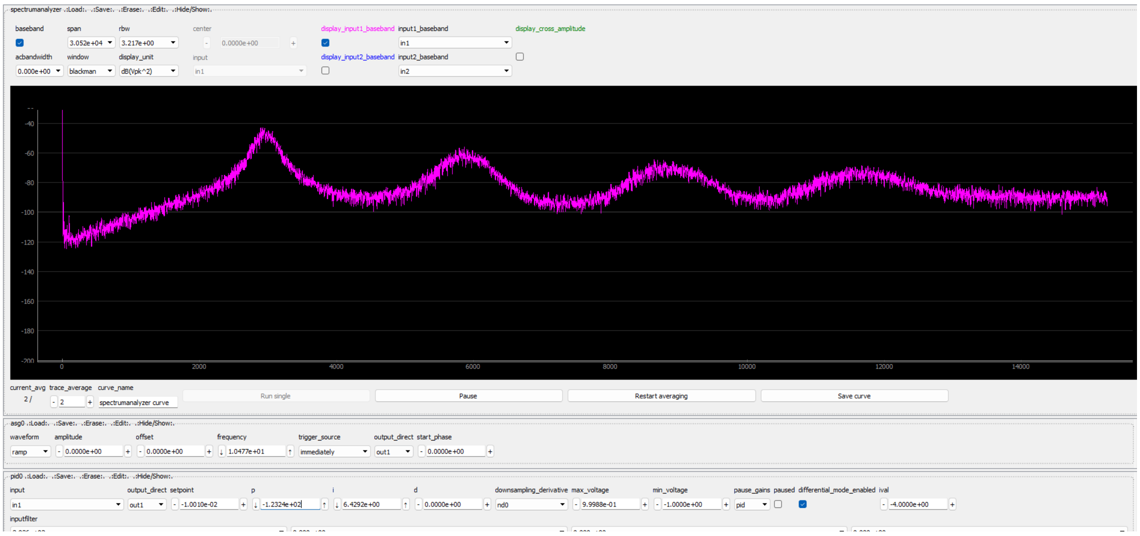


Figure 1.23: Closed-loop noise analysis and mechanical resonance identification using the Red Pitaya.

1.5 Spin Measurement Scheme

Total atom numbers are determined via standard absorption imaging on the broad ${}^1S_0 \to {}^1P_1$ transition at 461 nm using a Vortex laser. State-resolved spin detection follows the momentum-transfer ("spin-kick") technique detailed in [11].

The measurement sequence proceeds as follows:

1. Atoms are illuminated by two counter-propagating laser beams with opposite circular polarizations (σ^+/σ^-), as illustrated in Fig. 1.24(a).

2. A state-selective pulse tuned to a specific ${}^1S_0 \to {}^3P_1$ transition adiabatically transfers targeted spin populations (Fig. 1.24(b)).

3. Each absorption-stimulated emission cycle imparts a coherent $2\hbar k$ momentum kick to the selected m_F state.

4. Following a time-of-flight (TOF) expansion duration $t = 6.6$ ms (set for the gas to stay in camera plane), spatial separation between spin components is observed in the absorption image (Fig. 1.24(d)).

Unaddressed spin states remain at the central position of the cloud. In contrast, states subject to the momentum transfer form distinct spatial wavepackets shifted to the left or right depending on the direction of the resonant momentum kick (e.g., $m_F = -3/2$ and $m_F = -7/2$ in Fig. 1.24(d)).

To ensure efficient population transfer, the optical pulse sweep rate must satisfy the adiabaticity criterion across the avoided crossing between coupled adiabatic states (Fig. 1.24(c)).

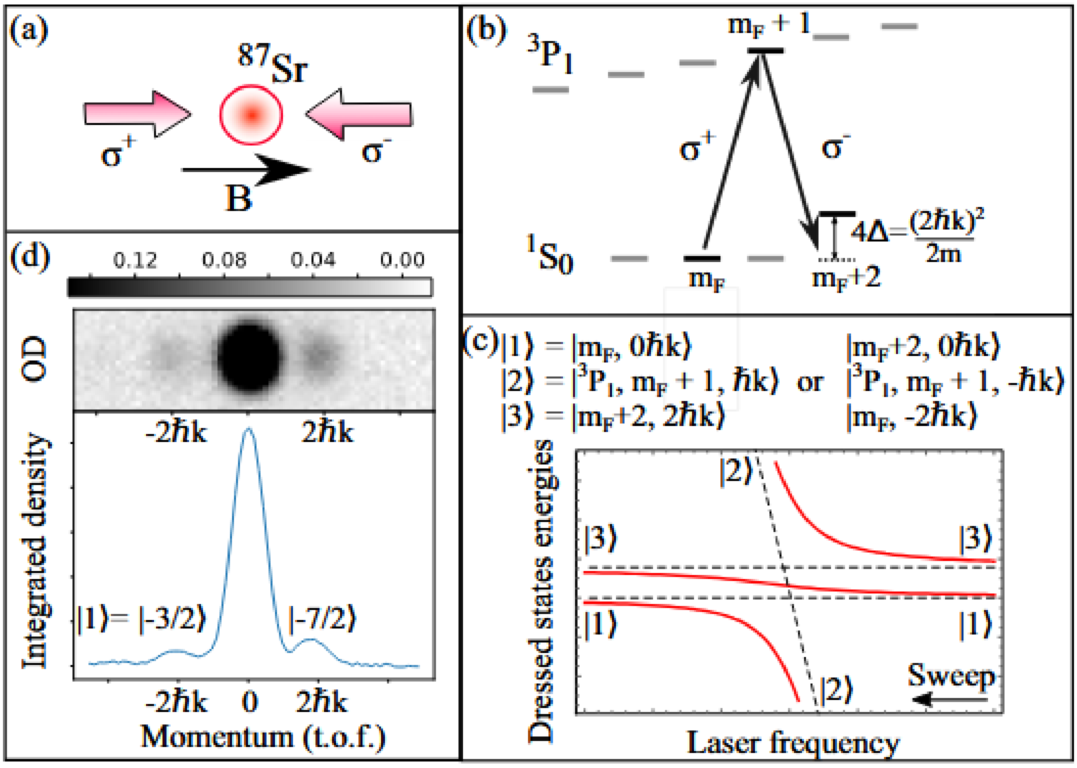


Figure 1.24: Principle of spin-selective detection via coherent momentum transfer (”spin kick”). (a) Counter-propagating beam geometry. (b) Level scheme. (c) Avoided crossing structure for adiabatic transfer. (d) Resulting TOF absorption image showing spatially separated spin components.

Chapter 2

Quantum state measurement and characterization in qudit systems

This chapter presents the experimental results published in Ref. [17], centered on the coherent manipulation of the 10-dimensional nuclear spin Hilbert space ($2I+1 = 10$) of ^{87}Sr atoms. By exploiting a tensor light shift to isolate tailored subsystems, we demonstrate control ranging from robust pseudo-spin qubits to high-dimensional qudits ($D = 3$ to 10) displaying multi-second coherence times. Based on this platform, we implement multi-state interferometric schemes over four nuclear levels for multi-parameter sensing. Following the reproduced article, Section 2.7 provides an extended theoretical derivation of the Positive-Operator-Valued Measures (POVMs) implemented in our non-commuting observables measurement scheme.

2.1 Gas preparation and spin control

Large-spin particles open new possibilities for quantum technologies. In the context of quantum many-body physics and quantum simulators, the enlarged spin degrees of freedom correspond to a significantly larger Hilbert space, which leads to qualitatively new physics [18, 19, 20, 21, 22, 23, 24]. Large spins can also enhance quantum information processing [25] and quantum metrology [26, 27, 28, 29]. Conveniently, they emerge from several physical platforms, such as superconducting devices [30, 31], molecular magnets [32], ions [33], multi-mode photons [34], and atoms [35, 36].

Among large-spin atoms, fermionic alkaline-earth and alkaline-earth-like atoms (AEA) are particularly interesting for quantum metrology because they display a weak sensitivity to magnetic fields and possess narrow optical transitions that make them prominent for optical clocks with cold atoms [37]. These atomic species are also remarkable for quantum many-body physics, as the purely nuclear nature of their spin in the ground state generates an $\mathfrak{su}(\mathrm{N})$ symmetry that introduces frustration and should thus strongly impact quantum magnetism [38]. This possibility has attracted considerable attention from both theoretical [39, 40, 41] and experimental [20, 42, 43, 44] standpoints. To exploit the full potential of large-spin fermionic alkaline-earth atoms and to characterize the $\mathfrak{su}(\mathrm{N})$ phases at low temperatures, it is crucial to develop tools that coherently control the spin states of atoms beyond

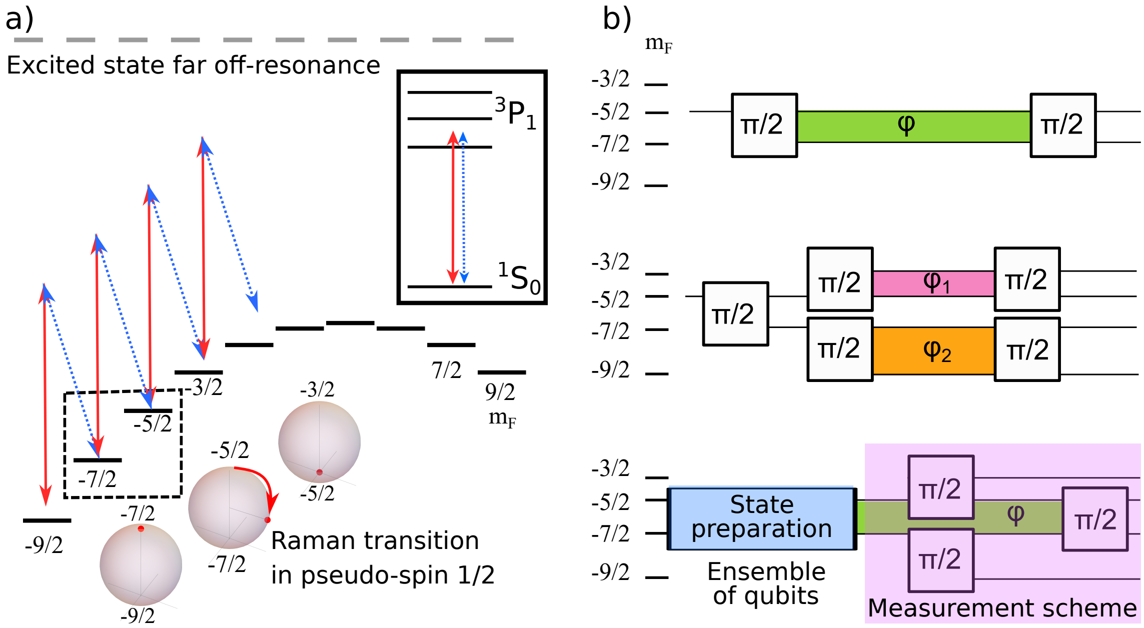


Figure 2.1: **Principle of the experiment.** (a) Atomic levels and laser coupling scheme. A π-polarized light beam tuned within the hyperfine structure of the $^1S_0 \to {}^3P_1$ transition (red arrows) creates an energy shift in the electronic ground state, with significant tensor component $U_{TLS}(m_F) = q\, m_F^2$. A second laser (here σ^- polarized) enables Raman transitions. Thanks to the tensor light shift, the Raman resonance conditions differ from each other, making it possible to drive Rabi oscillations between two isolated spin states (here $|-5/2\rangle$ and $|-7/2\rangle$), all other processes being off-resonant. The Raman operation can be understood as a rotation in a sub-Bloch sphere. A potential linear energy shift $U_{linear}(m_F) = b\, m_F$ (in the illustration, $b \simeq -3q$) does not directly affect the capacity to isolate a pair of states. (b) The three experimental schemes in this paper. Top: We operate a Ramsey interferometer between states $|-5/2\rangle$ and $|-7/2\rangle$, decoupled from the other spin states. Middle: We operate two interferometers in parallel, demonstrating multi-parameter simultaneous sensing of the fields acting on the atoms. Bottom: we characterize the collective state of atoms in a superposition of two Zeeman levels ($|-5/2\rangle$ and $|-7/2\rangle$), identified as qubit states. The measurement sequence starts by applying two pulses that map the populations of the qubit states into ancillary states. Combining this with a final rotation in the qubit sub-Bloch-sphere, one can simultaneously measure two orthogonal components of the pseudo-spin of the ensemble of qubits.

simple spin precessions (see [45, 13, 46]).

In this paper, we use a tensor light shift [47] to control the resonance conditions of Raman transitions within the manifold of $2F + 1 = 10$ Zeeman states of the spin-$F = 9/2$ ground state of ^{87}Sr. The quadratic energy shift $\propto m_F^2$ enables us to perform coherent manipulations between isolated pairs of m_F states with a fidelity better than 99 %. These operations demonstrate control over unitary transformations deriving from $\mathfrak{su}(N = 2F + 1)$ generators, beyond the usual precession operations deriving from the spin-F representation of the $\mathfrak{su}(2)$ group. Taking advantage of this new possibility and the atoms' insensitivity to magnetic fields, we operate a Ramsey interferometer between two isolated hyperfine states and demonstrate Ramsey fringes with no discernible loss of contrast over 3 s. We further exploit the large spin of the atoms with two new types of interferometers. In the first implementation, we operate two Ramsey interferometers simultaneously and in parallel, using two independent pairs of spin states within the hyperfine structure. This configuration allows simultaneous measurement of two independent parameters (here corresponding to the effective quadratic and linear Zeeman shifts). In the second implementation, we mirror two physical spin states that make a qubit into two ancillary spin states. Selective unitary transformation ("rotation") of the two qubit states and the measurement of all four states then permit the simultaneous measurement of two non-commuting observables of an ensemble of qubits.

2.2 Implementation of $\mathfrak{su}$(N) generators.

One key idea of this paper is to implement coherent rotations between two and only two arbitrary spin states chosen among the N spin states of an $N-$component Fermi mixture. Rabi oscillation between such a selected pair of states corresponds to rotations driven by infinitesimal generators of the $\mathfrak{su}$(N) spin algebra, described by an operator $\sigma^x_{m_F,m'_F}$. The restriction of $\sigma^x_{m_F,m'_F}$ to the Hilbert space $\{|m_F\rangle, |m'_F\rangle\}$ is the Pauli matrix σ^x, while all its other elements are zero. The evolution operator after a sequence reads $\sim \exp\left(-i\theta\sigma^x_{m_F,m'_F}/2\right)$, where $\theta = \int \Omega_R(t)dt$ is called pulse area and Ω_R is the Rabi frequency. The evolution operator describes a rotation in the restricted Hilbert space associated with the two selected spin states, and is the identity in the orthogonal space consisting of all other levels. Thus, it is distinct from the more usual manipulation of a spin $\vec{F}$ by precession around a magnetic field, $\exp(-i\vec{\alpha}.\vec{F})$, which involves a combination of the three generators $\vec{F} = (F_x, F_y, F_z)$ of the spin-F representation of the $\mathfrak{su}(2)$ algebra. This new tool enables complete control over the spin degrees of freedom, which is necessary to characterize phases of $\mathfrak{su}$(N)-symmetric quantum matter at low energy (see e.g. [48]) and for computing applications leveraging the vast Hilbert space provided by large spins (see e.g. [25]).

It is challenging to intuitively represent the state of large-spin atoms and the $\mathfrak{su}$(N) generators of rotations we implement since a spin $F > 1/2$ possesses a rich inner structure that the usual Bloch sphere fails to represent fully. This structure can be captured by, for example, the Majorana stellar representation [49] (see also [50, 51, 52]). However, the motion of the Majorana stars driven by coherent manipulations is generally quite complicated and does not always allow a clear physical interpretation. Qualitatively, $4F$ angles ($2F$ points on a sphere) are sufficient to describe the state. A simple representation, that can be well-suited to visualize some of our interferometric experiments, is to consider each two-level sub-space of adjacent Zeeman levels $(m_F, m_F + 1)$ in a Bloch sphere representation and consider the set of $2F = N - 1$ Bloch spheres as a representation of the state. For each Bloch sphere, the latitude represents the relative populations while the longitude represents the relative phase of the spin wavefunction in the two relevant states.

Experimentally, we perform rotations on a sub-Bloch sphere between selected pairs of nuclear-spin states, either $m_F \longleftrightarrow m'_F = m_F + 1$ or $m_F \longleftrightarrow m'_F = m_F + 2$ - see Fig. 2.1. To this end, we introduce two-photon Raman couplings. We rely on a quadratic dependence of the light shift as a function of m_F (tensor light shift, TLS), to resonantly drive only one of the two-photon resonances, whereas the other couplings are off-resonant and have negligible impact on the atomic state evolution. Provided the Raman coupling is much smaller than the TLS, we can spectrally resolve any pair of m_F states (with $\Delta m_F = 1, 2$), beyond what was achieved by the approach in Ref. [46], which only allowed coherent oscillations between $m_F = F$ and $m_F = F - 1$ (or $m_F = -F$ and $m_F = -F + 1$). Our work is thus analogous to the proposal in Ref. [53] but with Raman couplings instead of radio-frequency couplings between adjacent Zeeman sublevels. It is also very similar to the experiment on the solid-state platform in Ref. [32], where a hyperfine interaction, instead of an external field, lifts the degeneracy.

Pulsing the Raman coupling drives precession on one selected sub-Bloch sphere around the x axis (by phase convention). Furthermore, at any time, all of the sub-Bloch spheres experience σ_z precession at different rates determined by the detuning of the Raman beat note (controlled by an RF local oscillator) to the corresponding two-level transition. In this article, we explore three independent interferometric schemes restricted to a given Hilbert subspace (see the right panels of Fig. 2.1). They illustrate the new possibilities for quantum sensing or simulation that emerge when using simple combinations of coherent rotations between pairs of spin states, complementing previously demonstrated schemes using large-spin atoms [26, 29, 52]).

The intercombination line ${}^1\mathrm{S}_0 \to {}^3\mathrm{P}_1$ of the ${}^{87}\mathrm{Sr}$ isotope is particularly favorable to engineer large tensor shifts with only weak spontaneous emission, thanks to the high ratio of about 10^5 between the hyperfine splitting in ${}^3P_1$ and the inverse lifetime of these hyperfine states [54]. Using the much narrower ${}^1\mathrm{S}_0 \to {}^3\mathrm{P}_2$ transition might lead to even better performances. Our scheme should readily adapt to other alkaline-earth-like atoms and to other species with similarly rich electronic structures (Yb, Cd, Hg, Er, Dy, ...). Finally, compared to the recent results in Ref. [55, 56], where the spins of erbium and strontium atoms, respectively, are manipulated through coherent oscillations on a narrow optical transition, our scheme also benefits from *not* relying on a clock-like optical transition. This, we achieve because the Raman transitions negligibly populate lossy excited states.

2.3 Experimental setting

All the experiments described in the following sections start with a spin-polarized ultracold Fermi gas of ${}^{87}\mathrm{Sr}$. To produce this gas, we first laser cool and hold within a dipole trap a 5 µK-cold sample of atoms in the $5\mathrm{s}^{2}\,{}^1\mathrm{S}_0$ ground state, fully depolarized across the ten spin states of the $F = 9/2$ manifold. We then use optical pumping on the ${}^1\mathrm{S}_0 \to {}^3\mathrm{P}_1$, $F = 11/2$ intercombination line to empty all spin states but two (typically, the $m_F = -9/2$ and $m_F = -5/2$ states). We thus prepare a stable spin mixture since the $\mathfrak{su}(\mathrm{N})$ symmetry prevents spin-changing collisions. Ref. [13] gives more details on this optical pumping. We next use forced evaporative cooling to obtain an ultracold gas at a temperature of about 100 nK. Finally, we apply a state-selective radiation-pressure pulse to remove atoms in the $m_F = -9/2$ state, with no noticeable heating of the remaining $m_F = -5/2$ sample. We achieve such state selection by a 4.5 ms light pulse tuned to the $|{}^1\mathrm{S}_0, m_F = -9/2\rangle \to |{}^3\mathrm{P}_1, F = 11/2, m_F = -11/2\rangle$ transition in a magnetic field of 4.5 G. The polarized gas, of about 10^4 atoms, is then kept in a uniform magnetic field $B\vec{e}_z$ with $B = 5.2\,\mathrm{G}$ that defines the quantization axis and produces a linear Zeeman splitting of 960(5) Hz. With less than 1% of the atoms in each Zeeman state other than the one chosen, the residual spin entropy is below 0.5 (while a fully random spin state has entropy $\log(10) = 2.3$).

After this preparation, we coherently manipulate the atoms' nuclear spin using an external-cavity laser diode whose frequency is tuned within the hyperfine structure of the $^3\mathrm{P}_1$ state, 600 MHz to the red of the $|^1S_0, F = 9/2\rangle \rightarrow |^3P_1, F = 9/2\rangle$ line. We produce two phase-coherent laser beams with independently controlled frequencies from a single laser split into two paths with independent single-pass acousto-optic modulators. The first beam's primary role is to engineer a quadratic energy shift $U_{TLS}(m_F) = q\,m_F^2$ of the Zeeman states labeled by their spin projection m_F. We will refer to this beam as the "Tensor Light Shift (TLS)" beam. The second beam, labeled the "Raman" beam, acts together with the first (in most of the paper) to drive Raman transitions between two Zeeman states when their frequency difference matches the energy difference between states. We drive the AOMs using two AD9852 direct digital synthesizers synchronized on a shared 20 MHz clock. The beam frequency difference thus has a long term relative instability directly set by the shared clock (RIGOL DG1022), at the 5 ppm level. They are recombined on the same path with orthogonal linear polarizations through a single-mode optical fiber before being sent onto the atoms with a beam waist of 320 µm. This configuration also renders recoil effects negligible.

We set the TLS beam polarization almost co-linear to the magnetic field, corresponding to π transitions in this quantization axis. We ramp its power up to typically 5 mW in 3.5 ms, slowly enough that the atoms' spin state adiabatically follows the slight change of quantization axis. We stabilize the beam intensity to a sub-percent level at the millisecond timescale, using a photodiode positioned after the beam has crossed the atoms. The Raman beam polarization is linear and orthogonal to the magnetic field, allowing σ^+ and σ^- transitions. Throughout this article, we mainly apply $\delta m_F = 1$ spin-changing Raman transitions from absorption and stimulated emission of one π photon from the TLS beam and one σ^- photon from the Raman beam. By setting the two-photon detuning resonant with such transitions, the σ^+ Raman beam photons only drive widely off-resonant processes thanks to the TLS beam's quadratic shift $U_{TLS}(m_F)$ and ultimately remain spectators. Moreover, the σ^- transitions addressed by the Raman laser have higher Clebsch-Gordan coefficients and coupling strengths when manipulating $m_F < 0$ states. In an alternative scheme, the Raman laser is bichromatic, and drives alone $\delta m_F = 2$ two-photon spin-changing transitions using its two circular polarization components. The two optical frequencies are generated on the same AOM by a bichromatic RF drive, produced by frequency mixing.

At the end of an experiment cycle, we adiabatically ramp down the TLS beam and measure the spin-state distribution by spin-selective momentum transfer [13], simultaneously extracting and probing two spin states of choice (m_F-1 and m_F+1) for each measurement. Spin populations presented throughout the paper are relative to the total atom number. For this, we calibrate the detection efficiencies $\eta(m_F)$ for each spin state on clouds prepared in that single spin state. For $m_F = -7/2$, $-5/2$, and $-3/2$, we find $\eta(m_F) = 0.65$, 0.70, and 0.51, respectively [a]. However,

[a] The detection efficiencies are probably limited by a lack of optical power and by using rather

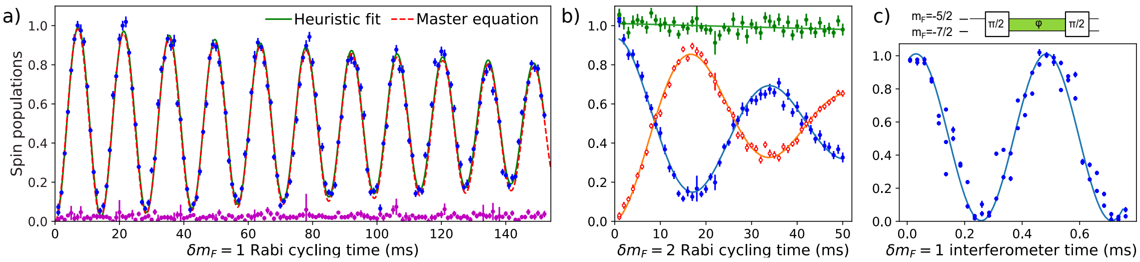


Figure 2.2: **Encoding a qubit on two nuclear spin states.** a) Rabi oscillations between $m_F = -3/2$ (blue) and $m_F = -5/2$ (not shown), driven by a $\delta m_F = 1$ Raman coupling. Here $\hbar\Omega_R/2q$ is small enough that the neighboring states $m_F = -7/2$ (magenta) and $m_F = -1/2$ (not shown) are unaffected. The green line is a heuristic fit to the data with a damped sine, yielding a $1/e$ damping time of 298 (17) ms. The dashed red line is the result of a master equation simulation with two adjustable parameters: the Raman coupling strength and the coherences decay rate. It is almost indistinguishable from the damped sine. (b) Rabi oscillations between $m_F = -3/2$ (blue, filled circles) and $m_F = -7/2$ (red, empty circles) driven by a $\delta m_F = 2$ Raman coupling. Green dots show the total population in these two states. Both populations are measured simultaneously. Solid lines are fits with damped sinusoidal oscillations, yielding a $1/e$ damping time of 40(5) ms. c) Ramsey interferogram realized between states $m_F = -7/2$ (blue dots) and $m_F = -5/2$, driven by a $\delta m_F = 1$ Raman coupling. The line is an optimal distance regression fit with a sinusoidal interferogram. In all of this figure, each data results from a single shot, and error bars reflect the population measurement statistical uncertainty. Furthermore, in plots (a) and (b), the detection efficiencies $\eta(-3/2)$ are slightly recalibrated to bring maximal $m_F = -3/2$ population estimates down to 1.

$\eta(-3/2)$ fluctuated throughout the experiments shown here, as evidenced by some population estimates larger than one. In such cases, we recalibrate $\eta(-3/2)$ by up to 6 % to bring extremal population estimates down to 1.

2.4 Coherent manipulation of a nuclear-spin qubit

Rabi oscillations between two isolated hyperfine states. We now turn to the experimental demonstration of coherent manipulations between selected pairs of states within the ^{87}Sr ground state manifold, which realizes evolutions driven by $\mathfrak{su}$(N) generators $\sigma^x_{m_F,m'_F}$ and $\sigma^z_{m_F,m'_F}$. We first show Rabi oscillations between two Zeeman sublevels, then characterize the coherence of quantum superpositions by Ramsey interferometry.

We first demonstrate Rabi oscillations with $\delta m_F = 1$. To this end, we prepare the gas in the $m_F = -5/2$ state and turn the quadratic light shift on $U_{TLS}(m_F) = qm_F^2$. For $\delta m_F = 1$ Raman transitions, this splits the resonance conditions of the various Raman processes by multiples of $2q/h$ (h is the Planck constant). Consequently, as illustrated in Fig. 2.1, tuning the Raman beam frequency to one of the resonance conditions (e.g., the process $-5/2 \to -3/2$) produces a state evolution that can be approximated as generated by a Pauli operator coupling two and only two states. To

warm clouds with momentum spread comparable to the optical recoil.

reach this regime, the separation between adjacent resonances $2q/h$ (proportional to the TLS beam intensity I_{TLS}) must be larger than the two photon Raman coupling strength ($\Omega_R \propto \sqrt{I_{TLS} I_R}$, where I_R is the Raman beam intensity), which implies $I_R \ll I_{TLS}$. In practice we use $I_R/I_{TLS} \leq 10^{-4}$.

In the presence of the TLS beam, we pulse the Raman beam at a fixed frequency and monitor the spin state evolution. In Fig. 2.2a, we demonstrate the Rabi oscillation dynamics driven between the states $m_F = -5/2$ and $m_F = -3/2$. Here $q/h = -320\,\text{Hz}$ and $\Omega_R/2\pi = 71\,\text{Hz}$. We observe well-pronounced coherent oscillations with a short-time contrast consistent with 1, while the population in the neighboring, not targeted state $-7/2$ grows by less than 1 % over almost 11 Rabi oscillations. The amplitude of the oscillations exhibits a slow exponential decrease with a characteristic time $\tau_{decay} = 298(17)\,\text{ms} \simeq 21\frac{2\pi}{\Omega_R}$. As we do not observe significant population diffusion to other states, this damping rate limits the fidelity of a $\pi/2$ pulse to 0.994. The shot-to-shot fluctuations of the Rabi pulse area away from $\pi/2$ (calibrated from complementary experiments) are even less limiting to the fidelity (> 0.998). This demonstration of an evolution well approximated by $\exp\left(-i\theta\sigma^x_{m_F,m_F+1}/2\right)$ constitutes a key result of this article, enabling the various experiments described in the following sections.

We also demonstrate Raman transitions with $\delta m_F = 2$ using a slightly different scheme: we detune the Raman beam 1 MHz below the TLS beam and modulate its frequency in two sideband components (suppressing the carrier frequency). In this manner, its two opposed $\sigma^\pm$ polarization components drive $\delta m_F = 2$ transitions, with a detuning controlled by the modulation frequency. We performed this experiment at a much lower $q/h = -95\,\text{Hz}$ (because of the aging of a laser component). In Fig. 2.2b, we observe the Rabi oscillation induced between the states $-7/2$ and $-3/2$. The two spin populations, monitored in parallel, exhibit complementary oscillations at frequency $\Omega_R/2\pi = 29\,(3)\,\text{Hz}$ with a damping time of $40\,(5)\,\text{ms}$. The total population in the two states decays by 3% within the damping time, so the damping mainly reflects a loss of coherence, not diffusion to other spin states. The limit imposed by this damping on the fidelity of a $\pi/2$ pulse can thus be estimated at 0.90, much lower than what we achieve on the $\delta m_F = 1$ Raman processes. The cause of this more significant damping compared to the $\delta m_F = 1$ coupling experiment is unknown - possibly the laser degrading, which also resulted in the lower value of q. Still, this experiment demonstrates a complementary set of two-level rotations. They are not required to have complete control over the atomic state, but, in principle, an extended set of generators of evolution can be advantageous for some manipulations (assuming similar fidelities). Furthermore, combining $\delta m_F = 1$ and $\delta m_F = 2$ transitions enables conceptually new kinds of experiments, such as engineering dynamics in a spin space with periodic boundary conditions, with phenomena related to Berry phases and non-Abelian gauge fields [57, 45]. In the remainder of the paper, we will use exclusively $\delta m_F = 1$ transitions as the basic building block for interferometers.

Coherence of the Rabi oscillations. Let us characterize the coherence properties of the nuclear qubit - first in the presence, then in the absence, of the Raman driving field. In the Rabi oscillations presented in Fig. 2.2a, each point is the outcome of an individual realization. The observed damping is seen without averaging different experimental realizations, which implies that it is associated with loss of coherence rather than fluctuations. This decoherence may result from a spatial inhomogeneity or from spontaneous emission. To establish which one dominates, we measure the heating rate of atoms exposed to TLS light in the dipole trap. We find it to be three times higher than the expected heating rate associated with spontaneous emission assuming the laser is monochromatic. We attribute this to a frequency noise pedestal in our laser spectrum due to amplified spontaneous emission (ASE) that causes *resonant* photon scattering. We then model the Rabi dynamics by a master equation describing the evolution of an atom in the laser and magnetic fields. We calculate the separate spontaneous emission terms from and to each spin state, such that the model captures decoherence and spin-state diffusion. To heuristically capture the effect of ASE, we scale these rates of coherence decay and spin diffusion by a factor of three. Typical coherence decay rates are $\sim 1\,\mathrm{s}^{-1}$, and population transfer rates $\sim 0.5\,\mathrm{s}^{-1}$. However, the simulation with these parameters does not produce the evolution observed in Fig. 2.2a, nor does artificially scaling the impact of spontaneous emission by a different factor. Instead, we note that there is TLS/Raman beam inhomogeneity, and empirically add a decay of coherences between spin states (m_1, m_2) with rate $\Gamma_q \times |m_1^2 - m_2^2|$. With $\Gamma_q = 1.2\,\mathrm{s}^{-1}$, we recover excellent agreement with the data, as shown in Fig. 2.2a. We estimate independently, by simulations with a random distribution of (q, Ω_R), that the observed dynamics is consistent with an inhomogeneity of q and Ω_R of about 1.2 % in standard deviation. Across a 5μm-radius cloud, this can be induced by a beam off-centered by $w/4$, where $w = 320\,\mu$m is the waist.

Coherence of the hyperfine qubit Ramsey interferometer. We then test the coherence of the qubit in the absence of Raman drive. For this, we implement an interferometer. Keeping the TLS beam on, we apply a set of two resonant $\pi/2$ pulses, separated by a temporal interval T - thus realizing a standard Ramsey sequence restricted to the Hilbert space of a qubit, see the top sequence schematic of Fig. 2.1b. During this time interval T, we alter the local oscillator frequency that controls the phase difference between TLS and Raman beams. The Ramsey interferometer is sensitive to the phase ϕ defined by $\frac{d\phi}{dt} = \Delta E_{m_F,m'_F}(t)/\hbar - \delta_{RF}(t)$, where $\Delta E_{m_F,m'_F}$ is the energy difference between the two qubit states and $\delta_{RF}(t) = \omega_R - \omega_{TLS}$. The quantities $\omega_{TLS}/2\pi$ and $\omega_R/2\pi$ are the (potentially time-dependent) TLS and Raman beam laser frequencies.

Figure 2.2c shows the fringe pattern as a function of T for qubits encoded in the two nuclear-spin states $m_F = -5/2$ and $m_F = -7/2$. Here, $q/h = 190\,\mathrm{Hz}$ and $\Omega_R/2\pi = 93\,\mathrm{Hz}$. The well-resolved oscillations come from the accumulated differential phase in the qubit states, resulting in a rotation (controlled by T) along the sub-Bloch sphere equator. We observe fully contrasted oscillations for the best data, which demonstrates that the coherence of the qubit is well preserved.

We now systematically study the contrast and noise of Ramsey interferometers as a function of T to characterize the sources of decoherence. We will highlight that the main detriment to the qubit's coherence is the TLS beam and that we can adiabatically remove it to recover coherence over seconds. We present this study in Fig. 2.3. We focus on two effects: decoherence, which reduces the interferometric contrast, and phase fluctuations. In principle, the sources of fluctuations in the data are (i) atomic quantum projection noise; (ii) measurement uncertainties associated with fitting the spin-separated optical density pictures; (iii) reproducibility of the

$\pi/2$ pulses; iv) fluctuations of the phase integrated by the interferometers. For our data, the quantum projection noise (i) is smaller than the measurement noise (ii) by typically a factor of 3, so we neglect it. Fluctuations of the area of "$\pi/2$" pulses (iii) are estimated from repeating a sequence without closing the interferometer. We find a standard deviation of the pulse area of 0.063(5) rad (which sets a fidelity upper bound for a $\pi/2$ pulse at 0.999). However, their effect in a *closed* Ramsey sequence remains smaller than the measurement uncertainty. On the interferogram shown in Fig. 2.3b, we notice that the residuals to a sinusoidal model are much larger than the fluctuations expected from combining the effect of (i), (ii), and (iii). We attribute this to uncontrolled fluctuations of the interferometer phase. To estimate their RMS amplitude, we numerically generate trial noisy models of the experiment, including all sources of fluctuations, and compare them with the data [b]. In this optimization, the interferometric contrast and the variance of phase fluctuations are the adjustable parameters.

Fig. 2.3c and d show the outcome of this analysis. As shown in Fig. 2.3c, the contrast (defined as peak-to-peak signal amplitude) decays as a function of T with a $1/e$ decay constant of 54 (5) ms. Like the damping of Rabi oscillation, this decay most likely results from the TLS inhomogeneity rather than photon scattering. The observed decay is significantly faster than that of the Rabi oscillations seen in Fig. 2.2a. This illustrates the sensitivity of Ramsey interferometers, which are directly sensitive to detunings Δ, in contrast to generalized Rabi frequencies, generically $\sqrt{\Omega_R^2 + \Delta^2}$.

A second observation is that a random phase is integrated for each realization of the interferometer. At the smaller dark time $T = 5\,\text{ms}$, its standard deviation is about 0.35 rad and gradually increases with T, as shown in Fig. 2.3d. It is about 1.4 rad after 100 ms. We attribute this to polarization fluctuations in the TLS beam, as we will demonstrate in Sec. 2.5.1. One could reduce these fluctuations by polarization filtering, although our setup, where we co-propagate the TLS and Raman beams in the same optical fiber, is inadequate for this.

Thus, simply removing the tensor light shift for most of the duration T of the interferometer should significantly mitigate decoherence and phase noise. After the initial Rabi pulse, we adiabatically turn off the TLS in 2 ms, and bring it back on in the last 2 ms before the second $\pi/2$ pulse. The atom's environment during the interferometer is now primarily the magnetic field and a dipole trap with negligible spin-dependent light shift. The observed decoherence is considerably suppressed, with no visible contrast decay after 3 seconds. The phase noise also grows with T at a much slower rate. Phase fluctuations of $\simeq 0.3$ rad are acquired during the 4 ms with TLS, but the phase-noise contribution of the remaining time without light only reaches 1 radian after about 2 seconds. Although the RMS phase fluctuations are not exactly linear with T, they can qualitatively be interpreted as an instability of the Raman detuning of $\sim 0.08\,\text{Hz}$ when integrated (averaged) over two seconds. The frequency instability of the DDS-controlled $\delta_{RF} = \omega_R - \omega_{TLS}$ is, in principle, one order of magnitude smaller. Thus, the phase noise probably reflects magnetic field fluctuations of about 0.4 mG at the second timescale, consistent with

[b] We look for a trial model for which a least-square sinusoidal fit estimates the same amplitude and with the same residuals as the data. Note that the peak-to-peak amplitude of the sinusoidal fit is generally smaller than the contrast of the interferogram: an interferogram with complete loss of control over the phase will be fitted with zero amplitude and large residuals.

the specifications of the current supplies for the coils.

In this section, we have demonstrated coherent control of a qubit defined over two selected Zeeman sublevels using $\sigma^x_{m_F,m_F+1}$, $\sigma^x_{m_F,m_F+2}$, and $\sigma^z_{m_F,m'_F}$ rotations. Our demonstrations have been restricted to the manifold $m_F \in \{-9/2, -7/2, -5/2, -3/2\}$. Note that, when driving atoms in other states, we observed uncontrolled population transfers. We attribute this to the value of our magnetic field $b \simeq -3q$, which causes quasi-degeneracy and sign reversal of the Raman energy differences, see Fig. 2.1a. One could lift this issue and acquire complete control over the entire spin manifold, provided $b > |q|(2F-1)$. Within the manifold of states used in this paper, we have shown the high sensitivity of the qubits' coherence and phases to the TLS beam fluctuations. Minimizing the exposure time to the TLS beam, we have demonstrated long-lived coherent superpositions over several seconds, without any magnetic shielding. Such coherence time is a feature of the minute Landé factor of the nuclear spins, basically immune to stray magnetic field gradients ($g_I \mu_0 \nabla B < 10^{-4}\,\mathrm{Hz}/\mu\mathrm{m}$), and of the extremely small vector and tensor polarisabilities in the closed-shell ground state, at the optical trapping wavelength (with spin-dependent shifts $\simeq 10^{-4}\,\mathrm{Hz}$). It can be a great asset, e.g., for high-precision sensing in long-interrogation-time interferometers or quantum information storage.

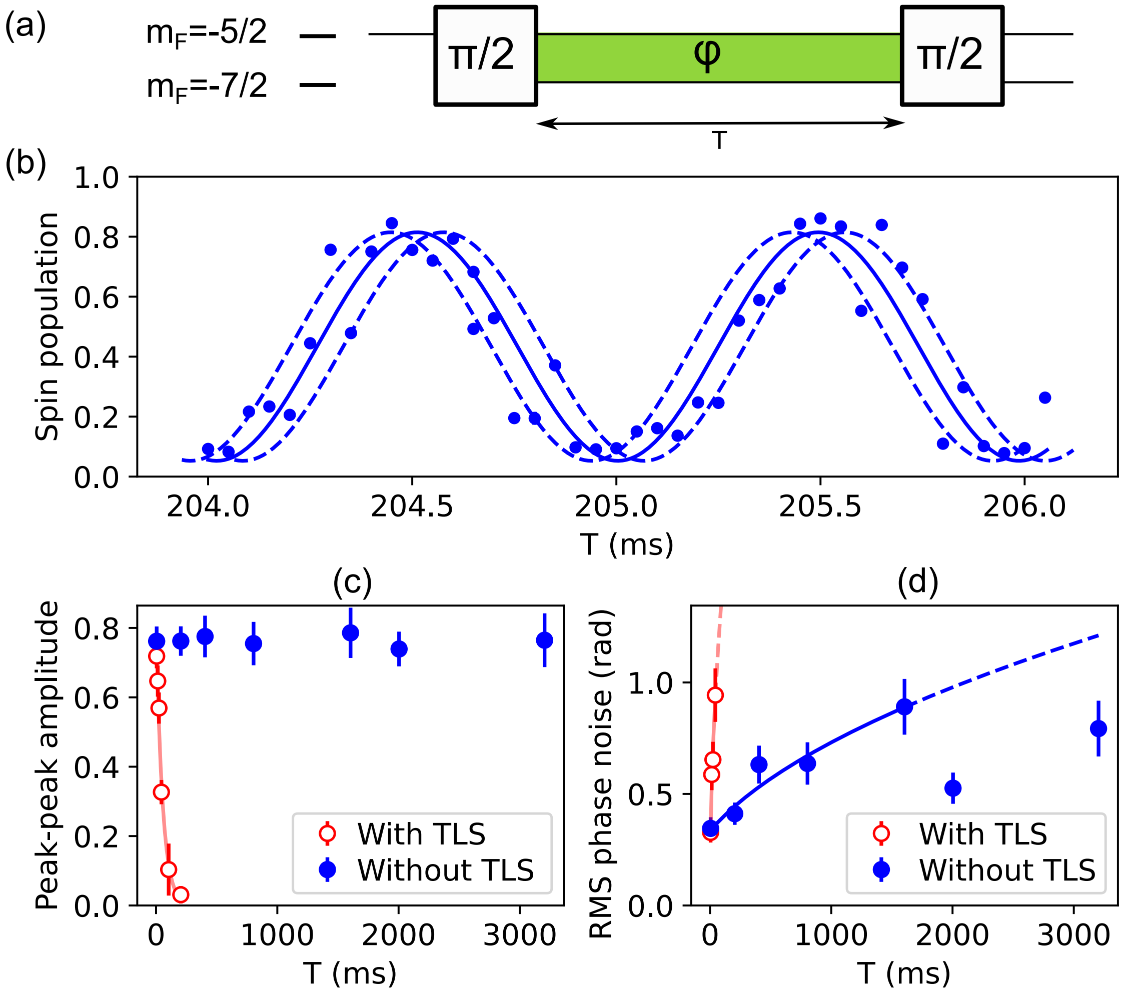


Figure 2.3: **Qubit coherence.** (a) Schematics of the interferometer. (b) Fringe pattern of a Ramsey interferometer operated between $m_F = -5/2$ (measured) and $m_F = -7/2$, with, between the two pulses, 4 ms of exposition to TLS light and $T-4$ ms of absence of any laser light. Error bars, combining population measurement uncertainty and expected impact of pulse area fluctuations, are comparable to the dots' size. The solid line is the underlying interferogram, barring phase fluctuations. The dashed lines interferograms are shifted by $\pm$ one standard deviation of the phase fluctuations. (c) Contrast as a function of T. The red line is a fit to an exponential decay for the data with TLS beam on (red, empty circles), with $1/e$ decay time of 54(5) ms. When the TLS beam is adiabatically turned off during the dark time (blue, filled circles), there is no observable contrast decay for at least 3 s. The shortest measured T (5 ms) corresponds to ramping down and almost immediately back up the TLS. (d) Phase noise of the interferometer as a function of T. Continuous lines are fits assuming $Var(\phi) = \delta\phi_0^2 + Dt$. We have $\sqrt{D} = 0.14(1)\,\mathrm{rad/ms}^{1/2}$ with TLS, and $\sqrt{D} = 0.021(2)\,\mathrm{rad/ms}^{1/2}$ without TLS.

2.5 Beyond pseudo-spin 1/2 schemes: qudit interferometry

The possibility to coherently couple two spin states among the $N = 10$ available ones in ground-state ^{87}Sr atoms enables coherent navigation over the whole ground-state manifold - for example, by simply concatenating a series of resonant pulses between isolated pairs of Zeeman states. Consequently, our experimental toolkit creates new possibilities that use more than two spin states in order to store or retrieve information. In this section, we experimentally investigate two of those new possibilities that use the large nuclear spin as a genuine qudit.

First, we will use well-defined linear-superposition states involving four spin states, in order to simultaneously measure more than one external field at each experimental realization. Such multi-parameter estimates are useful in the context of metrology [58]. Our work thus illustrates the potential of large-spin particles as original quantum sensors.

Second, we present a scheme that simultaneously measures collective observables that do not commute, in a single experimental realization. Building on Ref. [59], our scheme can characterize a many-body state where the coherences are stored in a subset of the fully available ground-state hyperfine manifold. The main idea is to extend the Hilbert space of the atoms for measurement purposes only; the measurement sequence involves mapping the state of the atoms into initially empty "ancillary" states and applying controlled rotations, which provides additional observables, and facilitates tomographic characterization of a quantum many-body state.

2.5.1 Parallel Ramsey interferometers for simultaneous multiple-field sensing.

Performing simultaneous measurements using the same local quantum sensor is a new possibility with large-spin particles that is particularly interesting, for example, because the measurements can then benefit from common noise rejection (see for example [60, 61, 62, 63]) or because it becomes possible to perform correlations analyses and thus learn from sources of noise (we will show an example in the following). Here, we present simultaneous measurements of two fields in each experimental realization. We drive two $s = 1/2$ Ramsey interferometers in parallel, using two independent sets of pairs of Zeeman states within strontium's ground-state hyperfine manifold. Our scheme using sets of qubits *internally distributed* within the ground-state spin manifold has similarities with the architecture of *spatially distributed* interferometers studied in Ref. [64].

The protocol is shown in Fig. 2.4a. From an initially pure spin state in $m_F = -5/2$ we apply a first $\pi/2$ pulse that creates a coherent superposition equally distributed over $\{-7/2, -5/2\}$ [65]. Next, each state enters the respective input port of two independent Ramsey interferometers, one engineered between $\{-5/2, -3/2\}$ extracting a phase ϕ_1, the other between $\{-7/2, -9/2\}$ extracting a phase ϕ_2. The experiment relies on a single RF local oscillator that controls the Raman detuning δ_{RF}. The two interferometers are thus opened sequentially at times t_1 and t_2 by switching the Raman detuning from one resonance to the other, after which they accumulate phase for a time together, then are finally closed sequentially again. Both interferometers are open for the same duration T. The first interferometer's populations are thus sensitive to the phase

$$\begin{aligned}\phi_1 &= \frac{1}{\hbar}\int_{t_1}^{t_1+T}\left(E_{-5/2}-E_{-3/2}\right)dt - \int_{t_1}^{t_1+T}\delta_{RF}(t)dt \\ &= \frac{T}{\hbar}(4q-b) - T\langle\delta_{RF}\rangle_1, \end{aligned} \tag{2.1}$$

and the second interferometer's populations, to

$$\begin{aligned}\phi_2 &= \frac{1}{\hbar}\int_{t_2}^{t_2+T}\left(E_{-9/2}-E_{-7/2}\right)dt - \int_{t_2}^{t_2+T}\delta_{RF}(t)dt \\ &= \frac{T}{\hbar}(8q-b) - T\langle\delta_{RF}\rangle_2. \end{aligned} \tag{2.2}$$

Note that $\langle\delta_{RF}\rangle_1 \neq \langle\delta_{RF}\rangle_2$. The parallel operation of the Ramsey interferometers provides a measurement of both terms contributing to the single-atom ground-state energies, that are, the linear energy splitting b (due to both the magnetic Zeeman effect and the vector light shift) and the quadratic energy splitting q (due to the tensor light shift). As $\langle\delta_{RF}\rangle_1$ and $\langle\delta_{RF}\rangle_2$ are controlled (here with an accuracy below 0.02 Hz), the difference between the two interferometers' phases $\phi_1 - \phi_2$ provides a measurement of q, which is insensitive to shot-to-shot fluctuations in b, while $2\phi_1 - \phi_2$ provides a measurement of b, which is insensitive to shot-to-shot fluctuations in q [c].

In Fig. 2.4b, we present the interferograms obtained by measuring the variation of the populations in $m_F = -7/2$ and $m_F = -3/2$, relative to the total atom number, as a function of the interferometer time T. Here, $q/h \simeq -320$ Hz (independently deduced by spectroscopy) and $\Omega_R/2\pi = 77$ Hz on the $(-5/2 \to -7/2)$ line. Both populations are measured simultaneously at each experimental realization. During the time when both interferometers are simultaneously opened ($\simeq T - 3.5$ ms), we set δ_{RF} to $2\pi \times 1$ Hz, off-resonant with both transitions. We observe that the periods of the two fringe patterns strongly differ, as $E_{-5/2} - E_{-3/2} \neq E_{-9/2} - E_{-7/2}$. From the two periods, we can deduce the mean quadratic and linear Zeeman splittings: $\langle q\rangle/h = -303(8)$ Hz and $\langle b\rangle/h = 1000(45)$ Hz - in agreement with independent estimates from Raman spectra. These are mean values over all realizations, as q, b can fluctuate from shot to shot. Upon knowing the fringe contrast and the two phases ϕ_1, ϕ_2 with better than π accuracy, single realizations of the experiment provide parallel individual simultaneous measurements of ϕ_1 and ϕ_2, from which we can deduce single-shot measurements of q and b. Given the interrogation time $T \simeq 4$ ms and noise level of these data (affected by Rabi pulse area fluctuations, mostly of the first pulse), the respective precisions in q, b from individual points when at maximal slopes are $h \times 1.8$ Hz and $h \times 11$ Hz.

To illustrate this, we operate the interferometer at a fixed T chosen to be close to the maximum phase sensitivity of the interferometer (*i.e.*, close to mid-fringe for both signals). Fig. 2.4c shows a set of 93 measurements of the two phases (ϕ_1, ϕ_2) taken consecutively. For each measurement, we directly reconstruct the deviations of b and q relative to the mean, as shown in Fig. 2.4d. These data demonstrate that

[c] Technically, q and b can also be time-varying during the interferometer sequence, and the interferometers are sensitive to their time-averages. Fluctuations of (q, b) at the timescale of T will be properly retrieved by the combined interferometers provided $T \gg 1/\Omega_R$, or if an additional laser frequency is used to engineer the Raman couplings simultaneously.

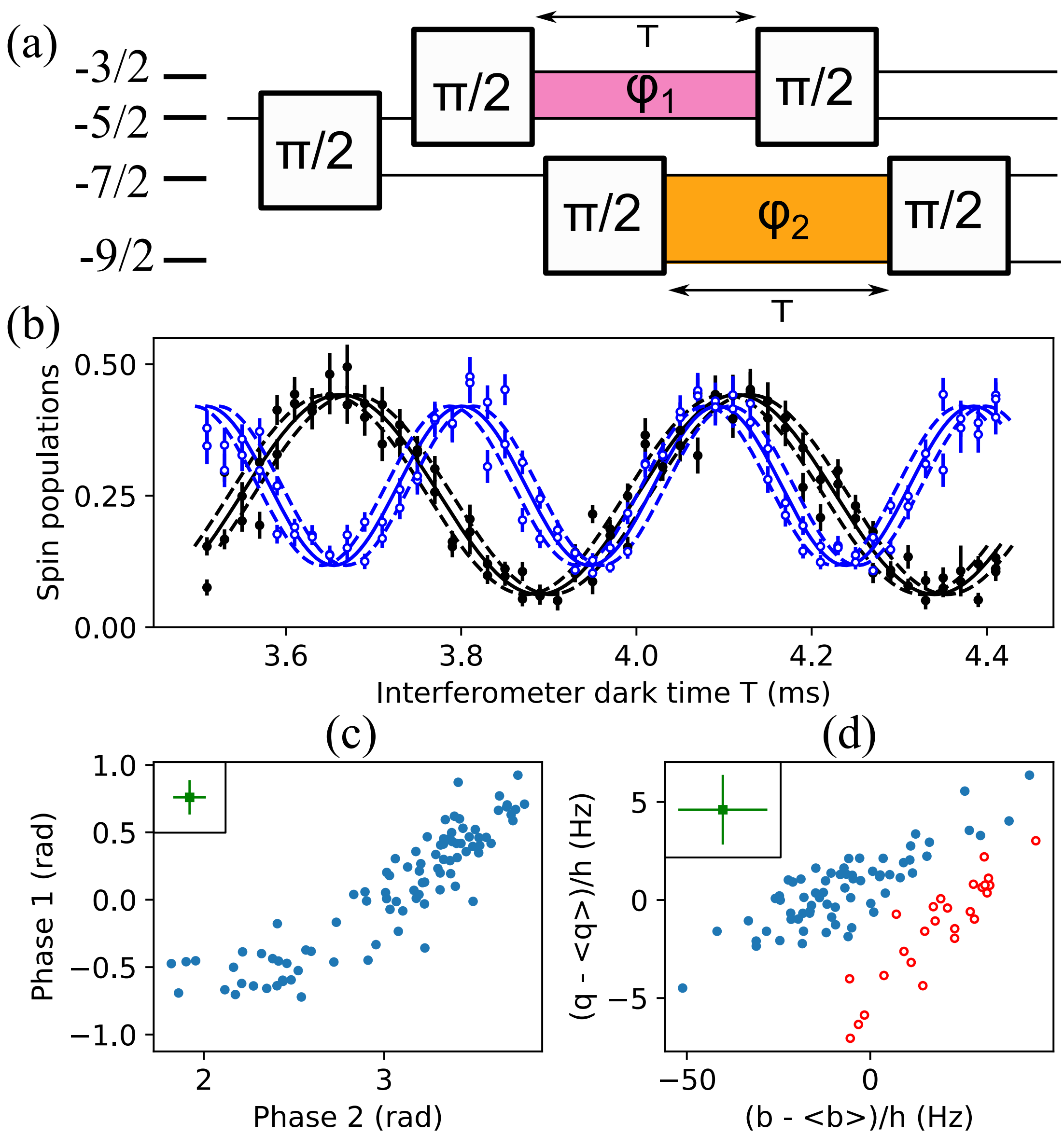


Figure 2.4: **Simultaneous sensing of two external fields.** a) Double interferometer pulse scheme. b) Measured interferogram. We vary the phases ϕ_1 and ϕ_2 together by tuning the interferometers' phase accumulation times T. We measure both states $|-3/2\rangle$ (black, full circles) and $|-7/2\rangle$ (blue, empty circles) in each realization. The differing periods highlight that two different energies are measured. The error bars combine population measurement uncertainties and the expected impact of Rabi pulse area fluctuations. The solid line is the underlying sinusoidal interferogram, barring phase fluctuations. The dashed-line interferograms represent shifts of $\pm$ one standard deviation of the phase fluctuations. c) Correlation analysis of the two phases ϕ_1 and ϕ_2, measured modulo 2π by repeating experiments at a chosen phase with the highest slope for both interferometric signals. We show the median of the error bars (including measurement uncertainty and pulse area fluctuations) in the top-left corner. d) Reconstruction of the two fields (b, q). The data split into two groups, identified by color and predominantly separated in b (the split corresponds to $\phi_2 \lesssim 2.8\,\text{rad}$). We cancel this experimental fluctuation for most data in the paper, including Fig. 2.4b, by protecting the TLS optical fiber from airflow.

the simultaneous nature of the measurement makes possible correlation analyses that would otherwise be impossible if the experiment were alternating between one interferometer and the other.

As an example of correlation analysis, we have used this data as follows. We observe that the parallel measurements of (q, b) do not seem 2D-Gaussian distributed. Instead, two classes of shots are identifiable. The two groups' separation in $\langle q \rangle$ is $-1.4(5)$ Hz, i.e., a small fraction (0.4%) of $\langle q \rangle$ and below the spread within a single group. In contrast, the separation in $\langle b \rangle$ of 23(3) Hz is significant and higher than the spread in a single group. We attribute this to an evolution of the TLS beam polarization that shifts between two configurations; and thus induces a fluctuating vector light shift with values clustering in two groups, while there should be no vector light shift for the ideal π polarization. By protecting the TLS optical fiber from airflow, only one group remains, and the fluctuations become consistent with measurement uncertainties (also impacted by the first pulse's area fluctuation). We acquire most of the data in this paper (including Fig. 2.4b) with fiber protection.

We now discuss the fundamental limit to the sensitivity of the measurements. In the experiments presented here, the dominant uncertainties on b and q arise from Rabi-pulse-area fluctuations and the atom number measurement precision. Nevertheless, the atomic quantum projection noise sets the fundamental limit to the achievable precision of measurements. A key question is what the tradeoff for parallelizing two measurements is. Each one uses, on average, half of the total atom number. In addition, we present in this paper a suboptimal measurement, as we measure only one output port of each interferometer at each realization. Not knowing the exact atom number fed into each interferometer in each realization increases the quantum-projection-noise limit for phase estimation by $\sqrt{3/2}$. However, in principle, we could measure all four states in each realization [66, 67, 13]. The fluctuations in the measurement of the two phases associated with the atomic quantum-projection noise are simply $\sqrt{2}$ times stronger than when using all atoms on the same interferometer. This mild disadvantage vanishes if one considers that at least two runs of the experiment would be required to measure sequentially the two interferometers.

We finally discuss the three main systematic effects that could affect the phase measurements. First, the pulses of the two interferometers are interleaved temporally. Consequently, between the splitting and recombining pulses of the interferometer (1), we realize one Raman pulse addressing interferometer (2), and vice-versa. This interleaving creates AC Stark shifts $\simeq \Omega_R^2/8q$ on the two levels probed by interferometer (1) of the same sign but not exactly compensating due to different Clebsh-Gordan coefficients in the Raman couplings. These shifts can be calculated, or minimized by ensuring the highest possible energy scale separation $2q/\Omega_R$. Second, when the Rabi frequencies are only slightly smaller than the energy difference between states, one should account for effects beyond the rotating-wave approximation, such as the Bloch-Siegert shift, and expect the interferometer to be sensitive to the initial temporal phase of the beat between the two Raman beams [68]. Third, in a scheme spanning many pulses and Zeeman states, off-resonant transfers between spin states can interfere with the resonantly driven population transfers. We will see in the next section evidence for this. Effectively, this affects the fringe patterns and phase estimation.

We have shown in this section an example of a sensor that parallelizes two measurements using an extended Hilbert space basis. Thanks to the simultaneity of the measurements, we have demonstrated that correlation analyses are possible. We illustrated it by tracking in parallel variations of the vector and tensor light shifts acting on the atoms. One can also interpret these experiments as a common-mode noise-rejection scheme, i.e., extracting precise values of q despite the fact that fluctuations of b affect all interferometers. Similar schemes for the simultaneous sensing of multiple fields could be devised, e.g., for measuring multiple spatial components of a vector (magnetic) field.

2.5.2 Simultaneous measurements of multiple collective atomic observables.

We will now explore a second possibility offered by the large spin-state manifold: increasing the number of observables accessible in one realization and characterizing the collective spin state of a quantum many-body ensemble.

Let us start by remarking that tracking two spatial components of a magnetic field by monitoring spin precession requires measuring the projection of the collective spin along at least two directions simultaneously. However, this is typically not possible due to the non-commutativity of the two spin-projection observables. More fundamentally, fluctuations of a collective spin in the two directions transverse to its mean orientation are critical to the precision of metrological applications [69, 70] but remain assessed along one given quantization axis at a time. Furthermore, measuring the collective spin properties enables tests of a large class of entanglement witnesses [71, 72, 73, 74]. We also point out that large-spin atoms are compelling due to numerous new scenarios involving entanglement witnesses [75, 76]. For large-spin atoms, only measuring the total population in each spin state can already reveal beyond mean-field dynamics [18, 77], and measuring collective spin fluctuations in a single axis can already be used to characterize correlations [24, 78]. Here, we will show that ensembles of large-spin atoms offer an opportunity for physics with ensembles of effectively smaller pseudo-spins: measuring two orthogonal spin-projection operators - or generically, two non-commuting observables - in one given experimental realization.

The general idea is to use a larger space of states for each atom in the measurement sequence than during the experiment that came before it, so that a higher number of observables is available. These observables (e.g., populations) in the large Hilbert space commute with each other and can be simultaneously measured; but their expectation values and even higher order statistics reflect those of non-commuting observables of the state produced by the experiment.

One such scheme emerges if one can coherently map the internal states involved in the experiment into other, initially empty, ancillary states used for measurement purposes only. Reference [59] demonstrated this idea, using the complete hyperfine structure of alkali atoms to probe spinor properties in the $F = 1$ manifold. We generalize this approach to the case of ground-state alkaline-earth atoms. In practice, we consider ensembles of effective qubits, i.e., pseudo-spins 1/2, initially restricted to two Zeeman sublevels. To simultaneously measure two pseudo-spin projections, we first map each state of this qubit into ancillary spin states accessible in the

$F = 9/2$ manifold, using a partial coherent transfer. Then, using again the capacity to perform manipulations targeted on specific pairs of states (i.e., rotations on a sub-Bloch-sphere), we perform different basis transformations on the qubit states than on the ancillary states. The final measurement of populations thus contains information formerly inaccessible in the initial populations of the qubit states. We show a typical sequence in Fig. 2.5a.

For our experiments, we define the two qubit states, $|\uparrow\rangle = |-5/2\rangle$ and $|\downarrow\rangle = |-7/2\rangle$. The initial state, which we produce to test the measurement protocol, is a coherent state of qubits $|\psi\rangle^{(N_{at})} = \left((|\uparrow\rangle - i\,|\downarrow\rangle)/\sqrt{2}\right)^{\otimes N_{at}}$, where N_{at} is the atom number. We produce it from a sample polarized in $|\uparrow\rangle$, by one $\pi/2$ Raman pulse between the qubit states. With our phase convention, this coherent state is polarized against the pseudo-spin quantization axis "y", i.e., the total pseudo-spin of the ensemble has projections $\langle \vec{s}_{\uparrow,\downarrow}\rangle = (0, -N_{at}/2, 0)$.

The measurement sequence then begins. First, two successive $\pi/2$ pulses duplicate the $|\uparrow\rangle$ amplitude to the ancillary state $|a\rangle = |-3/2\rangle$ and the $|\downarrow\rangle$ amplitude to the ancillary state $|b\rangle = |-9/2\rangle$, respectively. Second, we let the phases of all four states evolve in the presence of the TLS and magnetic field. In particular, the qubit sub-Bloch sphere effectively evolves by $\exp\left(-i\phi\sigma^z_{\uparrow,\downarrow}/2\right)$, where the angle ϕ is controlled by the duration of the rotation and by the detuning between the RF clock and the qubit transition. Third, we apply a final $\pi/2$ pulse between the two qubit states, corresponding to applying $\exp\left(-i(\pi/2)\sigma^x_{\uparrow,\downarrow}/2\right)$. At the end of the sequence, one would ideally measure the populations N_i of all four states. The population difference $N_a - N_b$ provides an estimator for the many-body operator $\hat{s}^z_{\uparrow,\downarrow}$. Similarly, $N_\uparrow - N_\downarrow$ provides an estimator for $\hat{s}^\phi_{\uparrow,\downarrow} = \cos(\phi)\hat{s}^y_{\uparrow,\downarrow} + \sin(\phi)\hat{s}^x_{\uparrow,\downarrow}$, the total pseudo-spin projection in one selected direction of the plane (x, y). Both quantities are thus accessible in a single experimental run.

Let us describe more formally the measurement, which will enable us to discuss its statistics. We label U the one-body evolution operator corresponding to the series of three $\pi/2$ pulses and phase evolution within the measurement sequence, outlined in Fig. 2.5, with $\phi = 0$ for simplicity. We apply the same evolution to all spins in parallel. Thus, the many-body evolution operator is factorizable onto all the modes of the atomic external degrees of freedom λ: $U^{(N)} = \Pi_\lambda U_\lambda$. We define two projective measurement operators,

$$\hat{O}^z = U^{(N)\dagger}(N_a - N_b)U^{(N)}, \text{ and} \tag{2.3}$$
$$\hat{O}^y = U^{(N)\dagger}(N_\uparrow - N_\downarrow)U^{(N)}. \tag{2.4}$$

They correspond to measuring all four atomic populations $(N_a, N_b, N_\uparrow, N_\downarrow)$ after the collective evolution described by $U^{(N)}$ and computing the appropriate number differences. Assuming that before the measurement sequence, the only populated states are those of the qubit $(\uparrow, \downarrow)$, one can easily show:

$$\left\langle \hat{O}^z \right\rangle = \left\langle \hat{s}^z_{\uparrow,\downarrow} \right\rangle, \left\langle \hat{O}^y \right\rangle = \left\langle \hat{s}^y_{\uparrow,\downarrow} \right\rangle, \tag{2.5}$$
$$\hat{O}^z\hat{O}^y = \hat{O}^y\hat{O}^z. \tag{2.6}$$

Note that $\hat{O}^z$ and $\hat{O}^y$ do commute with each other, which highlights the simultaneous nature of the two measurements. Thus, measuring the four populations after the outlined sequence allows us to estimate the expectation values $\langle \hat{s}^y_{\uparrow,\downarrow} \rangle$ and $\langle \hat{s}^z_{\uparrow,\downarrow} \rangle$ in a single physical realization. Furthermore, we have shown using second quantization formalism that

$$Var(\hat{O}^{\{y,z\}}) \;=\; Var(\hat{s}^{\{y,z\}}_{\uparrow,\downarrow}) + N_{at}/4, \tag{2.7}$$

provided that only the qubit levels $(\uparrow, \downarrow)$ are populated before the measurement operations described by $U^{(N)}$. It is remarkable that the scheme only introduces an additive fluctuation term $N_{at}/4$, which could then be subtracted in quadrature to estimate $Var(\hat{s}^{\{y,z\}}_{\uparrow,\downarrow})$ from the measurement of $Var(\hat{O}^{\{y,z\}})$. Note that $N_{at}/4$ corresponds to the transverse spin variance of a coherent state.

Figure 2.5b shows the measured interferograms. Here, $q/h = -330\,\text{Hz}$ and $\Omega_R/2\pi = 76\,\text{Hz}$ on the $(-5/2 \to -7/2)$ line. We control the phase ϕ by varying the Raman detuning during a fixed span of 0.51 ms just before the last Rabi pulse. Ideally, one would simultaneously measure the populations in all four states [66, 67]. In our setup, only two spin states, $m_F = -7/2$ $(\downarrow)$ and $m_F = -3/2$ (a), can be measured in the same experimental run, as it is the easiest for our spin-population measurement protocol [13]. We measure the spin state $m_F = -5/2$ $(\uparrow)$ separately, whereas, in the current implementation, the Clebsch-Gordan coefficient is too weak to efficiently measure the state $m_F = -9/2$. Since we operate in the regime where the Rabi pulses are well resolved, we can consider that $N_{-9/2} + N_{-3/2} = N_{at}/2$ and $N_{-7/2} + N_{-5/2} = N_{at}/2$ and thus relate each population to a spin observable,

$$\langle \hat{s}^z_{\uparrow,\downarrow} \rangle \;=\; \langle U^{(N)\dagger} \left(2N_{-3/2} - N_{at}/2\right) U^{(N)} \rangle, \tag{2.8}$$

$$\langle \hat{s}^\phi_{\uparrow,\downarrow} \rangle \;=\; -\langle U^{(N)\dagger} \left(2N_{-7/2} - N_{at}/2\right) U^{(N)} \rangle. \tag{2.9}$$

However, the fluctuation of these observables will be higher than those of $\hat{O}^z$ and $\hat{O}^\phi$, which are more optimal if all four states can be measured simultaneously.

The striking feature of Fig. 2.5b is that the populations in states $m_F = -7/2$ $(\downarrow)$ and $-5/2$ $(\uparrow)$ show pronounced oscillations, while the population in state $m_F = -3/2$ (a) barely evolves. The oscillations reflect that we effectively change the spin measurement basis with the interferometric phase. The approximately flat behavior of $N_{-3/2}$ demonstrates the capacity to rotate the spin in the sub-Bloch sphere set by the states $|-7/2\rangle$ and $|-5/2\rangle$ without significantly modifying the population $N_{-3/2}$. We still observe a weak dependence of $N_{-3/2}$ on the control phase ϕ. This signal is a signature of population transfers driven off-resonantly by the Raman couplings, which will be finite as long as $|\hbar\Omega_R/2q| > 0$. Interestingly, we did not see a signature of off-resonant transfers in the experiments restricted to a single pair of states. Indeed, in more complex pulse schemes, “leaked” atomic amplitudes have the opportunity to interfere with macroscopic amplitudes that are resonantly driven, which exacerbates their impact on populations.

The simple approach to tackle these parasitic population transfers is to increase the energy scale separation, $|2q/\hbar\Omega_R|$. We show in Fig. 2.5c,d how the stray signals fall with reduced Rabi couplings, using numerical simulations of the master equation. Here, the fundamental source of dissipation (*i.e.*, spontaneous emission due to a perfectly monochromatic laser) is included. The studied quantity is $\langle \hat{O}_z \rangle / N_{at}$ for a coherent state with $\langle \hat{s}^z_{\uparrow,\downarrow} \rangle = 0$. Provided the phase control is achieved in a time shorter than the Raman pulses, this quantity depends on the ratio $2|q|/\hbar\Omega_R$, irrespective of

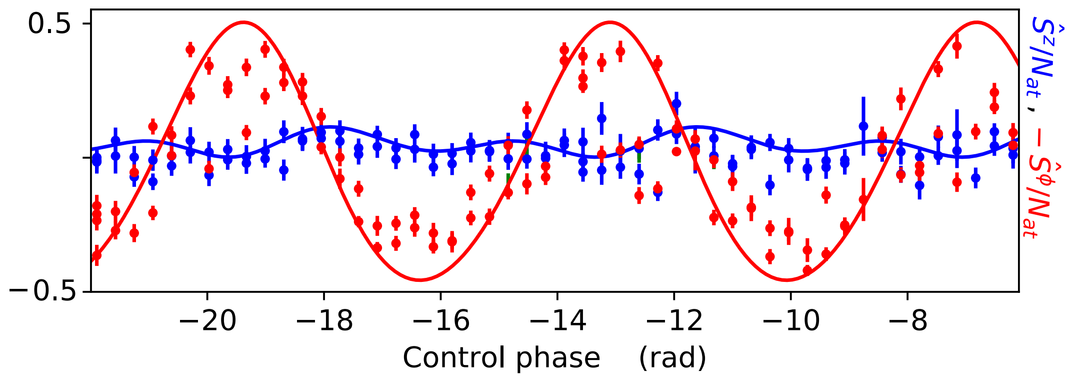


Figure 2.5: **Measuring two non-commuting observables in a single realization** a) Pulse scheme, highlighting one stage for preparing a well-defined state, and one measurement stage that includes unitary operations before the detection of populations. In frame (b), we monitor, as a function of the control phase, the fractional populations $N_{(-7/2,-5/2,-3/2)}/N_{at}$, in red, green, and blue, respectively. Error bars reflect only the population measurement uncertainty. Solid lines are the result of simulations of the master equation. We simultaneously acquire $N_{-7/2}/N_{at}$ and $N_{-3/2}/N_{at}$ during the same runs. From $N_{-3/2}$, we extract an estimate of the projection $\langle \hat{s}^z_{\uparrow,\downarrow} \rangle$ of qubit pseudo-spin, while from $N_{-7/2}$, we extract an estimate of $\langle \hat{s}^\phi_{\uparrow,\downarrow} \rangle$, all during the same runs of the experiment. We acquire the population $N_{-5/2}/N_{at}$ by independent runs. The solid lines result from a master equation simulation with a single adjustable parameter: an 18 Hz correction on the linear Zeeman splitting. (c) and (d) present the dependence of $\langle \hat{O}_z \rangle$ on the energy scale separation $2|q|/\hbar\Omega_R$ for Rabi pulses with a simple square temporal envelope, obtained from numerical simulations. We consider a coherent state with $\langle \hat{s}^z_{\uparrow,\downarrow} \rangle = 0$. We take the maximal, minimal, and mean values of $\hat{O}_z$ over a 2π variation of the control phase. We operate our experiment at $2|q|/\hbar\Omega_R \simeq 9$, providing the best balance between technically-limited decoherence and scheme fidelity.

the values of q and Ω_R. We see that the ϕ-dependence and mean value of $\langle \hat{O}_z \rangle / N_{at}$ both slowly approach the quantum-projection-noise limit ($\sqrt{1/2N_{at}} = 0.007$ for 10^4 atoms) when increasing $2|q|/\hbar\Omega_R$. However, at values $2|q|/\hbar\Omega_R > 10^2$, spontaneous emission introduces spin diffusion and degrades the scheme. Technical imperfections will bring the optimum to faster pulses, i.e., lower values of $2|q|/\hbar\Omega_R$.

To push this limitation, a most promising avenue is the temporal shaping of the Rabi pulses. By deviating from the square amplitude envelopes used here, even with simple shapes [79], one can drastically reduce off-resonant population transfers for a given Raman coupling strength, and thus at constant overall scheme duration. More dramatically, Ref. [53] demonstrates numerically that optimal control methods would enable performing arbitrary unitary $\mathfrak{su}(10)$ operations in a time $\sim 30\hbar/q$. This is about the same duration as the experiments that we present here, and relies on high inter-state couplings: $2q/\hbar\Omega_R \lesssim 1$. One could also use the ultra-narrow transition ${}^1\mathrm{S}_0 \rightarrow {}^3\mathrm{P}_2$ [80] to engineer the tensor light shift with dramatically reduced spontaneous emission.

Thus, expanding the internal Hilbert space in the measurement sequence enables the parallel estimates of several atomic observables. This follows the idea exposed in [81] for implementing positive-operator-valued measures (POVMs), and the literature on POVMs, e.g. to estimate measurement fluctuations [82], applies. Here, we have studied the case where two orthogonal pseudo-spin projections of an ensemble of qubits are measured by expanding the space to four states. This expansion comes at the cost of an increased impact from the quantum-projection noise, owing to smaller populations in each measured state and the stochastic nature of the "mapping" pulses. Given that there are ten possible spin states in strontium's ground state, one could, in principle, simultaneously measure all three spin projections of an ensemble of pseudo-spins $s = 1/2$ or even $s = 1$. Alternatively, one could simultaneously measure two projections of $s = 1/2$, $s = 1$, $s = 3/2$, or $s = 2$ pseudo-spin ensembles.

2.6 Conclusion

In this work, we have demonstrated the implementation of operations effectively driven by generators of the $\mathfrak{su}(10)$ group. We focused on selectively driving Rabi oscillations between adjacent ($m_F \longleftrightarrow m_F + 1$) and next-to-adjacent ($m_F \longleftrightarrow m_F + 2$) levels. We demonstrated these on four out of ten levels, with high fidelity > 0.99. We expect the generalization to the entire spin manifold to require moderate technical adjustments, such as a higher magnetic field bias. Being able to drive all $\Delta m_F = (1, 2)$ transitions effectively amounts to driving "rotations" around $2N - 3 = 17$ generators of the $\mathfrak{su}$(N) group. This number exceeds the $N - 1$ number of generators that can be simultaneously diagonalized (which correspond to the $N - 1$ quantization axes). Therefore concatenating $\Delta m_F = (1, 2)$ transitions is in principle sufficient to implement all the $N^2 - 1$ generators.

In practice, this set of rotations is sufficient to measure the relative amplitudes and coherences between all pairs of Zeeman levels, i.e perform tomography and thus reconstruct the complete state. Combining these schemes with a spin-selective detection capable of giving access to all ten spin components, as in Refs. [67, 66], one would gain comprehensive control of the spin state, with applications to quantum simulation, computation, or sensing schemes requiring operations within the $\mathfrak{su}(N)$ group [25, 53, 32, 33, 34]. Thanks to these tools, the spins of strontium 87 atoms are a fully exploitable qudit resource with $\mathfrak{su}(10)$ character, which one can even complement by other coherently controllable degrees of freedom in electronically-excited states, e.g., as in Ref. [83, 84].

These assertions also apply to species with similar atomic structures - alkaline-earth-like atoms and group IIB transition metals fermionic isotopes - to which the scheme could be implemented. Indeed, they have in common: a large ratio between the hyperfine splitting and the linewidth of the state ${}^3P_1$, favorable for tensor-shift engineering with low spontaneous-emission levels; a vast spin Hilbert space in the ${}^1S_0$ ground state; $\mathfrak{su}(N)$ symmetric interactions; extensive coherence times afforded by the Landé factor of nuclear spins.

We have seen that large-spin atoms open up the possibility of parallelized interferometers, which can be used for measuring multiple fields or for common-mode noise-rejection schemes. Similar sequences could readily explore other avenues, such as multiple-path interferometers and quantum random walks. More generally, being able to tailor the spin state by tensor light shifts enables the production of "non-classical" spin states, i.e., single-atom spin states that cannot be modeled by an assembly of uncorrelated spin-1/2 particles [85]. These "non-classical" spin states present specific advantages for quantum sensing [29].

Finally, following the original idea of Ref. [59], we have seen that the vast Hilbert space is an opportunity for the parallel estimation of multiple non-commuting observables of an atomic ensemble. We studied this in the case of two observables of an ensemble of pseudo-spins 1/2. However, it would be most interesting to study the parallel estimation, e.g., of three non-commuting observables or pseudo-spins $> 1/2$. It would come at the cost of more Hilbert space resources and increased quantum-projection-noise effects. The full potential of such measurements deserve investigation. The scheme is analogous to randomly partitioning the system into several sub-ensembles and measuring each along a different quantization axis. One question is whether this could be amenable to extracting signatures of specific many-body states (chiral states, cluster states [86, 87]), or to characterize general properties of the many-body states (entanglement, purity) in the spirit of, e.g., Refs. [88, 89].

2.7 Positive-operator-valued measures (POVMs)

In traditional quantum metrology, standard interferometers are bounded by Projective Valued Measurements (PVM), which restrict the device to sensing a single physical quantity per experimental run. Tracking multiple environmental parameters or reconstructing the full state usually requires sequential measurements across different configurations, heavily increasing the statistical overhead and cycle times. The high-dimensional ground state manifold of ^{87}Sr allows us to bypass these rigid constraints through parallelized architectures. Specifically, we demonstrated a parallel quantum sensor designed to simultaneously extract both the linear and the quadratic components of an external degeneracy-lifting field in a single shot. In the context of magnetic or optical field sensing, this capability is highly powerful: it enables the simultaneous measurement of a physical quantity such as a a transition (e.g optical clock transition) and the fluctuations on another quantity such as the $\vec{B}$ field, separating tracking signatures that would otherwise require separate, destructive measurement cycles.

To formally understand how a quantum system can extract such complementary pieces of information simultaneously, one look at the generalization of the measurement postulate via Positive-Operator-Valued Measures (POVM) [82, 90] of the reduced space of a two level object. Indeed, standard PVMs face a fundamental impossibility when trying to obtain complete information on a Hilbert space from a single outcome. Let us consider a system that produces either of these two non-orthogonal states, and try to extract from measruements which state is produced. For instance:

$$\left\{ |\psi_1\rangle = |0\rangle\,, \quad |\psi_2\rangle = \frac{1}{2}\,|0\rangle + \frac{\sqrt{3}}{2}\,|1\rangle \right\}$$

because $\langle\psi_1|\psi_2\rangle = 1/2 \neq 0$, a projection-based measurement in the $\{|\psi_1\rangle\,, |\psi_1^\perp\rangle\}$ basis—with $|\psi_1^\perp\rangle = |1\rangle$—yields a conditional probability of $P(\psi_1|\psi_2) = |\langle\psi_1|\psi_2\rangle|^2 = 0.25$. Unambiguous state discrimination (USD) addresses this limitation by guaranteeing error-free identification of non-orthogonal states at the expense of introducing a non-zero probability of obtaining an inconclusive result. Because of non-orthogonality, a single PVM outcome cannot deduce with certainty which state was prepared, and crucial information is completely lost.

A POVM overcomes this by defining an ensemble of positive hermitian operators $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$ ($\hat{E}_i \geq 0$, of positive eigenvalues) summing to the identity $\sum_i \hat{E}_i = \mathbb{I}$. This enforces orthogonality conditions in the same basis $\left\{|\psi_1\rangle\,, |\psi_2\rangle\,, |\psi_1^\perp\rangle\,, |\psi_2^\perp\rangle\right\}$:

$$\langle\psi_2|\,\hat{E}_1\,|\psi_2\rangle = 0 \tag{2.10}$$

$$\langle\psi_1|\,\hat{E}_2\,|\psi_1\rangle = 0 \tag{2.11}$$

with $|\psi_2^\perp\rangle = \frac{\sqrt{3}}{2}\,|0\rangle - \frac{1}{2}\,|1\rangle$. Probability conservation requires an inconclusive operator $\hat{E}_3$ such that $\mathbb{I} = \hat{E}_1 + \hat{E}_2 + \hat{E}_3$. Combined with the condition of semi-positivity of the eigenvalues, these constraints force the operators to be proportional to the dual basis projectors:

$$\hat{E}_1 = \lambda\,|\psi_2^\perp\rangle\,\langle\psi_2^\perp| \tag{2.12}$$

$$\hat{E}_2 = \lambda\,|\psi_1^\perp\rangle\,\langle\psi_1^\perp| \tag{2.13}$$

where the normalization $\lambda = 2/3$ validates the $\hat{E}_3 \geq 0$ condition. This leads to a success probability of $P(\hat{E}_1|\psi_1) = \lambda|\langle\psi_1|\psi_2^\perp\rangle|^2 = 1/2$ (equivalent for $P(\hat{E}_2|\psi_2)$), while the remaining 50% is routed to the inconclusive outcome $\hat{E}_3$ since projecting $\hat{E}_3$ onto ψ_1 or ψ_2 does not lead to a conclusive determination of the initial state.

Our second interferometric scheme, detailed in Section 2.5, constitutes a direct physical realization of this POVM framework. The protocol prepares a target qubit state $|\psi\rangle = \alpha\,|-5/2\rangle + \beta\,|-7/2\rangle$, and uses targeted coherent pulses to map a controlled fraction of these populations into two auxiliary ancillary states ($|-9/2\rangle$ and $|-3/2\rangle$).

2.7.1 Derivation of the experimental POVM framework

2.7.1.1 Four-level extended space dynamics

To model this sequence, we consider the four-level subspace $\mathcal{H}_4$ spanned by the basis states:

$$\mathcal{H}_4 = \left\{ \left|-\frac{9}{2}\right\rangle, \left|-\frac{7}{2}\right\rangle, \left|-\frac{5}{2}\right\rangle, \left|-\frac{3}{2}\right\rangle \right\}. \tag{2.14}$$

In this basis, the fundamental projective measurements (PVM) associated with state selective fluorescence detection are described by the diagonal projectors:

$$\hat{E}_1 = \left|-\frac{9}{2}\right\rangle\left\langle-\frac{9}{2}\right|, \quad \hat{E}_2 = \left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right|, \quad \hat{E}_3 = \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right|, \quad \hat{E}_4 = \left|-\frac{3}{2}\right\rangle\left\langle-\frac{3}{2}\right|. \tag{2.15}$$

The pulse sequence consists of a differential phase accumulation ϕ applied to state $|-5/2\rangle$, described by the phase operator:

$$\hat{\Phi}(\phi) = \mathrm{diag}\left(1, 1, e^{-i\phi}, 1\right), \tag{2.16}$$

followed by resonant $\pi/2$ pulse rotations acting simultaneously on the adjacent pairs $\{|-9/2\rangle, |-7/2\rangle\}$ and $\{|-5/2\rangle, |-3/2\rangle\}$. In matrix representation over $\mathcal{H}_4$, these beam-splitter-like operations take the block-diagonal form:

$$\hat{X}_{\pi/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i & 0 & 0 \\ -i & 1 & 0 & 0 \\ 0 & 0 & 1 & -i \\ 0 & 0 & -i & 1 \end{pmatrix}. \tag{2.17}$$

The full unitary transformation experienced by the system prior to detection is given by $\hat{U} = \hat{X}_{\pi/2}\hat{\Phi}(\phi)$. In the Heisenberg picture, the bare detection operators transform as $\hat{E}'_i = \hat{U}^\dagger \hat{E}_i \hat{U}$.

Carrying out the matrix multiplication for $\hat{E}'_2 = \hat{\Phi}^\dagger \hat{X}^\dagger_{\pi/2} \hat{E}_2 \hat{X}_{\pi/2} \hat{\Phi}$ explicitly yields:

$$\hat{E}'_2 = \frac{1}{2}\left[\left|-\frac{9}{2}\right\rangle\left\langle-\frac{9}{2}\right| + \left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right| + i\left|-\frac{9}{2}\right\rangle\left\langle-\frac{7}{2}\right| - i\left|-\frac{7}{2}\right\rangle\left\langle-\frac{9}{2}\right|\right]. \tag{2.18}$$

Similarly, evaluating the transformed operator for the fourth detection channel $\hat{E}'_4 = \hat{\Phi}^\dagger \hat{X}^\dagger_{\pi/2} \hat{E}_4 \hat{X}_{\pi/2} \hat{\Phi}$ accounts for the differential phase factor $e^{-i\phi}$ accumulated on the $|-5/2\rangle$ state:

$$\hat{E}'_4 = \frac{1}{2}\left[\left|-\frac{3}{2}\right\rangle\left\langle-\frac{3}{2}\right| + \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right| - ie^{i\phi}\left|-\frac{3}{2}\right\rangle\left\langle-\frac{5}{2}\right| + ie^{-i\phi}\left|-\frac{5}{2}\right\rangle\left\langle-\frac{3}{2}\right|\right]. \tag{2.19}$$

2.7.1.2 Reduction to the target qubit subspace

While the set $\{\hat{E}'_i\}$ forms a valid PVM over the full four-level space $\mathcal{H}_4$, the initial physical state $|\psi\rangle = \alpha|-5/2\rangle + \beta|-7/2\rangle$ is strictly prepared inside the two-level target qubit subspace $\mathcal{H}_{\mathrm{qubit}} = \left\{\left|-\frac{7}{2}\right\rangle, \left|-\frac{5}{2}\right\rangle\right\}$.

In the geometric state-discrimination framework described by Bergou [90], measurement probabilities correspond to state overlaps $P(i) = \langle\psi| \hat{E}_i' |\psi\rangle$. Because the prepared state has zero initial population in the auxiliary states $\left|-\frac{9}{2}\right\rangle$ and $\left|-\frac{3}{2}\right\rangle$, we have the boundary conditions:

$$\left\langle -\frac{9}{2}\middle|\psi\right\rangle = 0, \quad \left\langle -\frac{3}{2}\middle|\psi\right\rangle = 0. \tag{2.20}$$

As a consequence, any matrix elements of $\hat{E}_i'$ acting purely on $|-9/2\rangle$ or $|-3/2\rangle$ yield zero contribution when evaluated on $|\psi\rangle$. Restricting the transformed measurement operators to the active qubit manifold by projecting via $\hat{P}_{\text{qubit}} = \left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right| + \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right|$ defines the set of effective operators:

$$\hat{M}_i \equiv \hat{P}_{\text{qubit}} \hat{E}_i' \hat{P}_{\text{qubit}}. \tag{2.21}$$

Applying this projection to $\hat{E}_2'$ isolates the populated component:

$$\hat{M}_2 = \frac{1}{2}\left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right|. \tag{2.22}$$

Likewise, projecting $\hat{E}_4'$ eliminates the unpopulated state $|-3/2\rangle$ as well as the off-diagonal cross-terms involving $|-3/2\rangle$, leaving:

$$\hat{M}_4 = \frac{1}{2}\left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right|. \tag{2.23}$$

The set of restricted operators $\{\hat{M}_i\}$ on $\mathcal{H}_{\text{qubit}}$ are non-orthogonal ($\hat{M}_i^2 \neq \hat{M}_i$) and positive semi-definite ($\hat{M}_i \geq 0$), yet they fulfill the completeness relation $\sum_i \hat{M}_i = \mathbb{I}_{\text{qubit}}$. Restricting a global projective measurement on an extended Hilbert space to an active target subspace by eliminating unpopulated auxiliary channels provides a direct physical construction of a POVM.

ajouter ouverture sur mesure hétérodyne comme def d'un POVM [91] et l'application de l'article [81]

Chapter 3

Photoassociation : concept and spectroscopy of ^{87}Sr calibrated on ^{88}Sr

While the previous chapter presented the individual evolution of atoms superposed in multiple states, this third chapter focuses on the loss processes through which two atoms are photoassociated to explore the molecular formation of ^{87}Sr. Photoassociation (PA) consists of illuminating colliding atoms at a specific frequency to form a bound molecular state. With the advent of cold and ultracold atoms, PA has become a high-resolution spectroscopic tool, as the extremely low initial kinetic energy significantly reduces thermal broadening and reveals subtle structures [92].

However, the formation of ^{87}Sr molecules remains an open question. In contrast, many groups have reported molecular lines and quantum degeneracy in bosonic isotopes (^{88}Sr, ^{86}Sr, and ^{84}Sr), notably leveraging Bose-Einstein condensates (BECs) and ultracold mixtures as demonstrated by Schreck and co-workers [93] alongside other experimental works [5, 6, 7, 8]. Regarding the fermionic species, theoretical predictions for molecular potentials were established in [94], while experimental PA on the broad $^1\mathrm{S}_0 -^1 \mathrm{P}_1$ transition—which exhibits no hyperfine structure—was reported in [95]. Nevertheless, the intricate structure of the narrow intercombination line ($^1\mathrm{S}_0 -^3 \mathrm{P}_1$) has so far hindered clear detection of molecular signatures, despite its unique promise as a controllable interface for state manipulation.

Beyond fundamental spectroscopy, resolving these narrow molecular transitions is directly motivated by applications in quantum metrology. In this context, a fundamental benchmark for phase estimation with N independent particles prepared in a coherent spin state is the Standard Quantum Limit (SQL), defined by the shot-noise scaling $\Delta\phi_{\mathrm{SQL}} = 1/\sqrt{N}$ [96]. For an ensemble of particles with effective spin s where interferometry is performed on extreme projection states, this limit scales as $\Delta\phi_{\mathrm{SQL}} = 1/\sqrt{2Ns}$.

While standard protocols rely on coherent control, spin squeezing, or quantum gates to generate entangled states, the work of Foss-Feig et al. [97] inspired our team to pursue an alternative route. Rather than viewing dissipation as a decoherence mechanism to be avoided, we aim to harness the two-body reactive losses associated with these narrow molecular potentials as a dissipative engine. As a long-term prospect, such a process can drive the system toward a stationary Dicke state in a zero-projection spin ($m = J_z = 0$). In this configuration, the collective spin variance scales quadratically ($\propto N^2$) rather than linearly as for uncorrelated states ($\propto N$). Consequently, the resulting phase sensitivity bypasses the standard $1/\sqrt{N}$ shot-noise scaling of the SQL and approaches the fundamental Heisenberg limit ($\Delta\phi \propto 1/N$), offering the ultimate precision allowed by quantum mechanics [96].

To pave the way toward this goal, the work presented in this chapter focuses on an essential prerequisite: the spectroscopic identification of the molecular states involved in this spin-selective dissipative process. The main objective here is to benchmark the photoassociation mechanism itself, laying the necessary spectroscopic foundation for future state-purification protocols.

This chapter opens with the fundamental concepts of scattering theory required to describe photoassociation and atomic density dynamics via one- and two-body loss processes. Next, we present preliminary experimental results obtained with ^{88}Sr, taking advantage of its well-documented molecular potentials to calibrate and benchmark our experimental setup. Building upon these foundations, we then investigate the fermionic isotope ^{87}Sr. We present the experimental spectra and discuss the technical challenges and systematic effects that could account for the low contrast observed in the molecular lines. Finally, we conclude with an analysis of the surprisingly rich line spectrum and line shapes, which suggest yet unexplained physical mechanisms at play.

3.1 Introduction to photoassociation

3.1.1 Scattering theory

Understanding the PA process necessitates an introduction to scattering theory to define key quantities such as the scattering length and the optical length. These parameters emerge naturally from the asymptotic analysis of the radial Schrödinger equation describing the relative motion of two colliding atoms.

In the study of inter-atomic interactions, the Born-Oppenheimer approximation provides a natural simplification. Due to the large mass ratio between the nuclei and the electrons, the nuclei move on a much slower timescale. One can therefore solve the electronic problem for fixed nuclear positions. The resulting electronic energy curve as a function of the inter-nuclear distance r defines the adiabatic potential $\hat{U}(r)$ experienced by the two atoms.

Assuming a spherically symmetric potential $\hat{U}(r)$, the treatment of the two-body collision in the center-of-mass frame allows the total relative wavefunction to be factorized into radial and angular parts:

$$\psi(r, \theta, \phi) = \frac{u_l(r)}{r} Y_l^m(\theta, \phi) \tag{3.1}$$

where the spherical harmonics $Y_l^m(\theta, \phi)$ encode the angular dependence and l denotes the partial wave quantum number. By introducing the reduced mass of the two-atom system $\mu_N = m_{\rm at}/2$, the Schrödinger equation reduces to a one-dimensional radial equation for each partial wave l:

$$\left(-\frac{\hbar^2}{2\mu_N}\frac{d^2}{dr^2}+\frac{\hbar^2 l(l+1)}{2\mu_N r^2}+\hat{U}(r)\right)u_l(r)=Eu_l(r) \tag{3.2}$$

E the eigenvalue associated to the eigenfunction $u(r)$. This one-dimensional structure allows a direct analogy with the 1D quantum framework [98]. To extract the scattering properties of the system, one studies the asymptotic behavior of the solutions at large distances. Where the inter-atomic potential $\hat{V}(r)$ vanishes, the general solution for $u_l(r)$ takes the form [99]:

$$u_l(r)\xrightarrow{r\to\infty}a_l\sin\left(kr-\frac{l\pi}{2}+\delta_l\right) \tag{3.3}$$

where a_l is a normalization factor, $k=\sqrt{2\mu_N E}/\hbar$ is the collision wavenumber, and δ_l is the phase shift acquired by the l-th partial wave due to the short-range interaction potential.

For inter-atomic distances where $1/r^2$ and $V(r)$ are important [99]

$$u_l(r)\propto r^{l+1}+\frac{a}{r^l} \tag{3.4}$$

a a constant. Then continuity of the wavefunction in the full space imposes a constraint on the phase shifts which at low momenta k that finds:

$$\delta_l\propto k^{2l+1} \tag{3.5}$$

For s-waves ($l=0$), this leads to the definition of the scattering length:

$$a=-\lim_{k\to 0}\delta_0/k \tag{3.6}$$

The full wavefunction is recovered by summing over all partial waves. At large distances, this superposition can be rewritten as an incoming plane wave and an outgoing scattered spherical wave:

$$\Psi\xrightarrow{r\to\infty}e^{i\mathbf{k}\cdot\mathbf{z}}+f(\theta)\frac{e^{ikr}}{r} \tag{3.7}$$

where $f(\theta)$ is the scattering amplitude which can model both elastic and inelastic collisions and θ is the angle between the incident and scattered wavevectors. This amplitude expresses for a specific partial wave l as

$$f_l=\frac{1}{2ik}(e^{2i\delta_l}-1) \tag{3.8}$$

The assymptotical expression of Ψ and e^{ikz} is expressed in a spherical basis as a sum over all the partial waves. Then we can give as in 3.7 a decomposition of Ψ:

$$\Psi\simeq\frac{i}{2kr}\sum_{l=0}^{\infty}(2l+1)\left((-1)^l e^{-ikr}-S_l e^{ikr}\right) \tag{3.9}$$

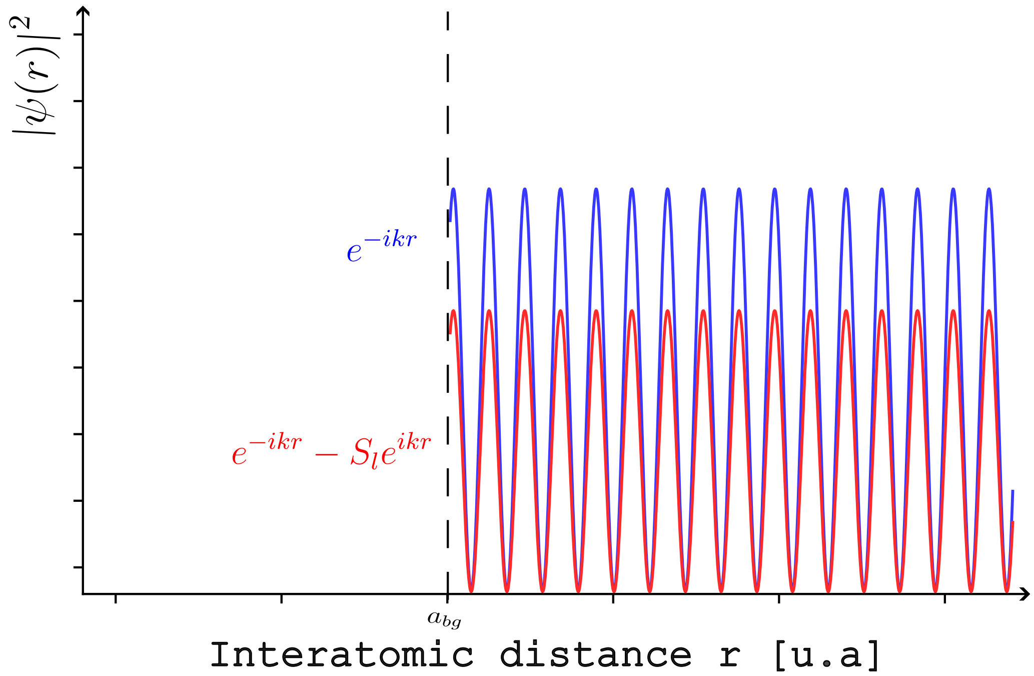


Figure 3.1: The wavefunction Ψ at large inter-atomic distances r behaves as a summation of an incoming plane wave and an outgoing spherical wave with attenuated amplitude $f(\theta)$. Here is a representation of this wavefunction for s-wave collisions at $l = 0$. The scattering occurs at background scattering length $r = a_{bg}$ the distance where the elastic collisions properties start to be modified.

where we identify a spherical part and a scattering term with $S_l = e^{2i\delta_l}$ a coefficient characterising the weight of the inelastic collisions.

In the presence of inelastic processes such as photoassociation, the colliding atom pair can be irreversibly lost from the scattering channel-either by forming a bound molecular state or by releasing kinetic energy. This loss of flux from the incoming channel cannot be captured by a purely real phase shift, as $|S_l| = 1$ implies full conservation of the wavefunction amplitude. To account for this absorption of flux, δ_l must be extended to the complex plane, allowing $|S_l| < 1$ and introducing the possibility of inelastic scattering.

The associated elastic cross section representing the surface where the scope of the potential is effective writes as

$$\sigma_{el} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)|1 - S_l|^2 \tag{3.10}$$

But we see in equation 3.9 that $|S_l|$ needs to be < 1 to allow dissipation of the wavefunction. Defining δ_l as a complex number satisfies this condition and the inelastic cross section is

$$\sigma_{inel} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(1 - |S_l|^2) \tag{3.11}$$

Then from the condition $\delta_0 << 1$ we deduce an approximated scattering amplitude S_0

$$S_0 = e^{2i\delta_0} \simeq 1 + 2i\delta_0 \tag{3.12}$$

Added to the result of 3.6,

$$1 + 2i\delta_0 \simeq 1 - 2ika \tag{3.13}$$

Because δ_0 is complex, a direct result of these equations is that we can write a general scattering length as

$$a = a_{bg} + ia'' \tag{3.14}$$

At low temperatures we can re-express equations 3.10 and 3.11 as

$$\sigma_{el} = 4\pi|a|^2 \tag{3.15}$$

$$\sigma_{inel} = 4\pi|a''|^2 \tag{3.16}$$

As an illustration a usual case of the elastic collisions is atoms in the ground state. For inelasticity photoassociation is a good example of dissipative process where two atoms in the ground state are coupled to a molecular state by a laser.

3.1.2 PA and complex scattering length

Photoassociation (PA) is a process in which a pair of colliding free atoms absorbs a photon to form a bound molecule in an excited electronic state. Effectively, because the short-lived molecule quickly decays into deeply bound states or dissociates with high kinetic energy, it escapes the optical trap. Consequently, the bound molecule is typically seen as lost, and the process is perceived as a macroscopic two-body loss effect .

In the atomic pair frame, the electric dipole interaction drives the system from the ground electronic state ψ_g of potential $U_g(r)$ to the excited molecular state ψ_e of potential $U_e(r)$. In the absence of a resonant laser, the internal evolution of the system is purely elastic and expresses as:

$$i\hbar\frac{d}{dt}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix} = \begin{pmatrix}-\frac{\hbar^2}{2\mu}\nabla^2 + U_g & 0\\ 0 & -\frac{\hbar^2}{2\mu}\nabla^2 + U_e - i\hbar\frac{\Gamma_{mol}}{2}\end{pmatrix}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix}. \tag{3.17}$$

In the frame of [100] one treat the case when a laser is tuned near a specific vibrational level of the excited potential $U_e(r)$, coupling the free-atom continuum to the bound molecular state, as illustrated in Fig. 3.2.

This coupling introduces inelastic channels into the system. The effective coupled-channel equation becomes:

$$i\hbar\frac{d}{dt}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix} = \begin{pmatrix}-\frac{\hbar^2}{2\mu}\nabla^2 + U_g & \hbar\Omega/2\\ \hbar\Omega/2 & -\frac{\hbar^2}{2\mu}\nabla^2 + U_e + \hbar\Delta - i\hbar\frac{\Gamma_{\rm mol}}{2}\end{pmatrix}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix} \tag{3.18}$$

where $\Gamma_{\rm mol}$ is the natural molecular spontaneous emission linewidth, $\Delta = \omega_e - \omega_L$ is the laser detuning from the excited bound state, and $\hbar\Omega/2$ represents the coherent optical coupling.

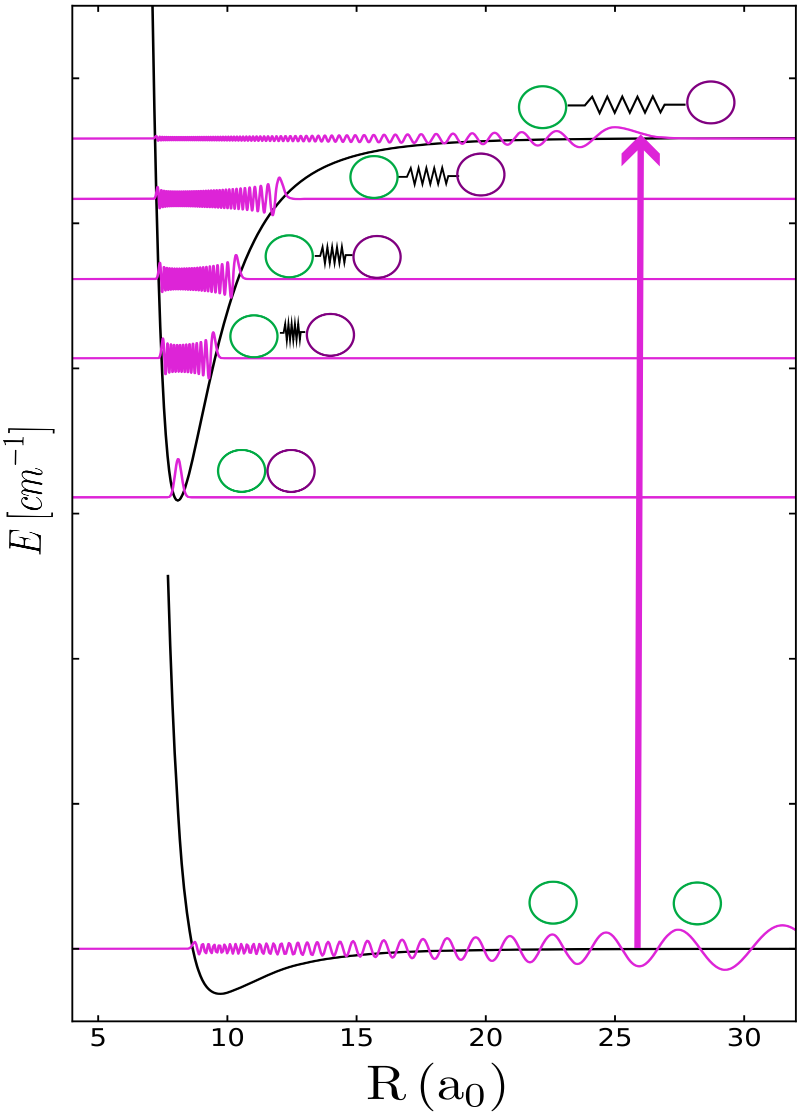


Figure 3.2: Adapted from [4]. General concept of photoassociation: a resonant laser couples a pair of colliding atoms from the free-atom continuum to a bound state of an excited molecular potential. Within this potential, the molecule occupies quantized vibrational states, described by the solutions of the 1D radial Schrödinger equation plotted for each level.

This matrix equation shows that the laser creates a bridge between the two manifolds. During a collision, two atoms approaching each other on the ground potential $U_g(r)$ have a non-zero probability to absorb a photon and transition to the excited molecular potential $U_e(r)$. Once populated, this state decays via spontaneous emission at a rate Γ_{mol}. To account for this particle loss within the quantum mechanical framework, the molecular decay appears as an imaginary term $-i\hbar\frac{\Gamma_{\text{mol}}}{2}$ in the Hamiltonian matrix. In quantum scattering theory, such an imaginary component acts as a sink that absorbs probability flux, directly translating the physical disappearance of the atoms from the elastic channel.

Since the laser is tuned far from any bare atomic resonance, the excited molecular state remains weakly populated, meaning the atoms spend most of their time interacting on the ground-state potential. To isolate the response of the excited molecular channel, we expand the closed-channel wavefunction $\psi_e(r)$ onto the bound vibrational eigenstates $|v_e\rangle$ of the excited Hamiltonian $\hat{H}_e = -\frac{\hbar^2\nabla^2}{2\mu} + U_e(r)$, which satisfy $\hat{H}_e|v_e\rangle = E_{v_e}|v_e\rangle$. Projecting the second line of Eq. 3.18 onto a specific isolated vibrational state $|v_e\rangle$ yields:

$$E\psi_e = \frac{\hbar\hat{\Omega}}{2}\psi_g + \left(\hat{H}_e + \hbar\Delta - i\hbar\frac{\Gamma_{\text{mol}}}{2}\right)\psi_e \tag{3.19}$$

$$\langle v_e|\,(E-\hat{H}_e)\psi_e = \langle v_e|\,\frac{\hbar\hat{\Omega}}{2}\,|\psi_g\rangle \tag{3.20}$$

$$\psi_e(r) \approx \frac{\left\langle v_e\middle|\frac{\hbar\hat{\Omega}(r)}{2}\middle|\psi_g\right\rangle}{E - E_{v_e} + \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}}v_e(r) \tag{3.21}$$

Substituting this spatial projection back into the ground-state channel (first line of Eq. 3.18) yields:

$$i\hbar\frac{d}{dt}\psi_g = \left(-\frac{\hbar^2\hat{\nabla}^2}{2\mu} + \hat{U}_g\right)\psi_g + \frac{\hbar\hat{\Omega}}{2}\psi_e \tag{3.22}$$

$$\left(E + \frac{\hbar^2\hat{\nabla}^2}{2\mu} - \hat{U}_g\right)\psi_g = \frac{\hbar\hat{\Omega}}{2}\psi_e \tag{3.23}$$

$$\left(E + \frac{\hbar^2\hat{\nabla}^2}{2\mu} - \hat{U}_g\right)\psi_g = \frac{\hbar\hat{\Omega}}{2}\frac{\frac{\hbar\hat{\Omega}}{2}\,|v_e\rangle\,\langle v_e|\,\frac{\hbar\hat{\Omega}}{2}}{E - E_{v_e} + \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}}\psi_g \tag{3.24}$$

$$\left(-\frac{\hbar^2\hat{\nabla}^2}{2\mu} + \hat{U}_g(r) + \hat{V}\right)\psi_g(r) = E\psi_g(r) \tag{3.25}$$

One want to isolate the term that concerns the ground state to observe the modifications of it from the coupling to the molecular state. Then by projecting 3.25 we define $\langle\hat{V}\rangle$ as:

$$\langle\psi_g|\hat{V}|\psi_g\rangle = \frac{\left|\left\langle v_e\middle|\frac{\hbar\hat{\Omega}(r)}{2}\middle|\psi_g\right\rangle\right|^2}{E - E_{v_e} + \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}} \tag{3.26}$$

Following Bohn and Julienne [101], one can define the stimulaled rate Γ_{stim} from $|v_e\rangle$ to $|\psi_g\rangle$ by the Fermi-Golden rule as

$$\Gamma_{stim} = 2\pi\left(\frac{2\pi I}{c}\right)d^2\,|\langle v_e|\,\psi_g\rangle|^2 \tag{3.27}$$

with I the laser intensity, c the light velocity and d the transition dipole moment. On equation 3.26 the $\frac{\hbar\hat{\Omega}(r)}{2}$ comes from the dipolar interaction of the atoms with the eletric field $\vec{E}$ of the laser, also expressing as $\frac{\hbar\Omega(\hat{r})}{2} = \vec{d}.\vec{E}$. One identify the numerator of 3.26 as the stimulated linewidth Γ_{stim}. This rate depends on the **Franck-Condon factor** that is the overlape between the ground state wavefunction and the excited molecular state.

Also integrating the optical potential $\hat{V}$ within the asymptotic boundary conditions for $r \to \infty$ modifies the scattering phase shift, leading to the complex scattering length $a(\Delta)$:

$$a(\Delta) = a_{\mathrm{bg}} + \frac{l_{\mathrm{opt}}\Gamma_{\mathrm{mol}}}{\Delta + i\frac{\Gamma_{\mathrm{mol}}}{2}} \tag{3.28}$$

where a_{bg} is the real background scattering length representing native elastic collisions [102]. The parameter l_{opt} is the temperature-independent optical scattering length, which encapsulates the strength of the light-induced interaction and scales strictly linearly with the laser intensity I. The light-induced stimulated linewidth $\Gamma_{\mathrm{stim}}(k)$ accounts for the laser coupling and scales linearly with the wavevector k at low temperature according to [103]:

$$\Gamma_{\mathrm{stim}}(k) = 2kl_{\mathrm{opt}}\Gamma_{\mathrm{mol}} \tag{3.29}$$

where l_{opt} is the temperature-independent **optical length**, a central parameter characterizing the strength of the light-matter coupling which scales linearly with the laser intensity I.

To distinguish how the laser independently alters the elastic interactions and the loss mechanisms, we multiply the resonant fraction in Eq. 3.28 by the complex conjugate of its denominator. This algebraic separation successfully decouples the scattering length into its real (dispersive) and imaginary (absorptive) components, $a(\Delta) = a_{\mathrm{bg}} + \delta a(\Delta) + ia''(\Delta)$:

$$a(\Delta) = a_{\mathrm{bg}} + \frac{l_{\mathrm{opt}}\Gamma_{\mathrm{mol}}\Delta}{\Delta^2 + \left(\frac{\Gamma_{\mathrm{mol}}}{2}\right)^2} - i\frac{l_{\mathrm{opt}}\Gamma_{\mathrm{mol}}\frac{\Gamma_{\mathrm{mol}}}{2}}{\Delta^2 + \left(\frac{\Gamma_{\mathrm{mol}}}{2}\right)^2} \tag{3.30}$$

The real light-induced shift $\delta a(\Delta)$ governs the modification of the elastic properties of the gas, whereas the imaginary part $a''(\Delta)$ directly quantifies the rate of inelastic two-body losses.

3.1.3 General scaling laws for PA

To connect this microscopic scattering amplitude to observable experimental quantities, one must translate these individual collision events into a population decay that writes as [104]:

$$\frac{dN}{dt} = -\beta(T)N^2 - \Gamma N \tag{3.31}$$

where $\beta(T)$ is related to the two-body loss rate K_2 and Γ is the one-body loss rate that depends experimentally mostly on the vacuum quality.

In quantum scattering theory, the fundamental two-body loss rate coefficient $K_2(k)$ is directly driven by the absorptive (imaginary) part of the scattering length a'' derived from Eq. 3.26:

$$K_2(k) = \frac{8\pi\hbar}{m_{\text{at}}} a''(k, \Delta) \tag{3.32}$$

Grounded on the systematic study of optical Feshbach resonances by Blatt et al. [105], substituting the expression of a'' into Eq. 3.32 yields the complete expression for the loss coefficient as a function of the collision wavevector k:

$$K_2(k) = \frac{8\pi\hbar}{m_{\text{at}} k} \times \frac{\Gamma_{\text{mol}} \Gamma_{\text{stim}}(k)}{\Delta^2 + \frac{1}{4}\left(\Gamma_{\text{mol}} + \Gamma_{\text{stim}}(k)\right)^2} \tag{3.33}$$

where m_{at} is the atomic mass, $\Delta = \omega_{\text{L}} - \omega_{\text{mol}}$ is the laser detuning from the molecular resonance, and Γ_{mol} is the natural molecular linewidth.

While K_2 characterizes the fundamental two-body collision process, the experimentally observed decay coefficient β depends on the spatial density distribution of the atomic cloud of width $\sigma_x, \sigma_y, \sigma_z$. The relation between K_2 and β is derived by integrating the local loss over a Gaussian density profile:

$$\frac{dN(t,T)}{dt} = -K_2 \iiint n^2(x,y,z,t,T)\, d^3r \tag{3.34}$$

$$\text{with } n(x,y,z,t,T) = n_0 \exp\left(-\frac{x^2}{2\sigma_x(T)^2} - \frac{y^2}{2\sigma_y(T)^2} - \frac{z^2}{2\sigma_z(T)^2}\right) \tag{3.35}$$

$$\text{and } n_0 = \frac{N(t,T)}{(2\pi)^{3/2}\sigma_x(T)\sigma_y(T)\sigma_z(T)} \tag{3.36}$$

$$\frac{dN(t,T)}{dt} = -K_2 \frac{N(t,T)^2}{2\sqrt{2}V_{\text{e}}(T)} \equiv -\beta(T) N(t,T)^2 \tag{3.37}$$

$$\text{where } V_{\text{e}}(T) = (2\pi)^{3/2}\sigma_x(T)\sigma_y(T)\sigma_z(T) \tag{3.38}$$

$$\text{Then } \beta(T) = \frac{K_2}{2\sqrt{2}V_{\text{e}}(T)} \tag{3.39}$$

consistent with the definitions found in [106].

For low temperatures, we neglect the contribution of the wavevector since $k \to 0$. In this threshold regime, the light-induced linewidth becomes negligible compared to the natural molecular width ($\Gamma_{\text{stim}}(k) \ll \Gamma_{\text{mol}}$). On resonance ($\Delta = 0$), substituting Eq. 3.29 into Eq. 3.33 allows the wavevector k to cancel out from the prefactor, and the microscopic PA rate coefficient simplifies to:

$$K_2 = \frac{16h}{m_{\text{at}}} l_{\text{opt}} \tag{3.40}$$

This fundamental result demonstrates that at low temperatures, the maximum photoassociation rate becomes independent of the temperature and scales purely linearly with the laser intensity I through its dependence on l_{opt}. Structurally, K_2 is maximized when the ground- and excited-state wavefunctions exhibit maximum overlap at the Condon radius. In a classical picture, this distance corresponds to the turning point where the colliding atoms spend the most time, maximizing the transition probability.

Finally, by combining Eq. 3.39 and the low-temperature limit of K_2, we obtain the total macroscopic evolution equation of the trapped population:

$$\frac{dN}{dt} = -\frac{4\sqrt{2}h}{V_{\mathrm{e}} m_{\mathrm{at}}} l_{\mathrm{opt}} N^2 - \Gamma N \tag{3.41}$$

which highlights that the global loss process is driven by the tunable optical length l_{opt} and enhanced by the peak atomic density $n_0 \propto 1/V_{\mathrm{e}}$.

3.1.4 Handling multiple molecular potentials

So far, the photoassociation (PA) process has been presented assuming a simplified picture with a single excited-state molecular potential U_e. However, the molecules created via PA exhibit a rich internal structure, making the identification of accessible excited states essential for predicting and optimizing the loss rate coefficient $\beta(T)$.

Unlike atomic spectroscopy, where selecting target states is straightforward, identifying specific molecular target states is significantly more involved due to the dense coexistence of multiple potential energy curves. Following the Born-Oppenheimer approximation, the fast electronic motion decouples from the slower nuclear motion. Each distinct molecular electronic configuration—labeled by its set of molecular quantum numbers—gives rise to an adiabatic potential energy curve $U_e(r)$. Determining these curves $U_e(r)$ is crucial, as they dictate the bound-state wavefunctions $\psi_e(r)$ and govern the overall PA dynamics.

3.1.4.1 Symmetry constraints and Pauli exclusion principle

In ultracold gases, ground-state collisions occur predominantly in the s-wave regime ($R_i = 0$), as higher partial waves ($l > 0$) are reflected by the centrifugal barrier before reaching the short-range interaction region. For fermionic ^{87}Sr atoms, Fermi-Dirac statistics impose a overall antisymmetric wavefunction $|\Psi\rangle$ under nuclear permutation.

For two identical atoms A and B with spatial states $|\beta_1\rangle$ and $|\beta_2\rangle$, the total molecular wavefunction is expressed as:

$$|\Psi\rangle = (|\beta_1\rangle_A |\beta_2\rangle_B - \rho \times p\, |\beta_1\rangle_B |\beta_2\rangle_A) \otimes |\phi\rangle \tag{3.42}$$

where $|\phi\rangle$ represents the total nuclear spin wavefunction. Since s-wave collisions ($R_i = 0$) feature a symmetric spatial wavefunction, Pauli's principle mandates that the ground-state nuclear spin wavefunction $|\phi\rangle$ must be antisymmetric. Because electric dipole ($E1$) transitions act purely on electronic and spatial degrees of freedom, the nuclear spin acts as a spectator; hence, $|\phi\rangle$ **remains antisymmetric throughout the PA process.**

Following [107], $\rho = p_1 p_2$ denotes the product of the intrinsic electronic parities of the two isolated atoms, while $p = (-1)^R$ reflects the rotational parity associated with the **molecular rotational quantum number** R**.**

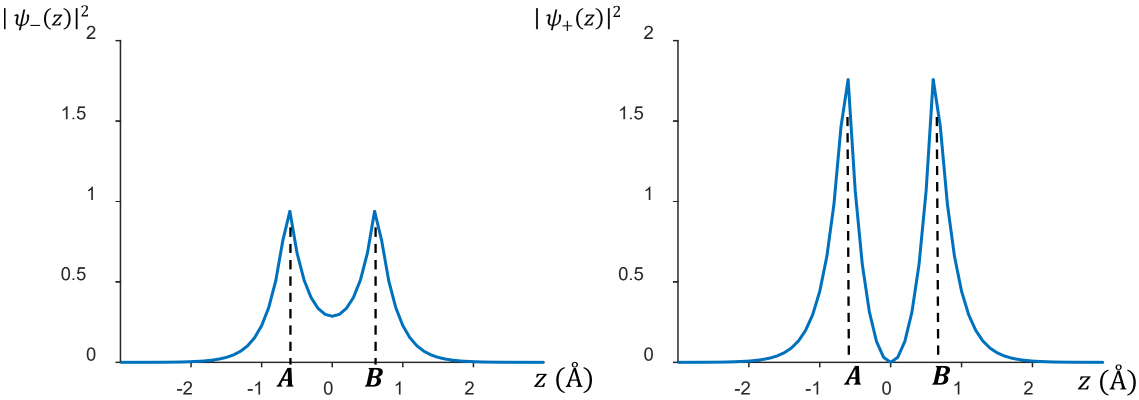


Figure 3.3: Gerade (g) and ungerade (u) spatial electronic wavefunctions for electrons orbiting nuclei A and B. Left: Square modulus $|\psi_-(z)|^2$ for an ungerade state, showing a non-zero probability density between the two nuclei. Right: For a gerade state, the electronic probability density vanishes at the midpoint between the nuclei.

3.1.4.2 Angular momentum coupling and state enumeration

Spin-orbit coupling plays a dominant role in shaping the excited molecular structure near the $^1\mathrm{S}_0 - {}^3\mathrm{P}_1$ intercombination asymptote. The total angular momentum of the molecule is defined by $T = R + F = R + I + J$, where $F = f_1 + f_2$ is the coupled atomic angular momentum, $J = j_1 + j_2$ is the total electronic angular momentum, and R is the rotational angular momentum.

Along the internuclear axis, the projection of the electronic angular momentum is quantified by:

$$\Omega = \Lambda + \Sigma \tag{3.43}$$

where Λ and Σ represent the internuclear projections of the orbital and spin angular momenta, respectively. Because spin-orbit interaction is strong relative to rotational coupling at short to intermediate ranges, the molecule is appropriately described within Hund's case (c) framework (see Fig. 3.4), where Ω remains a good quantum number.

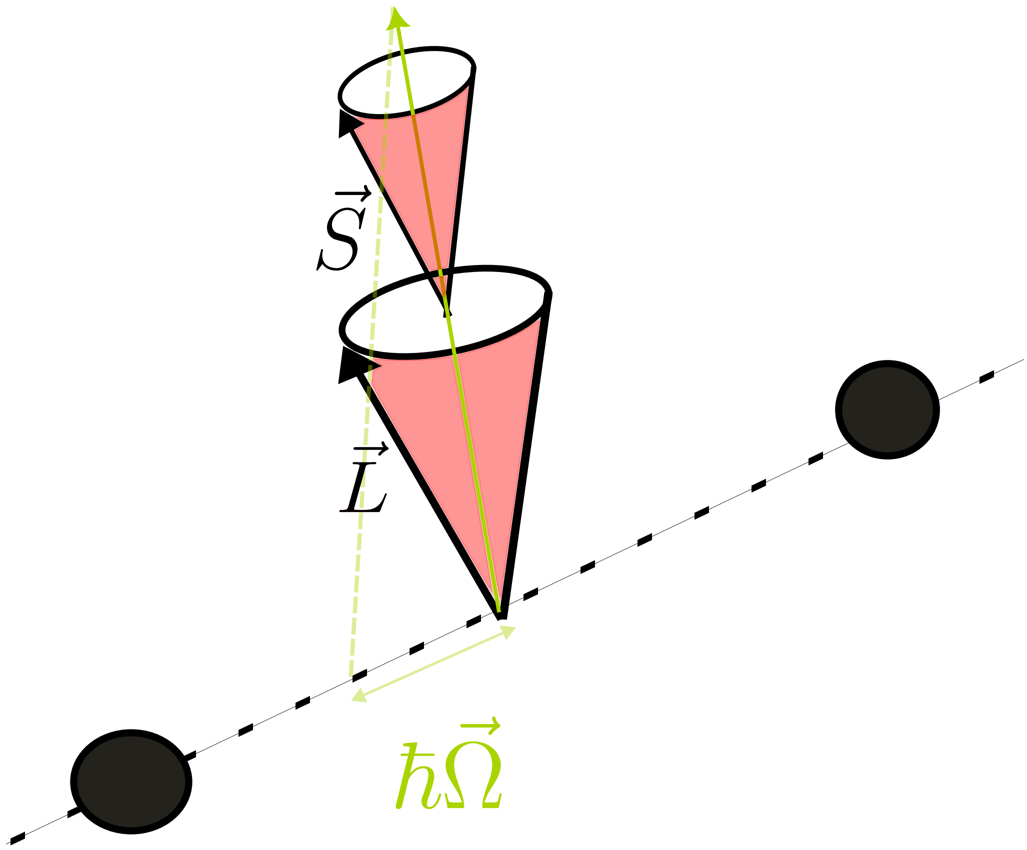


Figure 3.4: Hund's case (c) coupling scheme, where strong spin-orbit interaction couples the total electronic angular momentum to the internuclear axis with projection $\hbar\Omega$. In most ultracold photoassociation scenarios, the rotational quantum number is low ($R = 0, 1$).

To determine the accessible target molecular states, we analyze the parity requirements in both initial and final asymptotes:

- **Ground collisional state ($^1\mathbf{S}_0 + {}^1\mathbf{S}_0$):**
  Both atoms share identical, even electronic parity ($p_1 = p_2 = +1$), giving $\rho = +1$. To maintain an overall antisymmetric wavefunction with an antisymmetric spin state $|\phi\rangle$, we require $\rho \times p = +1$. This implies $p = +1$, which restricts ground s-wave collisions to even rotational numbers ($R_i = 0, 2\dots$).

- **Excited molecular state ($^1\mathbf{S}_0 + {}^3\mathbf{P}_1$):**
  The asymptotic electronic parities are opposite ($p_1 = +1, p_2 = -1$), leading to a constant electronic parity product $\rho = -1$. Because the nuclear spin state remains antisymmetric, the Pauli constraint $\rho \times p = +1$ requires $p = -1$. Consequently, the excited molecular states accessible from s-wave collisions must feature odd rotational quantum numbers ($R_f = 1, 3, 5\dots$).

Applying these selection rules to ^{87}Sr atoms photoassociated in the $F = 11/2$ manifold (with $f_1 = 9/2$ for $^1\mathrm{S}_0$ and $f_2 = 11/2$ for $^3\mathrm{P}_1$), the electronic angular momentum $J = 1$ couples with nuclear spin to yield total atomic angular momenta $F \in \{1, 2, \dots, 10\}$. Combining F with the allowed odd rotational modes ($R_f = 1, 3, 5\dots$) and taking into account electric dipole selection rules ($|\Delta T| \leq 1$

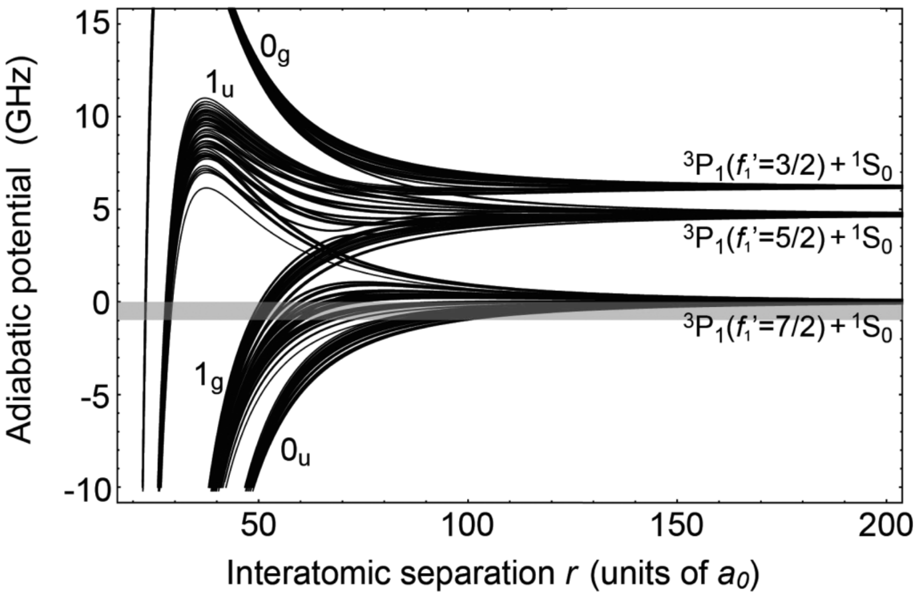


Figure 3.5: Density of molecular potentials for ^{173}Yb with three hyperfine states, giving rise to numerous triplets of quantum numbers (T, F, R). The spectral lines are identified as $\Omega_{u/g}$, where Ω is the total orbital momentum projection of the molecule and u/g denotes the electronic parity (see Fig. 3.3).

with $T = 0 \not\rightarrow T = 0$), the accessible final states T^f satisfy:

$$\text{For } R_f = 1, \quad T^f \in \{1, 2, \ldots, 10\} \tag{3.44}$$

$$\text{For } R_f = 3, \quad T^f \in \{1, 2, \ldots, 10\} \tag{3.45}$$

$$\text{For } R_f = 5, \quad T^f \in \{1, 2, \ldots, 10\} \tag{3.46}$$

3.1.4.3 Effective hamiltonian and potential modeling

Taking into account the hyperfine-induced mixing between g and u electronic manifolds $(0_u, 1_u, 0_g, 1_g)$ at long range—analogous to observations in ^{173}Yb [106]—we estimate a total of approximately 981 accessible (R_f, T^f) molecular configurations. This massive density of states illustrates the extreme complexity of the excited molecular spectrum to be mapped experimentally.

The high density of states expected for ^{87}Sr makes spectral assignment particularly challenging. To move beyond a statistical estimation and accurately predict the energy levels of these 981 configurations, precise modeling of the molecular interaction is required. This involves describing the coupling between the rotation of the nuclear frame, the electronic degrees of freedom, and the hyperfine structure.

Following the molecular picture introduced in Section 3.1.1, the energy of these states is governed by the total hamiltonian $\hat{H}$, which for a Sr-Sr system of reduced mass μ_N can be expressed as:

$$\hat{H} = \frac{p_r^2}{2\mu_N} + \frac{\hbar^2}{2\mu_N r^2} R(R+1) + U_{BO} + \hat{H}_{\text{hf}} \tag{3.47}$$

the first and second terms representing the kinetic energy and radial energy of the molecule. Higher-order electronic interactions have been neglected depending on the relevant Hund's case. For ^{87}Sr, the Born-Oppenheimer potential $\hat{U}_{\mathrm{BO}}$ is approximated using a long-range expansion similar to ^{173}Yb:

$$U_{\mathrm{BO}}(r) = -\frac{C_6}{r^6}\left(1 - \frac{\sigma^6}{r^6}\right) - s\frac{C_3^\Omega}{r^3} \tag{3.48}$$

where $s = +1$ and $s = -1$ correspond to g and u potentials, respectively. C_6 and C_3^Ω are Van der Waals and dipole-dipole coefficients [108], Ω still standing for projection of angular momentum, and σ is a free parameter to adjust with the experimental results.

The atomic hyperfine Hamiltonian $\hat{H}_{\mathrm{hf}}$ acts solely on the $^3\mathrm{P}_1$ atom (with nuclear spin $i_1 = 9/2$ and electronic angular momentum $j_1 = 1$), as the ground-state $^1\mathrm{S}_0$ atom has $j_2 = 0$ and thus no hyperfine interaction:

$$\hat{H}_{\mathrm{hf}} = A\,(\mathbf{i}_1 \cdot \mathbf{j}_1) + B\,\frac{3(\mathbf{i}_1 \cdot \mathbf{j}_1)^2 + \frac{3}{2}(\mathbf{i}_1 \cdot \mathbf{j}_1) - i_1(i_1+1)j_1(j_1+1)}{2i_1(2i_1-1)\,2j_1(2j_1-1)} \tag{3.49}$$

where A and B are the magnetic dipole and electric quadrupole hyperfine constants, respectively.

Among these potential curves, approximately 80 bound internal states were experimentally identified for ^{173}Yb [106]. Atom-loss spectra, recorded as a function of PA laser detuning, reveal the position of bound states relative to the atomic resonance $|^1\mathrm{S}_0, F = 5/2\rangle \to |^3\mathrm{P}_1, F = 7/2\rangle$. Near this threshold, atoms are illuminated for 30 ms at low intensity ($I = 37\,\mathrm{mW/cm^2}$) to avoid power broadening, resolving bound states closest to the dissociation limit. At larger red detunings, higher laser power and optimized exposure times are used to maximize signal depth.

Theoretically, these experimental loss lines are assigned by computing the eigenvalues of the effective Hamiltonian and adjusting parameters such as C_n coefficients to match observed vibrational levels [108]. Several semi-classical tools aid this assignment: the WKB approximation describes the near-threshold bound states, while LeRoy-Bernstein scaling predicts the spacing of consecutive vibrational lines near the dissociation limit without requiring detailed short-range potential data. Finally, mass scaling across strontium isotopes [8] serves as a robust benchmark to cross-check these energy assignments.

3.1.5 Internal energy states

3.1.5.1 WKB approximation

The Wentzel-Kramers-Brillouin (WKB) approximation is a semi-classical method used to estimate the vibrational eigenvalues of a diatomic system. By modeling the interatomic potential, the position of a first bound state can be determined, which subsequently enables the deduction of the following vibrational levels.

In this approach, the wavefunction is assumed to take the form of a modified plane wave of wavevector $k(x)$, $\psi(x) = \frac{C}{\sqrt{k(x)}} e^{i\int_{x_0}^{x} k(x')dx'}$ [109]. These waves are solutions to the one-dimensional Schrödinger equation for two particles with reduced mass μ and eigenenergy E:

$$-\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2} + U(x)\psi = E\psi \tag{3.50}$$

This leads to the expression of the local wavevector $k(x)$:

$$k(x) = \frac{p(x)}{\hbar} = \frac{\sqrt{2\mu(E - U(x))}}{\hbar} \tag{3.51}$$

The validity of this approximation is governed by the condition $\frac{1}{2\pi}\frac{d\lambda}{dx} \ll 1$, implying that the local de Broglie wavelength λ must vary slowly compared to the scale of the atoms' motion. This can be explicitly rewritten in terms of the potential gradient:

$$\left|\frac{\mu\hbar\frac{dU}{dx}}{[2\mu(E - U(x))]^{3/2}}\right| \ll 1 \tag{3.52}$$

This expression highlights two critical regimes where the semi-classical approach fails. First, near the classical turning points where $E \approx U(x)$, the kinetic energy vanishes and the wavelength diverges. Second, at short internuclear distances (Region I in Fig. 3.6), the steepness of the repulsive wall $|dU/dx|$ becomes too large for the atoms to follow the potential variations adiabatically. In these regions, connection formulas, typically using Airy functions, are required to match the solutions across the turning points.

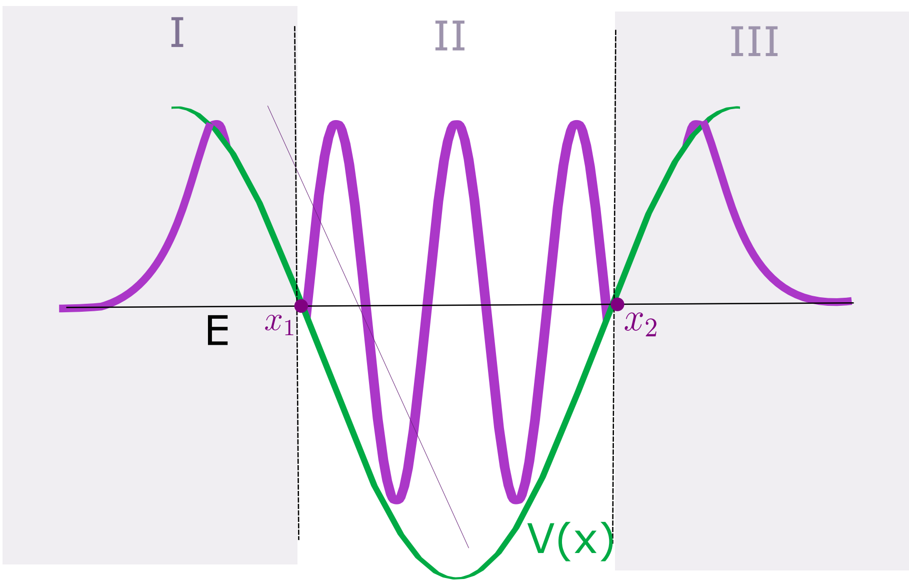


Figure 3.6: Semi-classical approximation at short range in an interatomic potential $U(x)$. The energy level E intersects the potential at the turning points x_1 and x_2. In the classically allowed Region II, the wavefunction oscillates with an amplitude proportional to $1/\sqrt{k(x)}$. In the forbidden Regions I and III, the wavefunction is evanescent.

At positions $x = x_1$ and x_2, the momentum reaches zero and becomes imaginary as the energy level E crosses the potential $U(x)$. In the classically forbidden regions

I and III, the wavefunction is evanescent and characterized by the decay constant $\kappa(x) = \sqrt{2\mu(U(x) - E)}/\hbar$. Around the turning points, it is standard to linearize the potential to connect the oscillatory and evanescent solutions. Region I is generally not studied in detail for photoassociation since the repulsive wall of the potential is extremely steep and diverges at very short distances.

To ensure a physical stationary state, the phase accumulated by the wave as it reflects between the barriers at x_1 and x_2 must satisfy a quantization condition. By ensuring the continuity of the solutions over $[0, +\infty[$, the phase integral must follow:

$$\int_{x_1}^{x_2} k(x)dx = \left(n + \frac{1}{2}\right)\pi \tag{3.53}$$

This formula, known as the Bohr-Sommerfeld quantization condition, enables the identification of the discrete energy eigenvalues E_n that satisfy the integral. these values correspond to the vibrational states of the molecular potential.

3.1.5.2 Leroy-Bernstein scaling

The LeRoy-Bernstein scaling law [110] builds upon the WKB approximation to identify the position of the high-lying vibrational states near the dissociation limit. This model is particularly powerful as it relies solely on the long-range behavior of the potential, bypassing the need for a detailed description of the complex short-range interactions.

Near the dissociation limit D_e, occurring at large internuclear distances r, the potential can be approximated by its leading long-range term:

$$U(r) \approx D_e - \frac{C_n}{r^n} \tag{3.54}$$

where C_n is the long-range coefficient (typically $n = 6$ for Van der Waals or $n = 3$ for resonant dipole-dipole interactions). For a vibrational state ν with energy E_ν, we define the binding energy as $\epsilon_\nu = D_e - E_\nu$. Substituting this into the Bohr-Sommerfeld quantization condition (Eq. 3.53), we seek solutions for:

$$\int_{r_{min}}^{r_{max}} \frac{\sqrt{2\mu}}{\hbar}\sqrt{\frac{C_n}{r^n} - \epsilon_\nu}\, dr = (\nu_D - \nu)\pi \tag{3.55}$$

where ν_D is a non-integer parameter representing the effective vibrational index at the dissociation limit. After integration, the relationship between the binding energy and the vibrational quantum number is given by:

$$\epsilon_\nu = \left[(\nu_D - \nu)\frac{\hbar n}{2\mu^{1/2}C_n^{1/n}}\frac{\sqrt{\pi}\Gamma(1/n + 1/2)}{\Gamma(1/n)}\right]^{\frac{2n}{n-2}} \tag{3.56}$$

Here, Γ denotes the Euler Gamma function. This expression is often rearranged to show the scaling of the vibrational spacing:

$$E_\nu = D_e - [(\nu_D - \nu)H_n]^{\frac{2n}{n-2}} \tag{3.57}$$

where H_n is defined as:

$$H_n = \frac{\hbar n \sqrt{\pi}}{2\mu^{1/2} C_n^{1/n}} \frac{\Gamma(1/n + 1/2)}{\Gamma(1/n)} \tag{3.58}$$

This notation highlights that the binding energy follows a power law relative to the "distance" $\nu_D - \nu$ from the dissociation limit, where the exponent is purely determined by the nature of the long-range interaction.

3.1.5.3 Mass scaling and isotopic shifts

Mass scaling is a powerful method used to predict the vibrational energy levels of different isotopes from a single known potential. By adjusting the reduced mass μ corresponding to the isotope considered, one can determine the phase constraint on the vibrational states. While this approach neglects the hyperfine structure, it provides a reliable estimation of the spectral range where vibrational resonances are expected.

In particular, ^{88}Sr is a bosonic isotope with a nuclear spin $I = 0$, meaning it possesses no hyperfine structure. Its photoassociation spectrum is significantly simpler than that of fermionic ^{173}Yb (see Fig. 3.5), involving only two accessible potentials in the first excited molecular state: 0_u and 1_u. Based on the results presented in [6], the potential can be modeled as:

$$U(r) \approx -\frac{C_6}{r^6} - \frac{C_3, \Lambda}{r^3} + \frac{\hbar^2 R(R+1)}{2\mu r^2} \tag{3.59}$$

for the two molecular symmetries $\Lambda = 0, 1$. Since the Born-Oppenheimer approximation remains valid across the strontium isotopes ($^{84}Sr,^{86}Sr,^{88}Sr$), the electronic potential $V_{BO}(r)$ is assumed to be identical for all species. Consequently, only the reduced mass μ in the WKB phase integral varies:

$$\int_{x_1}^{x_2} \frac{\sqrt{2\mu(E - U(x))}}{\hbar} dx = \left(n + \frac{1}{2}\right)\pi \tag{3.60}$$

The experimental positions of the first vibrational states for the 0_u^+ potential in $^{84}Sr,^{86}Sr$, and ^{88}Sr are shown as red crosses in Fig. 3.7. These data points allow for the refinement of the potential shape $U(r)$. The fitted functions (blue lines) then enable the prediction of the frequency ranges where ^{87}Sr resonances should occur. Nevertheless article [104] demonstrates the impossibily to determine scattering length on ^{87}Sr from mass scaling due to the hyperfine interaction of the species. Notably, discontinuities observed for the ^{84}Sr isotope correspond to avoided crossings resulting from couplings with the ground-state potential. In conclusion, combining semi-classical WKB theory with LeRoy-Bernstein scaling offers a practical framework to estimate vibrational spacings near the intercombination threshold. This approach serves as a reliable guide for bosonic isotopes like ^{88}Sr, presented in the next section. However, it cannot account for the ^{87}Sr photoassociation spectra, where instead of the dense manifold of lines, only a few narrow resonances are experimentally observed.

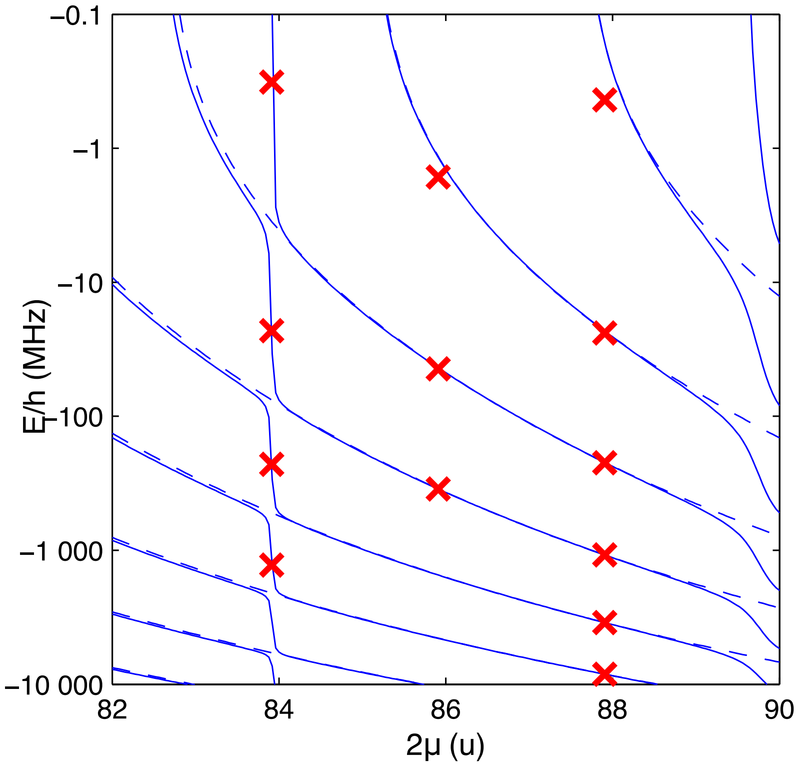


Figure 3.7: This graph - extracted from [8]- confronts the experimental internal states in red crosses to their theoretical predictions for different reduced mass of strontium isotopes. The theoretical model guarantees approximated scales for molecular lines of ^{87}Sr but not precisely because this model neglects the hyperfine interaction of the species.

3.2 Spectroscopy protocol and calibration of the PA laser

3.2.1 Gas preparation and isotope specificities

I will now present our experiments consisting in identifying photoassociation lines of ^{87}Sr yet unexplored, starting from a characterization of ^{88}Sr molecular lines abundantly published.
The photoassociation (PA) rate is strongly density-dependent, as detailed in Section 3.1.3. The experimental approach differs according to the quantum statistics of the isotope:

- **Fermions (^{87}Sr):** High densities are achieved by optimizing evaporative cooling ramps once the atoms are loaded into the crossed Optical Dipole Trap (ODT).
- **Bosons (^{88}Sr):** The near-zero scattering length of ^{88}Sr is detrimental to evaporative cooling, as it suppresses the elastic collisions necessary for thermalization. In this case, the final density relies almost exclusively on the efficiency of the preceding cooling and trapping stages (e.g., the MOT).

Once a dense cloud is obtained, the PA process is initiated by illuminating the atoms with a dedicated laser beam and all these datas are realised with a magnetic field of $B = 1.6\pm$ mazimized for π optical transitions.

3.2.2 Photoassociation lasers and their spectral characteristics

3.2.2.1 ECDL locking and frequency referencing

The PA beam is primarily generated using an ECDL laser from MOGlabs brand: the association of a cateye system to an etalon settles the wavelength of the laser within a precision of 10^{-4} nm (architecture outlined in 1.3.2). One use a beating procedure to lock this laser: as detailed in figure 3.9 the MOGLabs and "11/2 AOM" beams interfer on the same optical path intercepted by the "beating photodiode". The measured signal is then converted by the photodiode and mixed with an RF source to generate an error signal that is fed back to the MOGlabs laser current and piezo to stabilize the frequency.

The optical path of the figure 3.8 shows a design of the "11/2" beam that is shared in a first arm for the optical pumping of the atoms and in a second one for the beating procedure. In the same place the beams from repumper, MOGlabs and slave 1 are sent nearby to a wavemeter that is measuring one by one thanks to mechanical shutters.

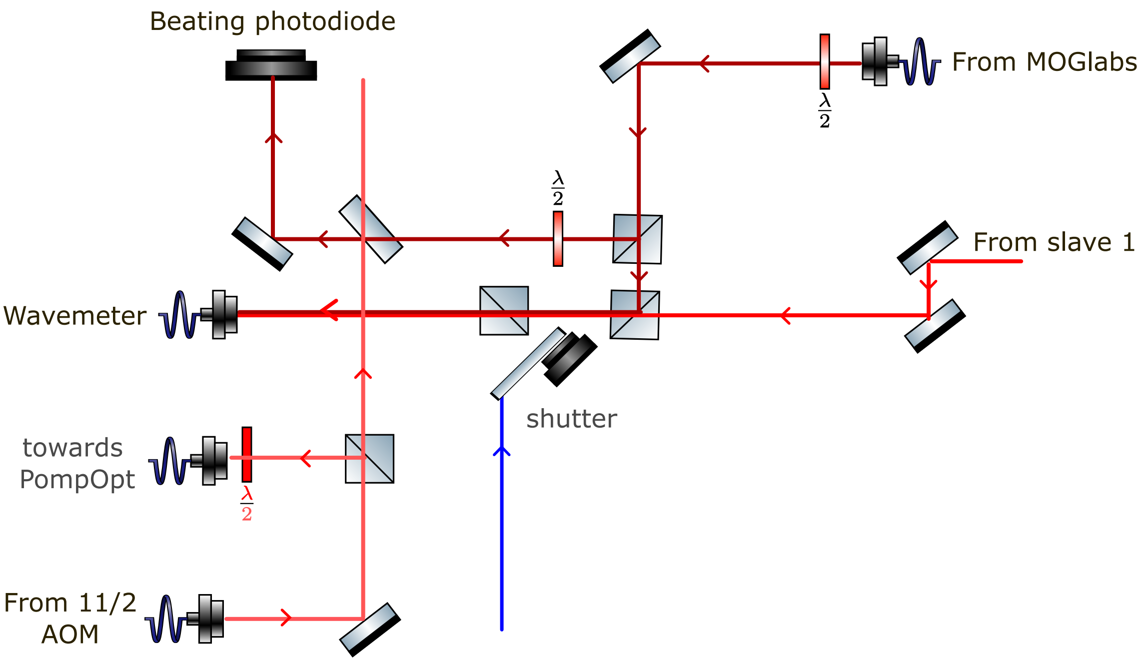


Figure 3.8: Beating scheme for PA process between the light from "AOM 11/2" and MOGlabs. The light is coming from 3.19 to split in two arms: optical pumping and PA. This last one serves for MOGlabs locking in a beating procedure where the beam is recombined by the 50/50 waveplate superposing with slave 1 beam.

The final frequency of the PA beam depends on multiple RF and optical frequencies:

- $f_{Red\ master}$ is the optical frequency of the head master ECDL laser
- $f^{RF}_{beating}$ the frequency used for demodulating the beating between 11/2 beam and MOGlabs. This RF source gives the error signal necessary for the wavelength setting and stabilization of the MOGlabs laser.
- f^{RF}_{Raman} the frequency of the "Raman AOM" used to shift the frequency beam of the PA beam in very last position of the scheme, also used for the Raman transitions used in the Ramsey interferometry schemes of Chapter 2
- f^{RF}_{slave1} is the frequency of the "Slave 1" AOM that is ahead of the lines for the ^{87}Sr MOT and the "11/2 AOM" responsible for the optical pumping and the beating. The shift of the diffracted order enables to go from the isotope 88 to the 87 (see 3.3.1)
- $f^{RF}_{11/2}$ is the frequency of the "11/2 AOM" used for optical pumping and beating.

The frequency of the beam sent to the atoms is then given by the following formula:

$$f_{PA} = f_{MOGlabs} + f^{RF}_{beating} - f^{RF}_{Raman} \pm 2 \times f^{RF}_{slave1} - 2 \times f^{RF}_{11/2} \tag{3.61}$$

The locking upper limit of the MOGlabs laser does not allow to reach frequencies above $f^{RF}_{beating} = 500$ MHz. A possibility to adress deeper bound vibrational states is to modify the different accessible frequencies -considering their RF frequency limits- and/or invert the diffracted order of the "Raman AOM" that allows to shift the PA beam to about ±260 MHz.

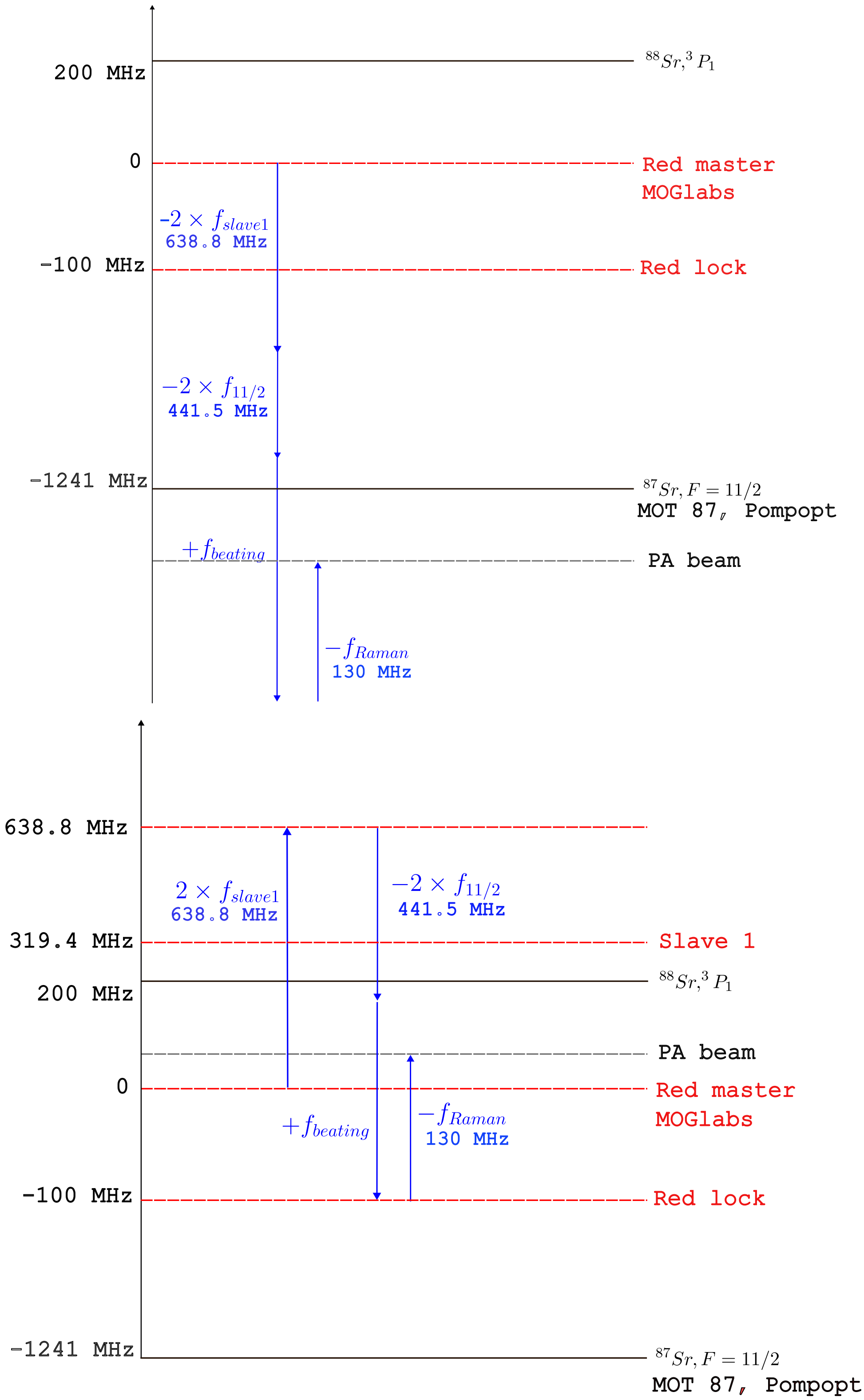


Figure 3.9: Frequency schemes for photoassociation (PA) depending on the targeted isotope: (top) scheme for ^{87}Sr and (bottom) scheme for ^{88}Sr. Switching the diffracted order of 'AOM Slave 1' allows for a seamless transition between isotopes, requiring only a minor realignment of the beam injection downstream of the AOM.

3.2.2.2 Spectral imperfections from the locking scheme

The beating spectrum of the PA laser locking is presented in figure 3.10 at a resonant bandwith RBW = 10kHz associated to a chosen video bandwith VBW = 300Hz. Expecting the molecular lifetime of ^{87}Sr to be $\Gamma_{mol} \simeq 15$ kHz this signal simulates the spectrum seen by the molecules. Side components around ±50 kHz carry nearly half the measured electrical power of the beating signal: then only half the measured optical power is adressing the particules at the beating center frequency. This observation limits the power at which one can photoassociate the atoms. While the laser width is fine (1kHz) it also means that the linewidth of the molecule or the atoms - depending on the frequency of the PA laser- is broaden by the beating spectrum with side components.

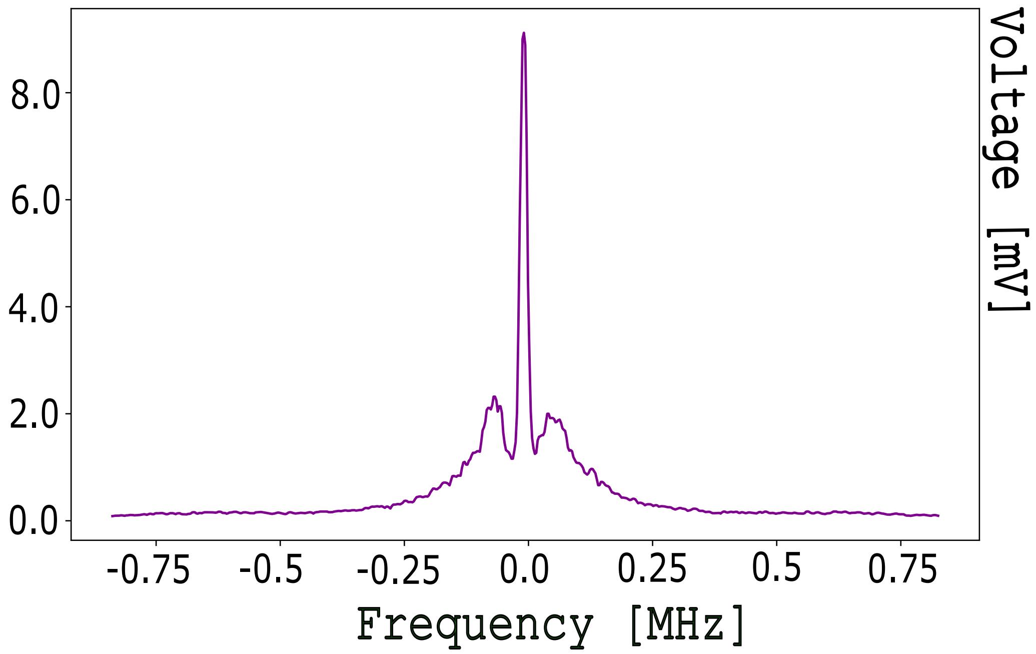


Figure 3.10: Beating spectrum between the MOGlabs laser and the "AOM 11/2" measured at the spectrum analyzer at RBW = 10 kHz and VBW = 300 Hz. The side components at around ±50 kHz induce a broadening of the natural linewidth of atoms/molecules adressed by the PA beam coming from the MOGlabs head beat locked.

3.2.2.3 Spectral imperfections from the filter cavity system

The laser passes through a Fabry-Perot cavity (see section 1.3.2) stabilized by a Pound-Drever-Hall (PDH) locking system containing an EOM. This latter modulates the incident electric field of frequency f_0 fixed at f_M = 40.48 MHz (see [3]). The phase modulation gives rise to interferences between the carrier at central frequency f_0 and the modulating signal at f_M:

$$E_{inc}^{mod} = E_0 \left(J_0(\beta) e^{if_0 t} + J_1(\beta) e^{i(f_0+f_M)t} - J_1(\beta) e^{i(f_0-f_M)t} \right) \tag{3.62}$$

The first term at amplitude $J_0(\beta)$ - the 0^{th} order Bessel function - is the incident electric field at frequency f_0 with effective amplitude $E_0 J_0(\beta)$. The second and third terms come from the interference of the carrier with the RF signal giving rise to two sidebands at $f_0 \pm f_M$ with amplitude $J_1(\beta)$ the first order Bessel function. Figure 3.11 shows the ratio between the carrier and sidebands depending on the settled β. Experimentally the modulation depth $\beta = 1$ gives around five times more intensity in the carrier relative to the sidebands which is convenient to these constraints and directly tunable by the amplitude of the RF signal.

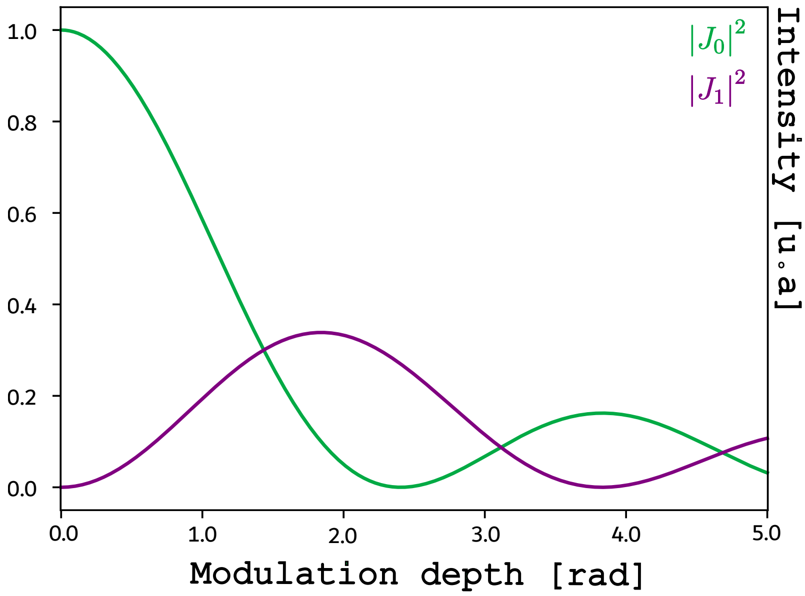


Figure 3.11: Modulus square of the 0^{th} and 1^{st} Bessel functions J_n governing the power distribution between the carrier and PDH sidebands.

We measured the effect of these sidebands on 500.000 atoms gas evaporated and recompressed to a temperature $T = 4.1\mu K$. It is tested on the two spin components $m_f = -9/2, +9/2$ optically pumped during 2 sec at optical power $I = 385\text{mW/cm}^2$ with the ECDL laser. Plot 3.13 shows the number of atoms adressed by the positive sideband: as schematized in 3.12 the presence of the sidebands triggers "ghost" resonances when the carrier is detuned by $f_0' = f_0 \pm f_M = f_0 \pm 40.48$ MHz from the atomic transition, leading to spurious one-body losses. The interfering components are also displaced by the same frequency. Then the sideband at $f_0' - f_M$ visible in the purple signal is adressing the atomic transition at f_0 with a reduced amplitude comparing to the dark signal previously centered in f_0.

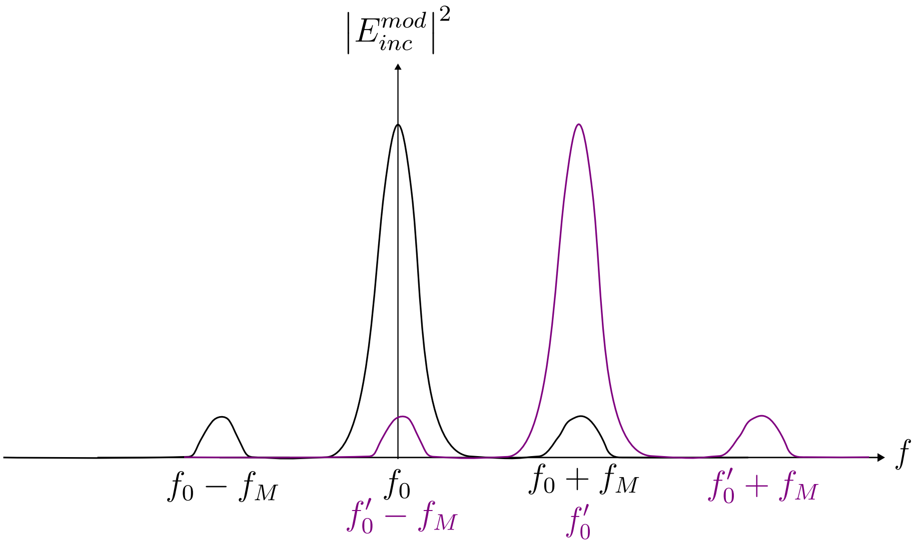


Figure 3.12: Scheme of the PDH sidebands addressing the atomic resonance. The purple plot is the dark signal -with the three interfering components between the carrier and the first harmonics- shifted by $f_0' - f_0$.

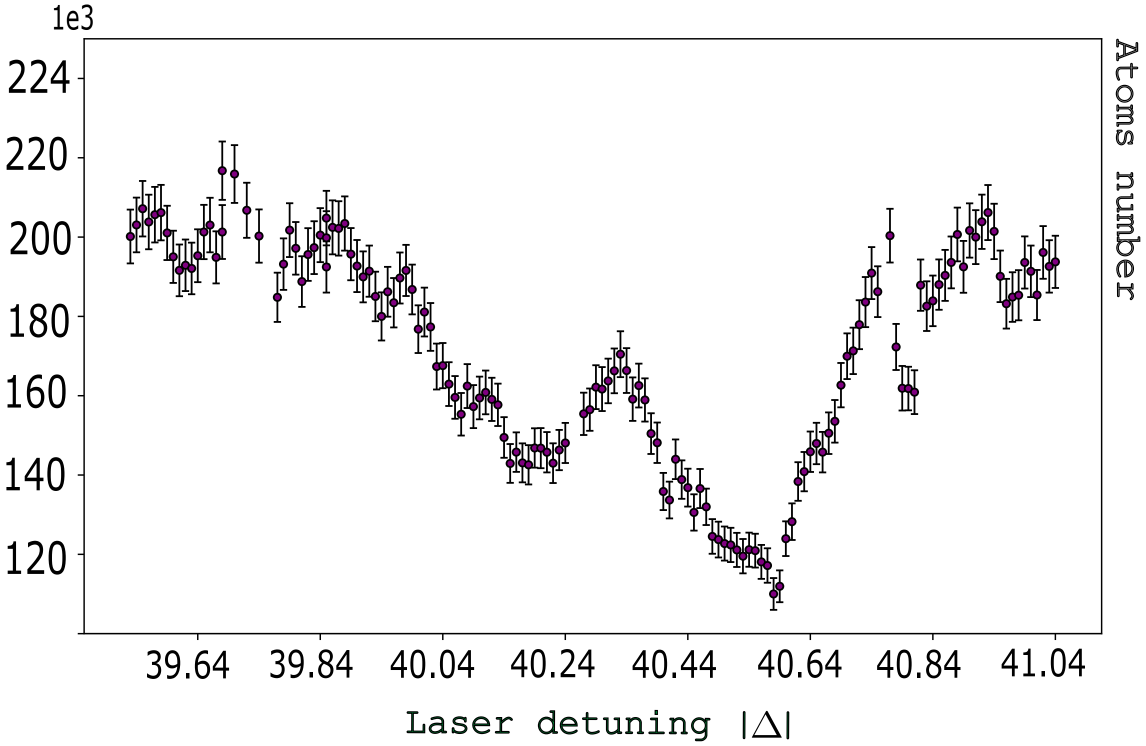


Figure 3.13: Atomic loss due to FP cavity PDH sideband that is adressing the $^1S_0 -^3P_1$ line at $\Delta = f_0 - f_M$ with f_M the modulation frequency of the PDH EOM.

The double-peak shape of this resonance comes from the recovering of the ten different m_F state that are populated by the PA laser imperfectly polarized: a small purcentage of σ polarization induces transitions to neighboring m_F states. Measurement of the same data for different magnetic field values confirmed this hypothesis since the splitting evolved with.

Settling lower modulation depth β decreased the absolute amplitude of this curve but it was not possible to completly cancel this effect: molecular line with small amplitude could be hiden in this signal.

Conclusion: Validation of the PA spectroscopic protocol

The purpose of this section was to establish and validate the operational procedure to tune the PA laser frequency. Characterization of the laser system highlighted two critical experimental insights:

- **Power calibration and spectral distribution:** Analyzing the beat-note signal revealed that approximately half of the total laser power is distributed into unwanted frequency components. This spectral dilution directly contributes to the observed broadening of both atomic and molecular spectroscopic lines.
- **Modulation-induced parasitic resonances:** Laser modulation can generate sidebands that induce unexpected atomic or molecular transitions. Specifically, sidebands at $f_M = \pm 40.48$ MHz from the main line can lead to a misinterpretation of atom loss signals. In our setup, this modulation originates from the electro-optic modulator (EOM) used for the Pound-Drever-Hall (PDH) lock, although similar effects can arise in the presence of an optical device that can be modulated by an close RF source frequency such as acousto-optic modulators (AOMs).

3.2.3 Atomic and molecular lines calibration

3.2.3.1 General experimental framework and baseline parameters

In this section the PA experiments on ^{88}Sr begins with an initial density of $n_0 = 5 \times 10^{13}$ atoms/cm^3 and a temperature of $T \simeq 1.7\ \mu$K. The atomic transition is addressed at $f^{\mathrm{RF}}_{\mathrm{beating}} = f_{\mathrm{Raman}}$ using a laser beam with an intensity of $I = 100\ I_{\mathrm{sat}}$ and a waist of $w_0 = 340\ \mu$m. The expected characteristic timescale for the illumination is defined by isolating the two-body loss term of Eq. 3.41:

$$\begin{aligned}
\frac{dn(t)}{dt} &= -\beta n(t)^2 \\
\int_{n_0}^{n(t)} \frac{dn}{n^2} &= -\beta \int_0^t dt \\
\left[-\frac{1}{n}\right]_{n_0}^{n(t)} &= -\beta t \\
\frac{1}{n(t)} - \frac{1}{n_0} &= \beta t \\
t &= \frac{1}{\beta} \frac{n_0 - n(t)}{n(t) n_0}
\end{aligned}$$

At half the initial density, the characteristic loss time τ is given by:

$$\tau = \frac{1}{\beta n_0} \tag{3.63}$$

Most of the spectroscopic lines presented below exhibit a two-body loss rate coefficient in the range of $10^{-12} < K_2 < 10^{-8}$ cm$^3 \cdot$ s^{-1}, implying that τ should typically be on the order of a few milliseconds or less. However, due to an unexpectedly low atom loss at resonance, it was subsequently determined that a spatial misalignment of the PA beam relative to the atomic cloud induced a significant spatial inhomogeneity. Consequently, all the data presented for ^{88}Sr were obtained with a prolonged illumination time of approximately 800 ms.

Furthermore at resonance an illumination time of 800 ms with saturating intensity corresponds a scattering rate $\Gamma_{at}/2$. Then the number of scattered photons is $N = \Gamma_{at}/2 \times t \simeq 10^4$ scattered photons per atom. This induces severe heating of the atomic cloud, leading to a significant thermal broadening of the molecular resonances.

The bottom panel of fig.3.9 displays the atom loss at this reference frequency. Then the laser MOGlabs robustly locked, we change $f^{RF}_{beating}$ to search the three first vibrational states visible in figure 3.14.

3.2.3.2 Fitting model

I deduced the number of atoms plotted in the following figures from the two-dimensional integration of the optical density, following the method demonstrated in Ref. [10]. Each data point of the presented atomic and molecular lines is extracted by fitting the absorption images of the atomic cloud with a 2D Gaussian function. The fit covariance yields a relative fitting uncertainty defined as:

$$\sigma_{\text{gaussian}} = \sqrt{\left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta \sigma_x}{\sigma_x}\right)^2 + \left(\frac{\delta \sigma_y}{\sigma_y}\right)^2} \tag{3.64}$$

where δA, $\delta \sigma_x$, and $\delta \sigma_y$ are the diagonal elements of the fit covariance matrix, corresponding respectively to the amplitude and the spatial widths along the x and y axes.

When sweeping the laser frequency across an atomic or molecular resonance, the evolution of the atom number exhibits a Lorentzian profile:

$$f(x) = y_0 + \frac{A}{(x - x_0)^2 + \left(\frac{\Gamma}{2}\right)^2} \tag{3.65}$$

characterized by a full-width at half-maximum (FWHM) Γ, centered at x_0, with a peak amplitude $A/(\Gamma/2)^2$ and an offset y_0.

To account for shot-to-shot experimental fluctuations, the RMS atom number variation over ten consecutive realizations was included. The total uncertainty associated with each atom number measurement is given by:

$$\sigma_N^{\text{tot}} = N\sqrt{\sigma^2_{\text{gaussian}} + \sigma^2_{\text{systematic}}} \tag{3.66}$$

The resonances exhibit an effective linewidth Γ_{tot} resulting from physical and technical broadening mechanisms:

$$\Gamma_{\text{tot}} = \sqrt{\Gamma_0^2 + \Gamma_{\text{thermal}}^2 + \Gamma_{\text{laser}}^2 + \Gamma_{\text{remainder}}^2} \tag{3.67}$$

where Γ_0 is the natural linewidth ($\Gamma_0 = 2\pi \times 7.5$ kHz for the ${}^1S_0 -{}^3 P_1$ atomic line of ^{88}Sr, and $\Gamma_0 \approx 15$ kHz on average for the molecular states). Due to small Franck-Condon factors, the effective saturation intensity $I_{\text{sat}}^{\text{eff}}$ for molecular transitions is significantly larger than for atomic ones, making power broadening completely negligible for molecular lines [6].

Furthermore, the thermal motion of Doppler-shifted atoms introduces a thermal broadening given by:

$$\Gamma_{\text{thermal}} = \frac{1}{\lambda}\sqrt{\frac{k_B T}{\mu}} \tag{3.68}$$

with $\mu = m_{\text{at}}/2$ for molecular resonances, $\mu = m_{\text{at}}$ for single atoms, λ the laser wavelength, T the gas temperature, and k_B the Boltzmann constant.

Finally, Γ_{laser} accounts for the intrinsic spectral width of the laser source, while $\Gamma_{\text{remainder}}$ gathers unassigned technical broadening sources. By subtracting the known physical contributions ($\Gamma_0, \Gamma_{\text{thermal}}$) in quadrature from the measured total linewidth Γ_{tot}, one can isolate the effective technical laser linewidth:

$$\Gamma_{\text{laser+rem}} = \sqrt{\Gamma_{\text{tot}}^2 - (\Gamma_0^2 + \Gamma_{\text{thermal}}^2)} \tag{3.69}$$

3.2.3.3 Experimental results on ^{88}Sr

The first molecular state shown in the middle panel of Fig. 3.14 was detected using an exposure time of $t = 1000$ ms and an intensity of $I = 826$ mW/cm^2. This state is located at a detuning of $f = -23.99 \pm 0.01$ MHz from the atomic transition (measured with the MOGlabs laser system), revealing a total effective linewidth of $\Gamma_{\text{tot}} = 192 \pm 9$ kHz. Accounting for the natural width $\Gamma_0 = 15$ kHz and a thermal broadening estimated at $\Gamma_{\text{thermal}} = 19$ kHz, the residual technical broadening is deduced to be $\Gamma_{\text{laser+rem}} = 190.47$ kHz$\pm$.

Subsequently, a second resonance was observed using the same laser system at a detuning of $f = -222.70 \pm 0.01$ MHz, with a beam intensity of $I = 303$ mW/cm^2 and an exposure time of 50 ms. The measured linewidth of this second resonance was $\Gamma_{\text{tot}} = 330 \pm 13$ kHz, with a thermal broadening contribution of $\Gamma_{\text{thermal}} = 21$ kHz. Subtracting these physical terms yields a technical laser width of $\Gamma_{\text{laser+rem}} = 328.99$ kHz$\pm$.

These experimental findings agree well with the results reported by Zelevinsky et al. [6], who identified vibrational states at $f_{\nu=-1} = -23.932(33)$ MHz and $f_{\nu=-2} = -222.161(35)$ MHz associated with the 0_u photoassociation potential. The tensorial light shift induced by the photoassociation (PA) beam causes a frequency shift in these molecular states, as investigated by Blatt et al. [7], where the $\nu = -2$ state exhibited a red-shift of 200 kHz for a laser intensity increase of approximately 800 mW/cm^2.

The measured linewidths ($\Gamma_{\text{tot}} \sim 192$–$330$ kHz) remain significantly larger than the expected intrinsic molecular linewidth of $\Gamma_0 \approx 15$ kHz. This substantial broadening stems from technical limitations, notably laser phase noise and sidebands (see section 3.2.2.2). To confirm that this broadening was technical, we measured the same molecular line with an alternative, higher-stability laser source.

3.2.3.4 Alternative source: the injection-locked laser

Spectroscopy measurements confirmed the significant broadening induced by the MOGlabs ECDL laser. To demonstrate its technical origin, we compared the MOGlabs results with a more stable laser source: the "11/2 AOM" beam. The measured parameters for the $\nu = -2$ molecular state are summarized in Table 3.1 and illustrated in Figure 3.15.

Table 3.1: Experimental comparison of the $\nu = -2$ molecular resonance linewidth measured with two different laser sources.

Parameters	**AOM 11/2 beam**	**MOGlabs laser**
Detuning f [MHz]	-222.70 ± 0.01	-222.70 ± 0.01
Intensity I [mW/cm^2]	[VAL_AOM]	303
Natural width Γ_0 [kHz]	15	15
Γ_{thermal} (calc.) [kHz]	21	21
Total measured width $\Gamma_{\textbf{tot}}$ [kHz]	115.5 ± 8.8	344.5 ± 27.3
Deduced technical width $\Gamma_{\textbf{laser+rem}}$ [kHz]	112.58	328.99

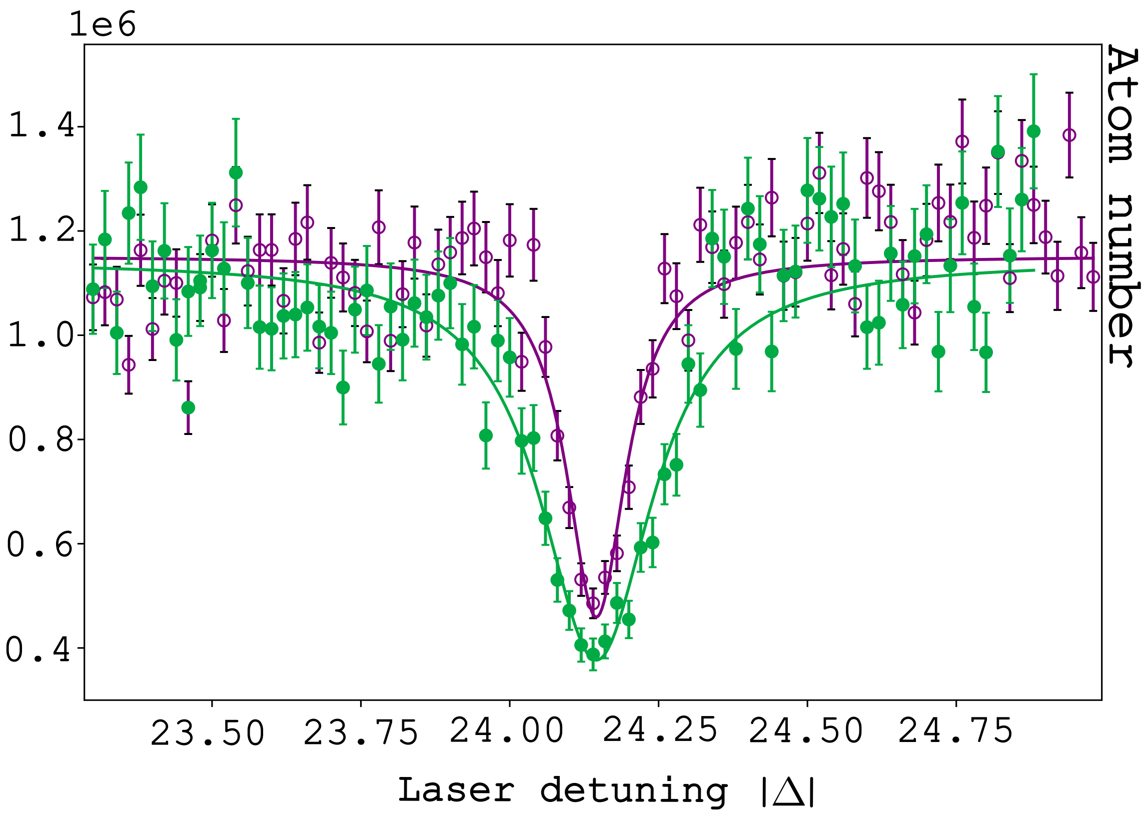


Figure 3.15: Comparison of the $\nu = -2$ molecular line profile. The MOGlabs dataset (deep purple) yields a line three times broader than the stable "11/2 AOM" source (light purple). MOGlabs frequencies were shifted for visualization clarity.

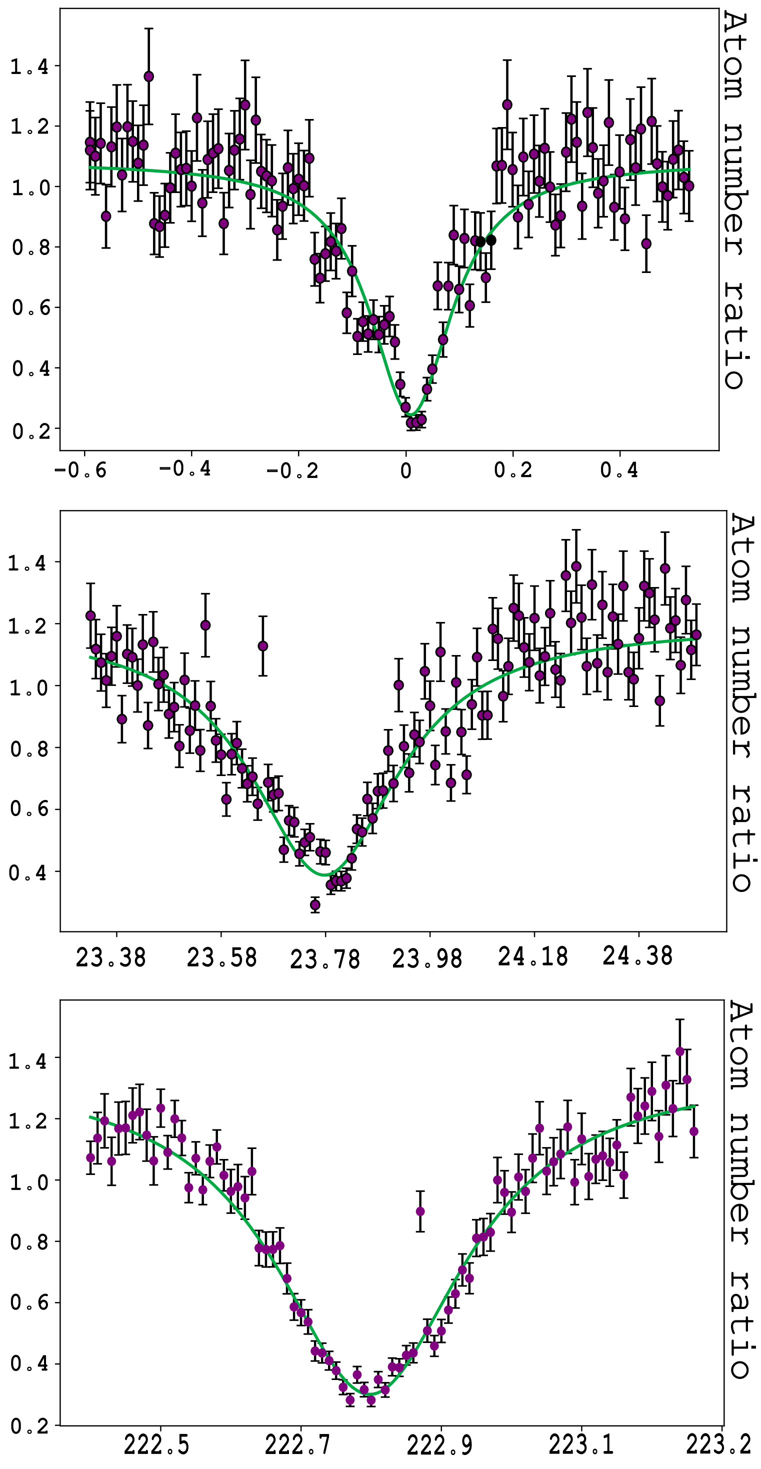


Figure 3.14: Normalized ^{88}Sr atom number across the intercombination line. (Top) Atomic resonance used as the PA frequency reference. Data points are extracted from 2D Gaussian fits of absorption images; error bars include fit covariance and unperturbed cloud RMS fluctuations. (Middle) $\nu = -1$ vibrational state (I = 826 mW/cm^2, 1000 ms). (Bottom) $\nu = -2$ vibrational state (I = 303 mW/cm^2, 50 ms). Solid lines are Lorentzian fits with errors accounting for covariance and systematic fluctuations.

The MOGlabs source yields a technical linewidth $\Gamma_{\text{laser+rem}}$ approximately three times larger than the "11/2 AOM" beam. Nevertheless, the MOGlabs laser remains essential for broad spectroscopic searches due to its wide frequency tunability.

3.2.3.5 Noise of the diode

While comparing the broadening of the molecular line with two different lasers the spectroscopy revealed from the Thorlabs red diode **HL6750MG**, noise at ±6 MHz and ±12 MHz from the central frequency f_0. This effect has been observed on the $F = 9/2$ hyperfine state of ^{87}Sr that has twelve states of total spin projection $-11/2 < m_F < 11/2$ and that we were capable of resolving as visible in 3.16. The same measurement blue detuned from the atomic resonance confirmed in fig. 3.17 the atomic nature of the losses. Starting from the atomic resonance at 0 MHz one decrease the frequency of the beating-note: centered in around -6 MHz and -12 MHz we see the presence of around twelve peaks that corresponds to the hyperfine state targeted. mean temperature of the gas, SU(10)... Looking for molecular lines around +6 MHz and +12 MHz with the 11/2 beam is then compromised by the presence of these peaks.

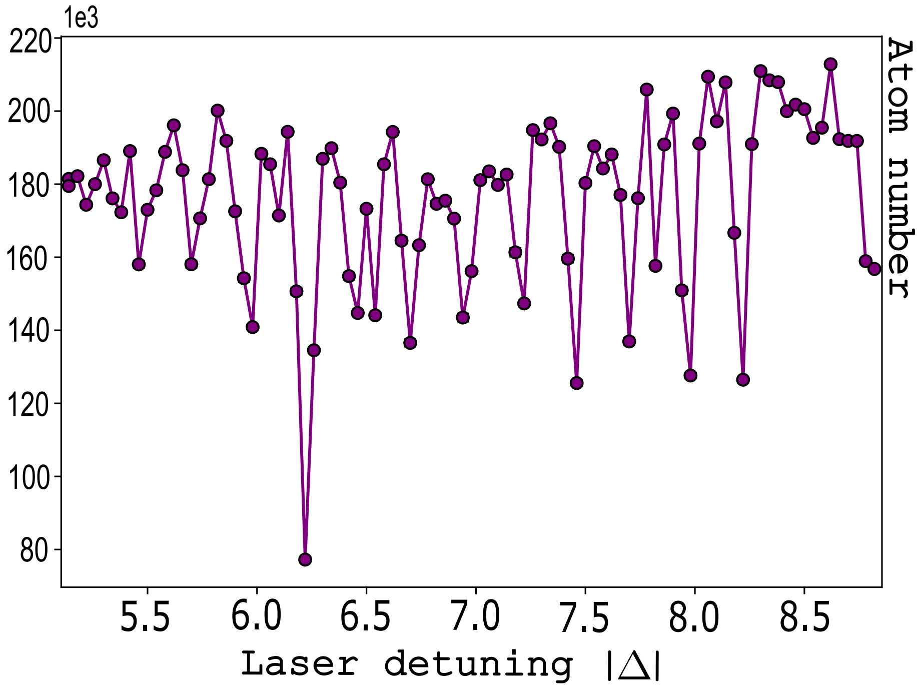


Figure 3.16: Twelve m_F states of $F = 11/2$ hyperfine state measured in an SU(10) gas at $\delta = +6\ MHz$ from the atomic resonance. The presence of these peaks comes from a noise at this frequency coming with the red diode.

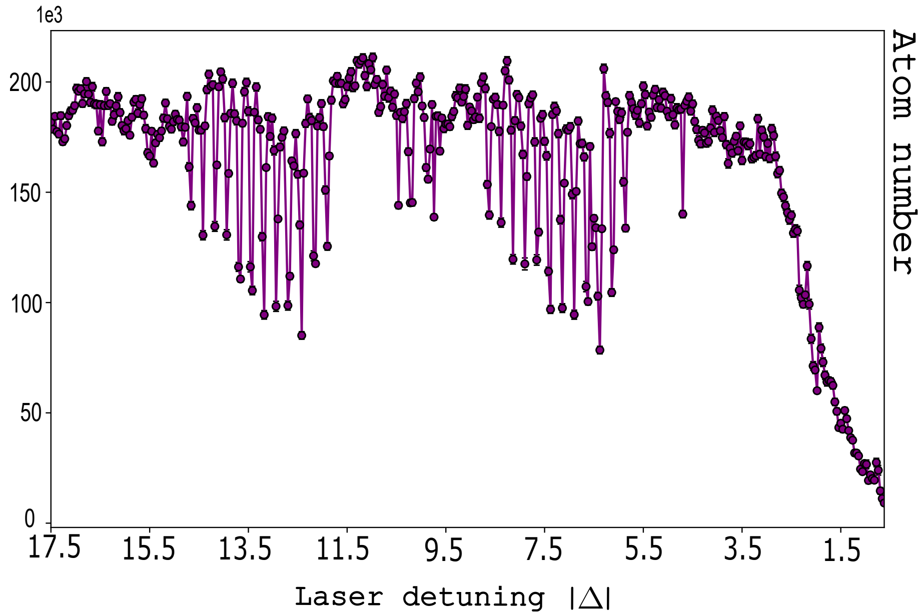


Figure 3.17: Twelve m_F states of $F = 11/2$ hyperfine state measured in an SU(10) gas at $\delta = -6\ MHz$ and $\delta = -12\ MHz$ from the atomic resonance. The presence of these peaks comes from a noise at this frequency coming with the red diode.

3.2.3.6 Decay dynamics (dN/dt)

The ultimate verification of inelastic nature of collisions is the time evolution of the atom number dN/dt discussed in Section 3.1.3. This curve allows one to (i) verify the presence of two-body losses and (ii) determine the value of l_{opt}/I which characterizes the strength of the loss rate β. The fitting function is obtained by integrating the rate equation:

$$\frac{dN(t,T)}{dt} = -\beta(T(t))N(t,T)^2 - \Gamma N(t,T) \tag{3.70}$$

$$\text{Setting } M(t,T) = N(t,T)e^{\Gamma t} \iff N(t,T) = M(t,T)e^{-\Gamma t} \tag{3.71}$$

$$\dot{M}(t,T)e^{-\Gamma t} - \Gamma M(t,T)e^{-\Gamma t} = -\beta(T(t))M(t,T)^2 e^{-2\Gamma t} - \Gamma M(t,T)e^{-\Gamma t} \tag{3.72}$$

$$\int_0^t \frac{\dot{M}(t',T)}{M(t',T)^2}\,dt' = -\int_0^t \beta(T(t'))e^{-\Gamma t'}\,dt' \tag{3.73}$$

$$-\frac{1}{M(t,T)} + \frac{1}{N_0} = -\int_0^t \beta(T(t'))e^{-\Gamma t'}\,dt' \tag{3.74}$$

$$M(t,T) = \frac{N_0}{1 + N_0\int_0^t \beta(T(t'))e^{-\Gamma t'}\,dt'} \tag{3.75}$$

$$N(t,T) = \frac{N_0 e^{-\Gamma t}}{1 + N_0\int_0^t \beta(T(t'))e^{-\Gamma t'}\,dt'} \tag{3.76}$$

Because $N(t,T)$ depends on the temperature, one wants to re-express Eq. 3.76 by explicitly showing the T-dependence of $\beta(T)$. The atoms are trapped in an ODT with a mean trapping frequency $\bar{\omega} = (\omega_x \omega_y \omega_z)^{1/3}$. According to the equipartition theorem, the spatial width σ_i (with $i = \{x, y, z\}$) of the Gaussian distribution relates to the temperature T via $\frac{1}{2} m_{\text{at}} \omega_i^2 \sigma_i^2 = \frac{1}{2} k_B T$. This yields the effective volume:

$$V_e(T) = (2\pi)^{3/2} \left(\frac{k_B T}{m_{\text{at}}} \right)^{3/2} \frac{1}{\omega_x \omega_y \omega_z} = (2\pi)^{3/2} \left(\frac{k_B T}{m_{\text{at}} \bar{\omega}^2} \right)^{3/2} \tag{3.77}$$

with $\bar{\omega} = 2\pi 50$ rad/s. By substituting $\beta(T(t')) = \frac{K_2}{2\sqrt{2} V_e(T(t'))}$ into the population evolution law, the final fitting equation becomes:

$$N(t, T) = \frac{N_0 e^{-\Gamma t}}{1 + \left[\frac{K_2}{8\pi^{3/2}} (m_{\text{at}})^{3/2} \omega_x \omega_y \omega_z \right] N_0 \int_0^t \left(\frac{1}{k_B T(t')} \right)^{3/2} e^{-\Gamma t'} \, dt'} \tag{3.78}$$

This fitting function is used to extract the two-body loss rate coefficient K_2 from the experimental data. The temperature $T(t)$ is numerically interpolated from the measured temperature. Then for each illumination time, the integrale of 3.78 is calculated and the experimental data are fitted to extract K_2.

From equation 3.70, one expects two distinct slopes on the decay curve: an initial fast two-body decay regime followed by a slow one-body regime as the atomic density drops.

Figure 3.18 illustrates these two regimes. The experimental data are adjusted using this model, from which the β value is extracted. For the resonance $f_{\nu=2} = -222.161(35)$ MHz, the deduced optical length is $l_{\text{opt}} = 13\, a_0/(\text{W/cm}^2)$, which can be compared to the value of $l_{\text{opt}} = 1\, a_0/(\text{W/cm}^2)$ reported in [6]. For the second line, we measure $l_{\text{opt}} = 7.6 \times 10^3\, a_0/(\text{W/cm}^2)$, which is consistent with the theoretical predictions from [105].

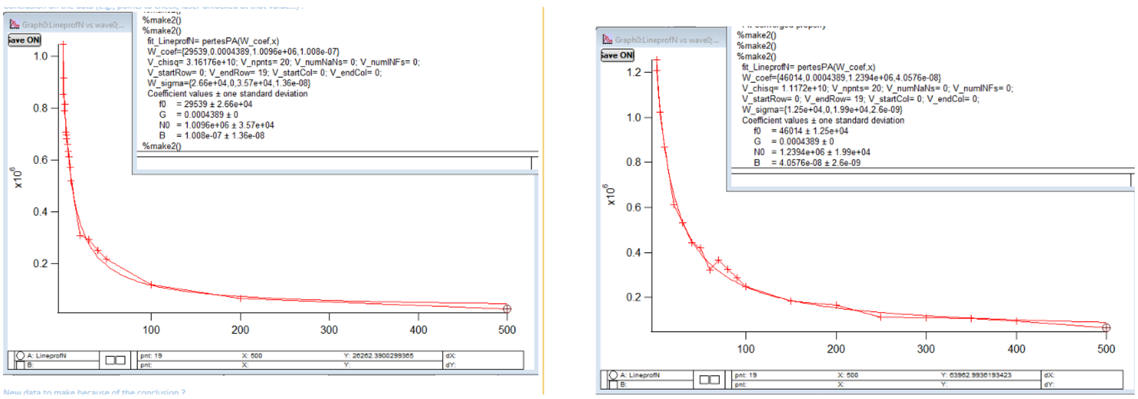


Figure 3.18: Needs to be replaced with data

The results obtained for the ^{88}Sr isotope are in good agreement with the literature. This calibration establishes a reliable baseline to search for ^{87}Sr molecular lines, provided the following experimental constraints are taken into account:

- **Parasitic RF sidebands:** Measurements performed with the 11/2 AOM beam exhibit noise components at ±6 MHz and ±12 MHz, which can drive parasitic transitions apart from the main carrier frequency.
- **Laser-induced broadening:** The MOGlabs laser broadens the molecular lines slightly more than the 11/2 beam, though this contribution remains minor.
- **Thermal and power effects:** Driving the PA transition at high intensity significantly heats the gas, resulting in substantial thermal broadening of the spectroscopic lines. While power broadening primarily affects the atomic reference line, an additional, yet unidentified, broadening mechanism persists.
- **Loss dynamics:** At a variable (or stable) temperature, the time evolution of the atom number is accurately described by Eq. 3.76, derived from the integration of the combined one- and two-body loss equation: $\frac{dN}{dt} = -\beta N^2 - \Gamma N$.

3.3 ^{87}Sr molecules

3.3.1 Results on ^{87}Sr

The molecular lines on ^{87}Sr have been reached only with the MOGlabs source. To adress the fermionic species I did change the diffracted order of the red optical chain for the opposite order to adress $F = 11/2$ hyperfine state 1.441 GHz from the bosonic species 88. (see Setup 3.19).
Datas have been obtained for different waist size in an effort to increase two-body losses, same for the density of the cloud.
For this species with hyperfine structure one expect to visualize hundreds of states as described in the theoretical part of this section each wide of about 15 kHz.

All the resonances presented in this section are done with a shot-to-shot normalisation method in order to suppress short time derivation due to experimental fluctuation. The first atomic number measured during the experiment is the measurement of the atomic density in absence of the PA laser. The second one is the measurement of the effect of the PA laser in the gas. This is repeated for the entire experiment and the final plots are the ratio of the perturbated gas on the non-perturbated one.

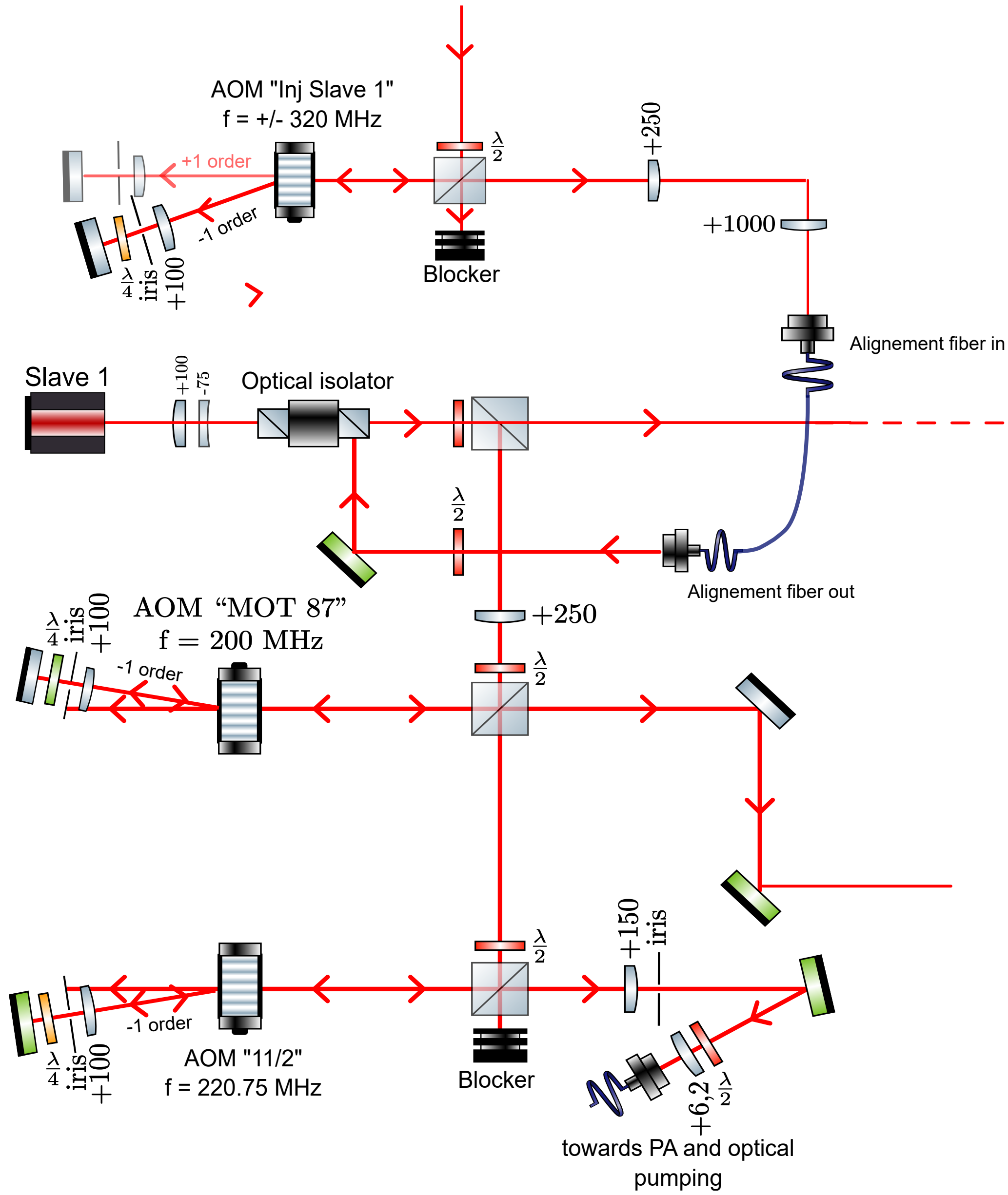


Figure 3.19: Zoom on the section of the red chain dedicated to the PA process. The frequency for both laser configurations is tuned by "AOM Inj Slave 1", adjusted by "AOM 11/2" before being sent to the atoms.

3.3.2 Verification of the molecular nature of the lines

3.3.2.1 dN/dt decay interpretation

Before presenting the molecular lines obtained for ^{87}Sr I want to discuss the reliability of the dN/dt test to identify the nature of the losses. While the two slopes are usually a signature of two-body losses I observed the same behavior for the atomic resonance obtained with the red diode noise at 6 MHz as visible in 3.20 in dark purple. This experiment has been obtained on a red detuned signal from $F = 11/2$ hyperfine state of ^{87}Sr. The atoms have been evaporated during 500 ms, recompressed and illuminated during $T = 2000$ms. In this figure the atom decay is clearly well described by equation 3.41 and presents a change of slope around 3s: the fit of this data leads to a misinterpretation of the collision nature. The same data in the absence of the PA laser plotted in the same figure shows a linear behavior on the atomic decay. As presented in 3.3.2.2 the exponential decay test is done in atomic resonance: this measurement is not reliable to identify with certainty the nature of the losses while no molecule has been enhanced on this experiment.

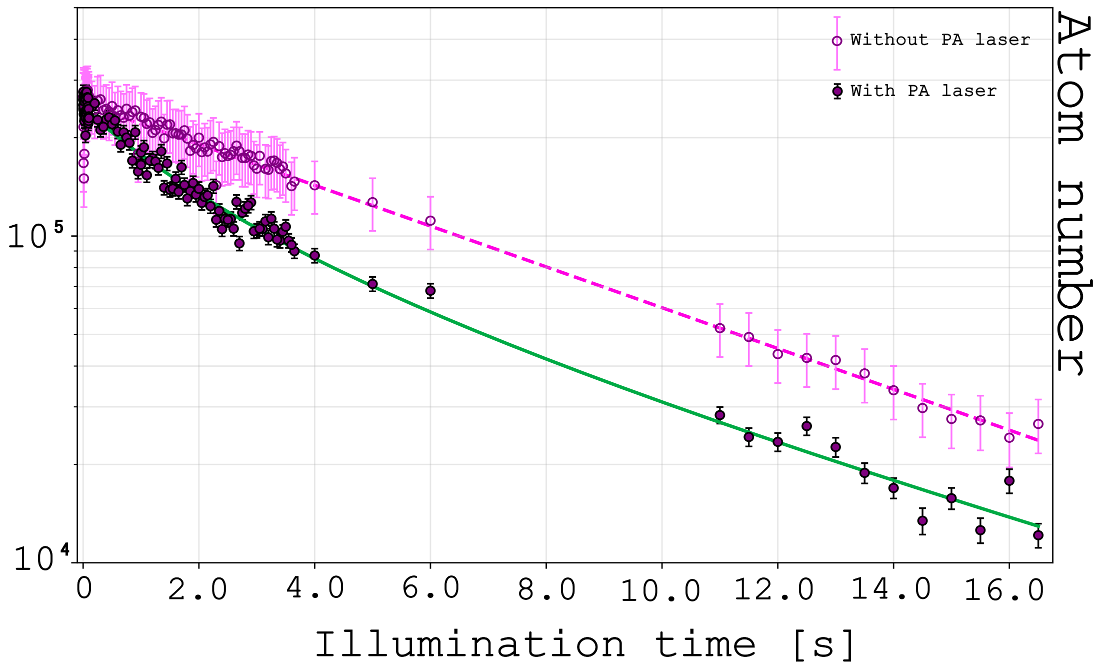


Figure 3.20: Evolution of the atom number (on a logarithmic scale) as a function of the illumination time, measured at a fixed laser detuning of $f = -6$ MHz from the atomic resonance (see Section 3.2.3.5). Filled circles represent the data in the presence of the photoassociation (PA) laser, adjusted with the fit function (Eq. 3.76) accounting for combined one- and two-body losses. Open circles display the reference measurement at the same frequency in the absence of the PA laser. The distinct decay behaviors observed between the two datasets highlight that relying solely on a single dN/dt decay measurement is insufficient to unambiguously characterize the loss mechanisms.

It appears that for non-recompressed gas it is possible to still have thermalization when the evaporation is not well optimized and that affects the two-body losses rate. For recompressed gas as in 3.20 collisions keep occuring for atoms contained in a smaller volume which explains the slope break on the plot.

3.3.2.2 Temperature as an ultimate proof

A last trustworthy test is the measurement of the temperature of the gas: when PA occurs we expect the gas to be hotter. The coldest atoms at the bottom of the trap are most likely to photoassociate. Then the resting atoms redistribute by elastic collisions leading to a hotter gas with a mean temperature per atom increased.

Figure 3.21 is a representation of the temperature of the gas during the same experiment as 3.16. This plot shows a linear increase of the temperature coming from the direct increase of illumination time: there is no exponential behavior of this curve as expected.

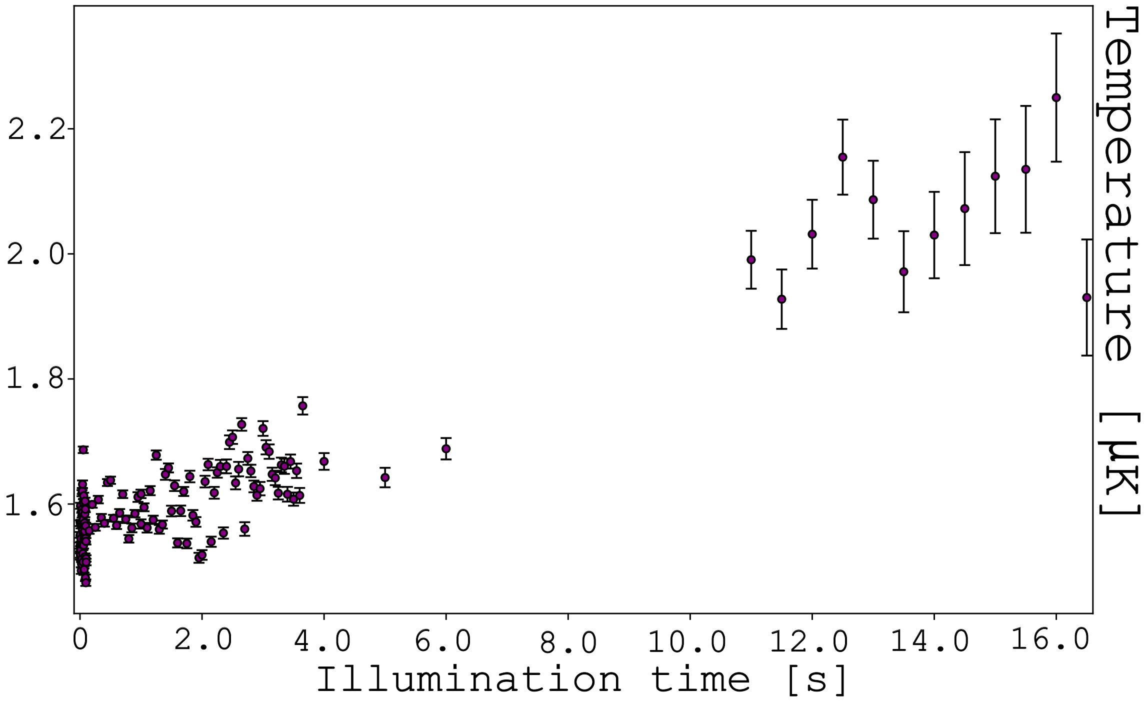


Figure 3.21: Temperature of the gas during the experiment described in 3.16 where we observe a number of atoms decaying with a two-body loss-like behavior while it is not the case. The temperature of the gas does not show any increase which is a proof of the elastic collisions only and the absence of molecule formation.

The cloud temperature is derived assuming an isotropic thermal velocity distribution. During time-of-flight (TOF), the cloud undergoes isotropic ballistic expansion, where the Gaussian width σ_y measured along a given axis evolves as:

$$\sigma_y^2 = \sigma_{0,y}^2 + v_{\mathrm{th}}^2\,\mathrm{TOF}^2 \tag{3.79}$$

where $\sigma_{0,y}$ is the *in-situ* RMS width of the gas in the optical dipole trap (ODT) intersection, and v_{th} is the thermal velocity.

Assuming a Maxwell-Boltzmann velocity distribution, the equipartition theorem links the thermal velocity to the isotropic temperature T along each degree of freedom via $\frac{1}{2}m_{\mathrm{at}}v_{\mathrm{th}}^2 = \frac{1}{2}k_B T$. The temperature is thus extracted from the Gaussian spatial width σ_y measured on the absorption image:

$$T = \frac{m_{\mathrm{at}}\left(\sigma_y^2 - \sigma_{0,y}^2\right)}{k_B\,\mathrm{TOF}^2} \tag{3.80}$$

Assuming independent uncertainties on both the *in-situ* size $\sigma_{0,y}$ and the expanded size σ_y, error propagation in quadrature yields:

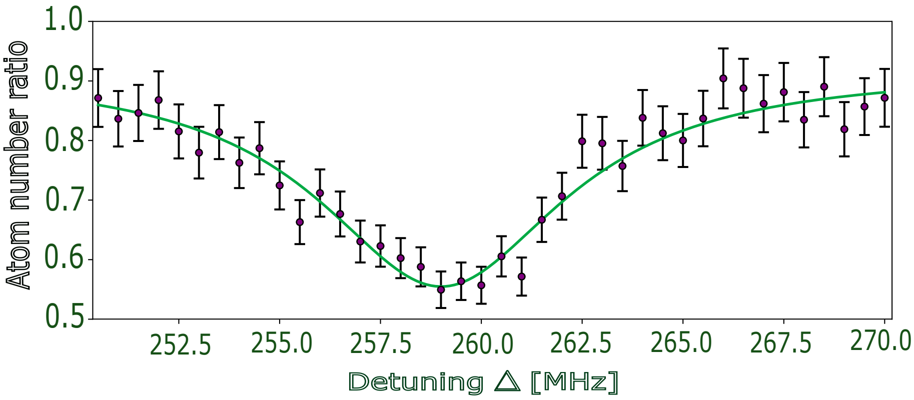
$$\delta T = \frac{2m_{\mathrm{at}}}{k_B\,\mathrm{TOF}^2}\sqrt{\left(\sigma_y\delta\sigma_y\right)^2 + \left(\sigma_{0,y}\delta\sigma_{0,y}\right)^2} \tag{3.81}$$

3.3.2.3 Molecular lines on ^{87}Sr

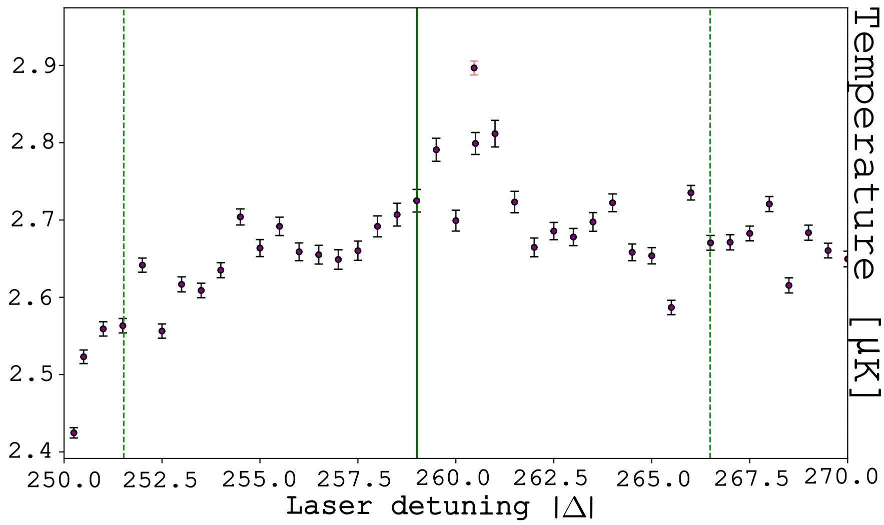
Line at -259.01 **MHz** I did find the closest molecular state red detuned from the atomic resonance at $f_1 = -259.01 \pm 0.12$: it has been obtained for illumination time of t = 1000 ms on a recompressed gas at intensity $I = 19$ W/cm^2. The lorentzian shape of the data 3.22a (see 3.2.3.2) reveals a surprising measured linewidth of $\Gamma = 2.53 \pm 0.17$ MHz with a maximized contrast of half of the atom number at resonance.

Fig.3.22c is a measurement of the temperature of the cloud from the same experiment that should validate the molecular nature of the collisions. While one expect an increase of the temperature at the center frequency of the molecular line the temperature appears to increase for about 0.5 μK symmetrically with the resonance, where the width of the lorentzian fit of the resonance is marked by the green dashed lines.

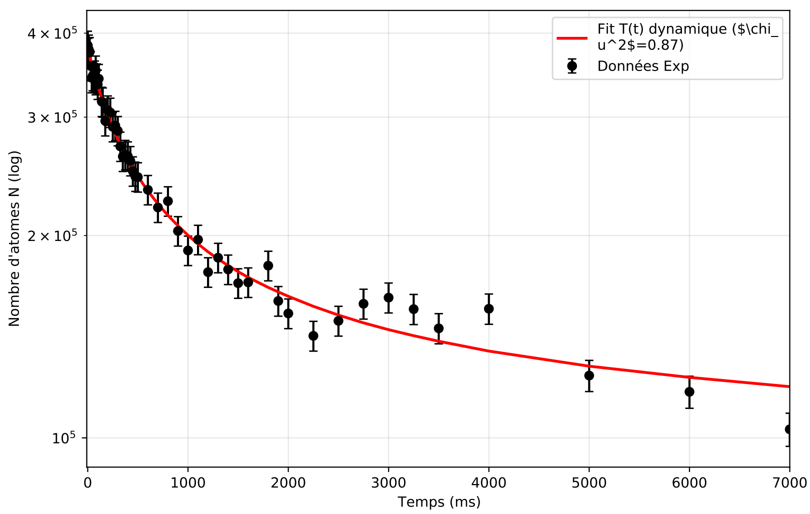
The analysis of the resonance is completed by the exponential decay observation in 3.23c: the data is fitted by the function demonstrated in 3.76 with an interpolation of the temperature during the illumination time with a unique value of $\beta(T)$ for each point. Figure 3.23b shows a lot of fluctuations on the temperature that is directly impacting the shape of the atomic number decay over the time.



(a) Resonance spectrum at 259.01 ± 0.12 MHz ($I = 19$ W/cm^2 for t = 1000 ms).



(b) Temperature of the gas at the 259.01 MHz resonance.



(c) at the 259.01 MHz resonance.

Figure 3.22: Characterization of the molecular line at 259.01 ± 0.12 MHz in ^{87}Sr ($I = 19$ W/cm^2). All detunings are measured from the ^{87}Sr atomic resonance.

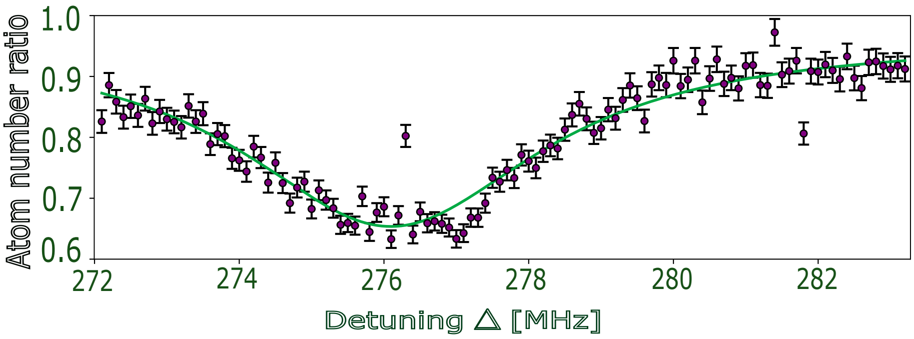
Line at 276.83 **MHz** The second line detected is at $f_2 = -276.83 \pm 0.03$ MHz from the atomic resonance plotted in 3.23a. The gas prepared of density 2×10^{13} atoms/cm^3 was evaporated in the crossing and illuminated for 1000 ms. in reservoir and dimple Once again the linewidth measured $\Gamma = 7.48 \pm 0.77$ is surprisingly large. The points out of the fit could come from a delocking laser or simply a mis-fitting of the gaussian shape of the gas.

The measurement of the temperature 3.23b is almost maximum at the center of the resonance but shows an unexplained oscilation just after of about 0.1 μK around 1.52 μK.

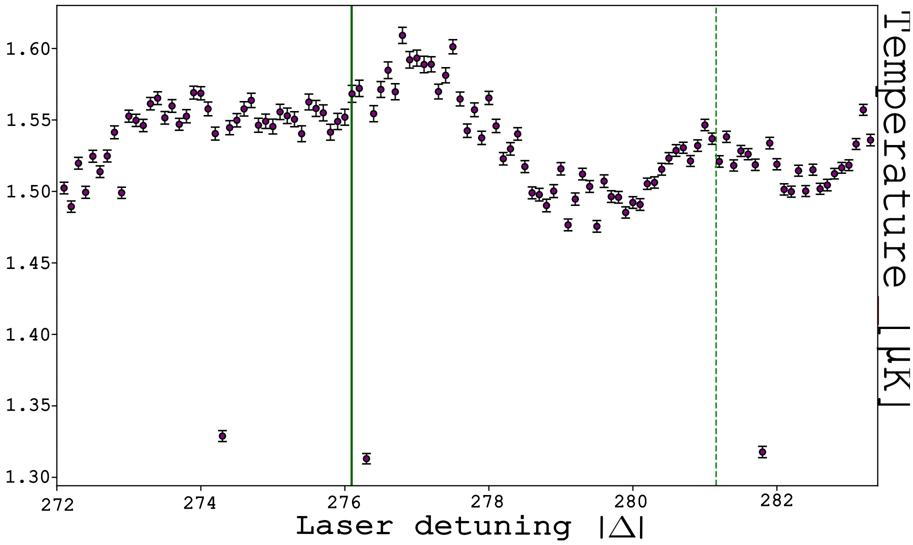
The exponential decay of this resonance, as for the previous one, has first been tested with an invariant temperature which gave an unsuccessful fit 3.23c with an associated really low value of $l_{opt}/I = (2.6 \pm 0.9) \times 10^{-6}\ a_0/(W/cm^2)$.

Then I solved the misfitting by considering the β factor dependant with the temperature since the gas is hotter when photoassociated and the plot of the temperature shows lots of fluctuations.

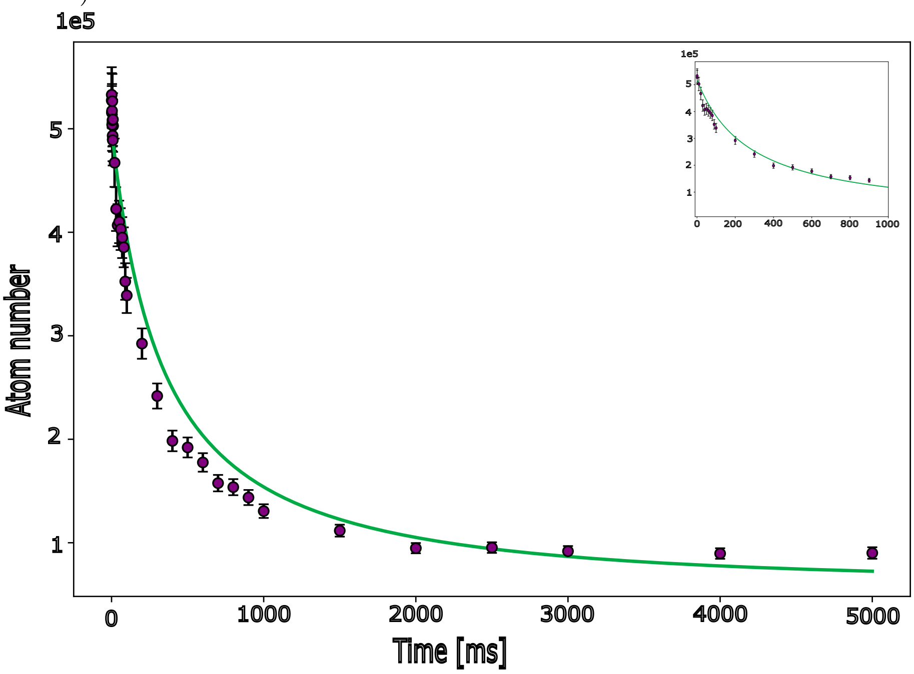
The interpolation of the temperature during the illumination time is then used to fit the data with a unique value of $\beta(T)$ for each point.



(a) Resonance spectrum at 276.83 ± 0.03 MHz ($I = 5.2$ W/cm^2 for t = 1000 ms).



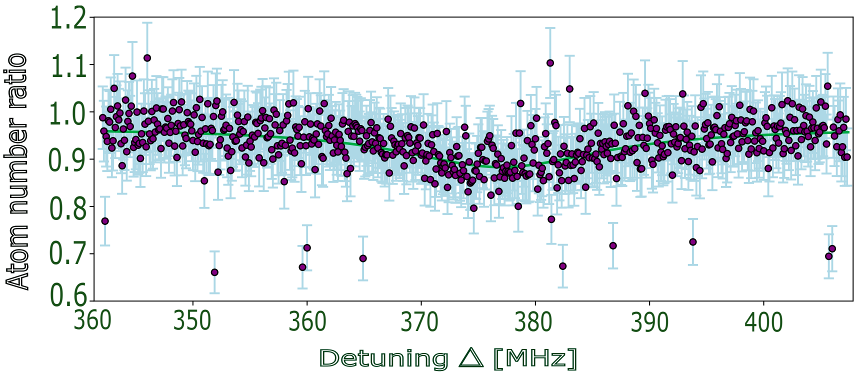
(b) Temperature of the gas at the 276.83 MHz resonance (figure to be added).



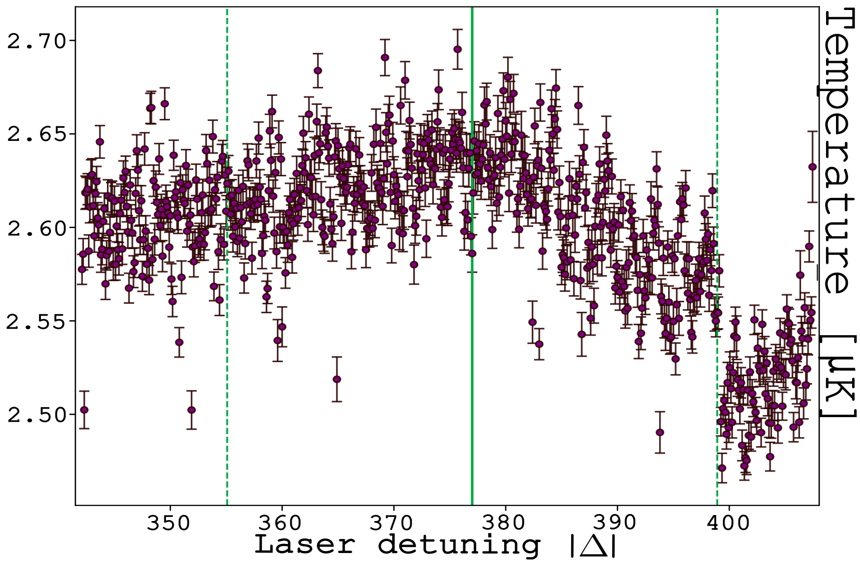
(c)

Figure 3.23: Characterization of the molecular line at 276.83 ± 0.03 MHz in ^{87}Sr ($I = 5.2$ W/cm^2). No exponential decay measurement is available for this line. All detunings are measured from the ^{87}Sr atomic resonance.

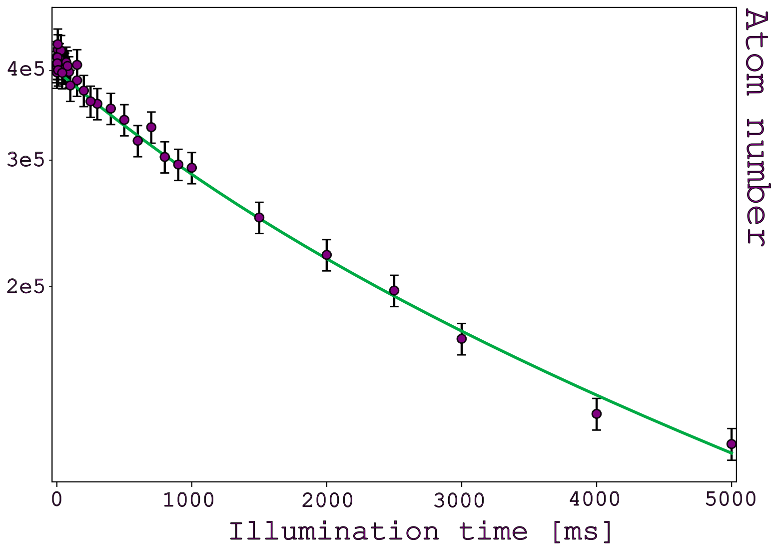
Line at 375.74 MHz The last measured molecular line at $f_3 = 375.74 \pm 0.48\,\mathrm{MHz}$ is obtained for a recompressed gas illuminated during 1000 ms with an intensity $I = 5.6\,\mathrm{W/cm^2}$. It has a really low contrast and still a large linewidth of $\Gamma = 10.96 \pm 2.68$ MHz. The temperature measurement in 3.24b shows a frank slope change between the resonance and the red detuned frequencies where it decreases to $\Delta T = 200$ nK. The temperature being mostly constant during the entire molecular resonance dN/dt measurement 3.24c is well described by the fitting function 3.76 valid for β independant on T. slopes ?



(a) attention premier trait 360 au lieu de 340 Resonance spectrum at 375.74 ± 0.48 MHz ($I = 5.6$ W/cm^2 for t = 1000 ms).



(b) Temperature of the gas at the 375.74 MHz resonance.



(c) Exponential decay (one- and two-body losses) at the 375.74 MHz resonance.

Figure 3.24: Characterization of the molecular line at 375.74 ± 0.48 MHz in ^{87}Sr ($I = 5.6$ W/cm^2 for t = 1000 ms). All detunings are measured from the ^{87}Sr atomic resonance.

The exponential decay of the second molecular line at $f = 276.83$ MHz is presented in figure ??:

Table 3.2: Summary of fit parameters and experimental conditions for data series 171, 195, and 184.

Freq. f **(MHz)**	259.01 ± 0.12	276.83 ± 0.03	375.74 ± 0.48
Depth	-0.307 ± 0.010	-0.364 ± 0.019	-0.083 ± 0.009
Width (MHz)	2.53 ± 0.17	7.48 ± 0.77	10.96 ± 2.68
Time t (ms)	1000	1000	1000
Intensity for resonance I (W/cm^2)	19	5.2	5.6
Intensity I for l_{opt} (W/cm^2)	19	31	19
l_{opt}/I $(a_0/(W/cm^2))$	$(6.11 \pm 3.34) \times 10^{-1}$	$(5.2 \pm 0.4) \times 10^{-1}$	$(3.24 \pm 2.25) \times 10^{-2}$

ajouter valeurs attendues

Defining l_{opt}

There are two ways to determine the optical length. One is to use the Lorentzian fit of the resonance. Following the formulation from [6], the inelastic two-body loss rate coefficient is expressed as:

$$K_2(\Delta) = \frac{4\pi\hbar}{\mu} \frac{\Gamma_{\text{mol}}^2 l_{\text{opt}}}{\left(\frac{E}{\hbar} + 2\pi\Delta\right)^2 + \frac{(\Gamma_{\text{stim}}+\Gamma_{\text{mol}})^2}{4}} \tag{3.82}$$

where μ is the reduced mass $m_{at}/2$, Γ_{mol} is the natural molecular linewidth, and Γ_{stim} is the light-induced stimulated width.

Experimentally, the loss profile is fitted using a standard empirical Lorentzian function where the frequency offset is explicitly written to match the resonance condition:

$$f(\Delta) = \frac{A}{1 + \left(\frac{\Delta + \frac{E}{2\pi\hbar}}{\Gamma_{\text{fit}}/2}\right)^2} = \frac{A(\Gamma_{\text{fit}}/2)^2}{(\Gamma_{\text{fit}}/2)^2 + \left(\Delta + \frac{E}{2\pi\hbar}\right)^2} \tag{3.83}$$

where A is the peak amplitude and Γ_{fit} is the full-width at half-maximum (FWHM) extracted by the fitting software.

By dividing both the numerator and denominator of Eq. 3.82 by $(2\pi)^2$, the identity with Eq. 3.83 becomes direct and allows us to map the physical parameters to our fit observables:

$$\Gamma_{\text{fit}} = \frac{\Gamma_{\text{stim}} + \Gamma_{\text{mol}}}{2\pi} \tag{3.84}$$

$$A = \frac{4\hbar}{\pi\mu} \frac{\Gamma_{\text{mol}}^2 l_{\text{opt}}}{\Gamma_{\text{fit}}^2} \tag{3.85}$$

Remarkably, the product of the amplitude and the square of the linewidth eliminates the temperature-dependent broadening terms. The optical scattering length l_{opt} can therefore be straightforwardly extracted from the experimental fit parameters via:

$$l_{\text{opt}} = \frac{\pi m_{at}}{8\hbar\Gamma_{\text{mol}}^2} \times \left(A \cdot \Gamma_{\text{fit}}^2\right) \tag{3.86}$$

The second way to define l_{opt} is straightforward : dN/dt fit gives an estimation of K_2 that is related with l_{opt} as $K_2 = \frac{16h}{m_{at} l_{opt}}$.

PA lines found on ^{87}Sr

These results show multiple surprising behaviors of our atoms :

- First, we have not been able to detect two-body losses before 259.01 MHz which is surprising taking into acount the expect frequency ranges on 87 isotope from mass scaling discussion of [8].
- Then we hardly exceed 30% of loss: all three datas present small depths from $8.3 \pm 0.9\%$ to $36.4 \pm 1.9\%$ for intensites greater than many experiments like [6] and similar atomic densities.
- In addition the linewidth of each line is large : 2.53 ± 0.17, 7.48 ± 0.77 MHz and 10.93 ± 2.68 MHz while we expect $\Gamma = \gamma_{mol} \simeq 2 \times \gamma_{at} \simeq 15$ kHz.

The plots of molecular lines added to the really low estimated l_{opt} values for high intensity expose multiple unexpected behaviours.
In the next subsection I will discuss the possible interpretation of these caracteristics.

3.3.3 Interpretation of low contrast

The surprising low number of the ^{87}Sr PA lines coupled to the large width and low strength of the lorentzian shape have been observed in [111]. where $^{87}Rb^{170}Yb$ bi-alkali molecules are photoassociated on the intercombination line. Only two spectroscopic lines have been observed with a low contrast while the intensity of the laser reaches $I = 40$ mW/cm^2 corresponding to a two-body loss rate $\beta = 5 \times 10^{-14}$ cm^3/s corresponding to a really low optical length.

Authors give two interpretations for this result : the first one is the strong dipole at low internuclear distance coming from the rubidium potentials, leading to atomic decay to $^1S_0$. Nevertheless this effect is estimated to be less than 10% probable.

The second interpretation is the predissociation process illustrated in Fig.3.25. Two free atoms from potential A are photoassociated to a vibrationnal state of a molecular potential B. Atomic potential C crosses potential B above its dissociation limit which creates a coupling between the two potentials ([112]). In the case of ^{87}Sr this potential is one of the hyperfine structure of the ^3P_J state. The bound atoms then decay to the continuum of the potential C that impacts the lifetime of the excited molecular state as

$$\Gamma_{eff} = \Gamma_{stim} + \Gamma_{predissociation} + \Gamma_{mol} \tag{3.87}$$

with $\Gamma_{predissociation} \propto \rho(\epsilon_f)$ where $\rho(\epsilon_f)$ is the kinetic energy continuum of potential C that dilutes the probability of transfering to the excited molecular state B. The predissociation process is then a loss channel for the molecular state that decreases the lifetime of the molecule and increases its linewidth.

In 3.5 the curves do not seem to be affected by the predissociation from the linewidth values. We were thinking that PA on the lowest hyperfine state as for Yb would avoid this effect but the data on ^{87}Sr show the contrary: for a reason we ignore this atom is affected by this process but the Yb with hyperfine structure is not.

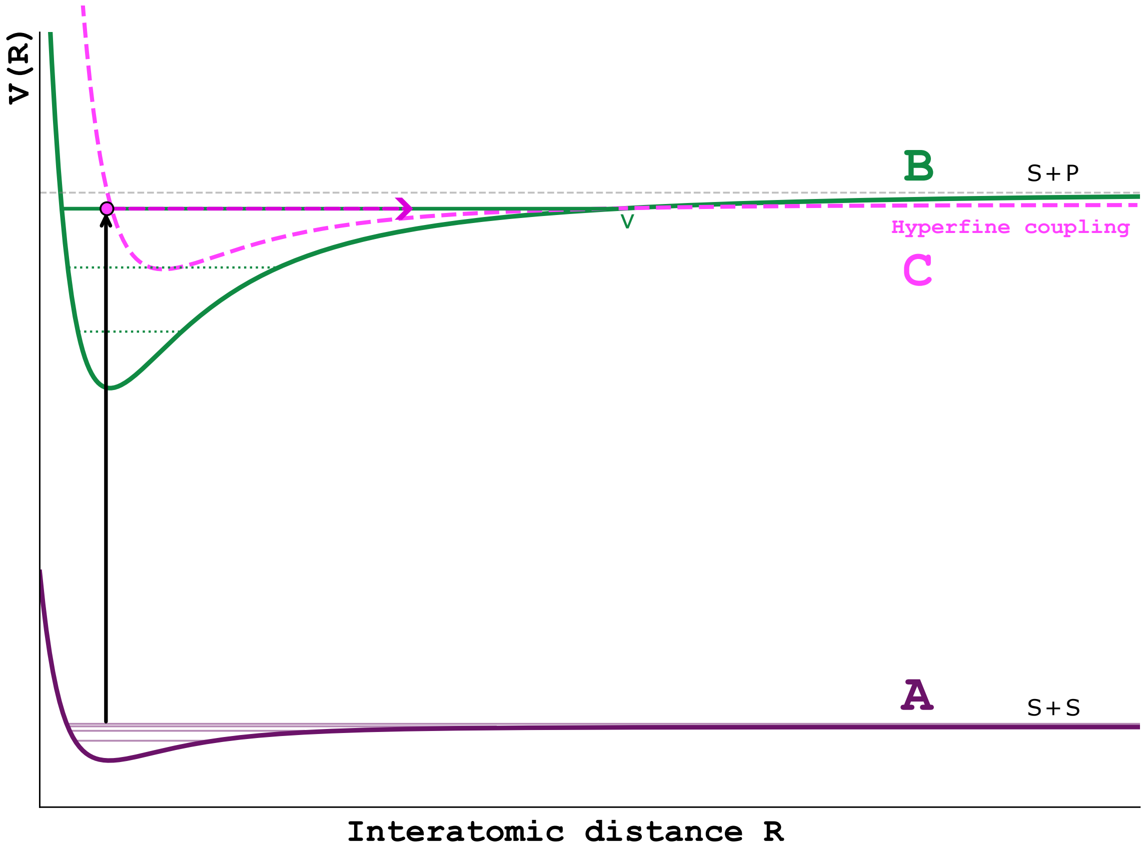


Figure 3.25: Scheme of predissociation which is a phenomenon that occurs when dissociation limit of an excited molecular potential C is under the closed channel of the excited molecule of potential B. Then the ν_B state is coupled to a density of states of potential C that modifies the linewidth $\Gamma_m ol$ of the molecular state

Changing the polarization of the PA laser could change the coupling strength with the atomic continuum. Same as for the magnetic field that could change the energy difference between the hyperfine states and so decrease the density of states. Both of these possibilities have been tested following this work I presented but no significant change has been observed on the linewidth and the contrast of the molecular lines.

3.3.3.1 Coupling to more energetic state from the IR

ODT can be a distructive source of PA process if it is resonant with high energy levels at $\lambda = 1064$nm from the target state of the first molecular excited state. The experimental sequence of 3.26 implements a chopping of optical trap in opposite phase with PA laser that suppresses eventual coupling to much higher levels.

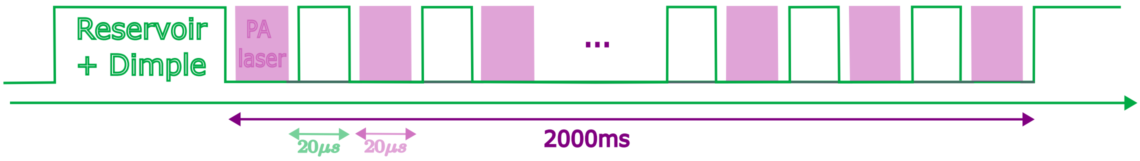


Figure 3.26: Experimental sequence of IR and PA laser chopping to cancel possible coupling to higher energy levels around $\lambda = 1064nm$ that would destruct or decrease collision process. The RF signal of ODT and Raman AOMs is modulated by a GBF at frequency 25 kHz in opposite phaseso the atoms in the trap are not lost during the 100μ s falling time when photoassociated.

3.3.4 Conclusion and outlook

To conclude on this chapter I started to present theoretical aspect of PA process by introducting fundamental material such as scattering length, optical length, atoms evolution time for two-body collisions... Then I presented the experimental results on ^{88}Sr on intercombination line as a reference to experimentally characterize the process. This chapter ends up with the measurement of molecular lines on an unpolarized gas of ^{87}Sr that comes with low contrast and wide linewidth, explained by predissociation process higly present for numerous molecular potentials such as ^{87}Sr being an atom with hyperfine structure.

Predissociation could be proved by implementing again the optical pumping on the experiment: the number of potentials will decrease and the density of states $\rho(\epsilon_f)$ also. Then the contrast of the molecular lines and their associated linewidth will be improved.
Optical pumping tests have already been operated to enhance SU(2) gas with maximum value $m_F = 9/2$ and $m_F = -9/2$ in order to have the maximum value of T, m_T and

Conclusion

Ouverture sur retournement champ pour adresse les mF extremes

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Appendix A

Algorithms

The variance

$$\mathrm{Var}(\hat{J}_x) = \frac{1}{2}\left(J(J+1) - m^2\right) \tag{A.1}$$

is maximized for the projection $m = 0$ with $J = NS$, S the total spin, N the atomic number. This configuration defines the **Dicke states**, powerful resources for quantum-enhanced metrology. The sensitivity of such a state is quantified by the **Quantum Fisher Information** (QFI). For a $m = 0$ Dicke state, the QFI reaches:

$$F_Q = 4\mathrm{Var}(\hat{J}_x) = 2NS(NS+1) \tag{A.2}$$

According to the **Cramér-Rao inequality**, which relates the phase uncertainty $\Delta\phi$ to the QFI:

$$\Delta\phi \geq \frac{1}{\sqrt{F_Q}} \tag{A.3}$$

it becomes evident that the precision on ϕ is improved by a factor of approximately $\sqrt{N}$ compared to the SQL, scaling as $1/NS$, and approaching the **Heisenberg Limit**. This represents the ultimate precision allowed by the laws of quantum mechanics.